

The First Response to Antigen

5

What the USMLE Requires You to Know

- The meaning of antigen, epitope, antigenic determinant, and hapten
- The sequence of inflammatory events and their role in the initial response to invasion
- The mechanism and chemoattractive molecules involved in phagocyte extravasation
- The steps of phagocytosis and mechanisms of intracellular killing
- The meaning of opsonization and the molecules involved
- The clinical sequelae of defects in chemotaxis and phagocytosis (LAD and CGD)

Historically, the word **antigen** was meant to designate a substance capable of inducing the formation of a specific antibody: an **antibody-generating** substance. Today the word is used almost interchangeably with the word **immunogen**—an **immune response-generating** substance—to encompass any substance capable of activating and generating a response from any committed lymphocyte. For a molecule to be an immunogen, it needs to fit three basic criteria:

- It must be recognized as **foreign**.
- It must have a certain degree of **chemical complexity**.
- It must have a **molecular weight** of at least 5,000 to 10,000 K.

B lymphocytes are capable of recognizing molecules of almost any chemical composition. The portion of the foreign molecule that actually fits into the idiotype of the B-cell receptor is quite small: 5 to 6 amino acids in the case of a protein or 4 to 5 hexose units in the case of a carbohydrate. This portion of the molecule that has three-dimensional complementarity with the idiotype is called the **epitope** or **antigenic determinant** of the immunogen. Because most of the microbes that challenge the immune system are quite large, most naturally occurring antigens have many copies of the same epitopes. The size of a molecule is a critical factor in its immunogenicity because B lymphocytes can only be activated when their antigen receptors are **crosslinked** by the accommodation of more than one identical epitope. This serves as the first signal to then send the message through the signal transduction complex to activate the cell. Molecules that possess only one epitope are called **haptens** because they can occupy only one idiotype of the double-armed B-cell receptor. T lymphocytes recognize peptides of 10 to 20 amino acids in length only when presented to them in the groove of an MHC molecule on the surface of an antigen-presenting cell.

In A Nutshell

- An antigen or immunogen is a substance that can elicit an immune response.
- It must be foreign, chemically complex, and large.

Note

The idiotype of a cell receptor fits the epitope (antigenic determinant) of the antigen.

Note

Haptens are single antigenic determinants.

Clinical Correlate

Drug allergies with penicillin and other agents such as streptomycin, aspirin, sulfa drugs, succinyl choline, and some opiates can be induced by small doses of the drug and are not consequences of the pharmacologic or physiologic effects of the drugs. Typically, an allergic response occurs 7 to 14 days following exposure, and the first symptoms may be mild. Subsequent drug exposures can result in severe and life-threatening anaphylaxis (see Chapter 13). Most drugs are low molecular weight compounds that are not capable of inducing immune responses by themselves—they act as **haptens**. Inside the body, however, these agents can become conjugated to body proteins (the **carrier**), and the hapten-carrier conjugate serves as the immunogen for the ensuing allergic response.

In A Nutshell

- The acute inflammatory response is a first response to invasion.
- Cytokines increase expression of selectin-type adhesion molecules.

In A Nutshell

Extravasation of phagocytes involves rolling, activation by chemoattractants, arrest/adhesion, and transendothelial migration.

Antigens are normally introduced into the body across the mucosa or the epithelia. The **acute inflammatory response** is often the first response to this invasion and represents a response of the innate immune system to block the challenge. The first step in the acute inflammatory response is activation of the vascular endothelium in the breached epithelial barrier. **Cytokines** and other inflammatory mediators released in the area as a result of tissue damage induce expression of **selectin-type adhesion molecules** on the endothelial cells. **Neutrophils** are usually the first cell to bind to the inflamed endothelium and extravasate into the tissues, peaking within 6 hours. Monocytes, macrophages, and even eosinophils may arrive 5 to 6 hours later in response to neutrophil-released mediators.

The extravasation of phagocytes into the area requires four sequential, overlapping steps:

Step 1: Rolling

Phagocytes attach loosely to the endothelium by low-affinity, selectin-carbohydrate interactions. E-selectin molecules on the endothelium bind to mucin-like adhesion molecules on the phagocyte membrane and bind the cell briefly, but the force of blood flow into the area causes the cell to detach and reattach repeatedly, rolling along the endothelial surface until stronger binding forces can be elicited.

Step 2: Activation by chemoattractants

Chemokines released in the area during inflammation, such as interleukin 8 (IL-8), complement split product C5a, and N-formyl peptides produced by bacteria bind to receptors on the phagocyte surface and trigger a G-protein-mediated activating signal. This signal induces a conformational change in integrin molecules in the phagocyte membrane that increases their affinity for immunoglobulin-superfamily adhesion molecules on the endothelium.

Step 3: Arrest and adhesion

Interaction between **integrins** and **Ig-superfamily cellular adhesion molecules** (Ig-CAMs) stabilizes adhesion of the phagocyte to the endothelial cell.

Step 4: Transendothelial migration

The phagocyte extends pseudopodia through the vessel wall and extravasates into the tissues.

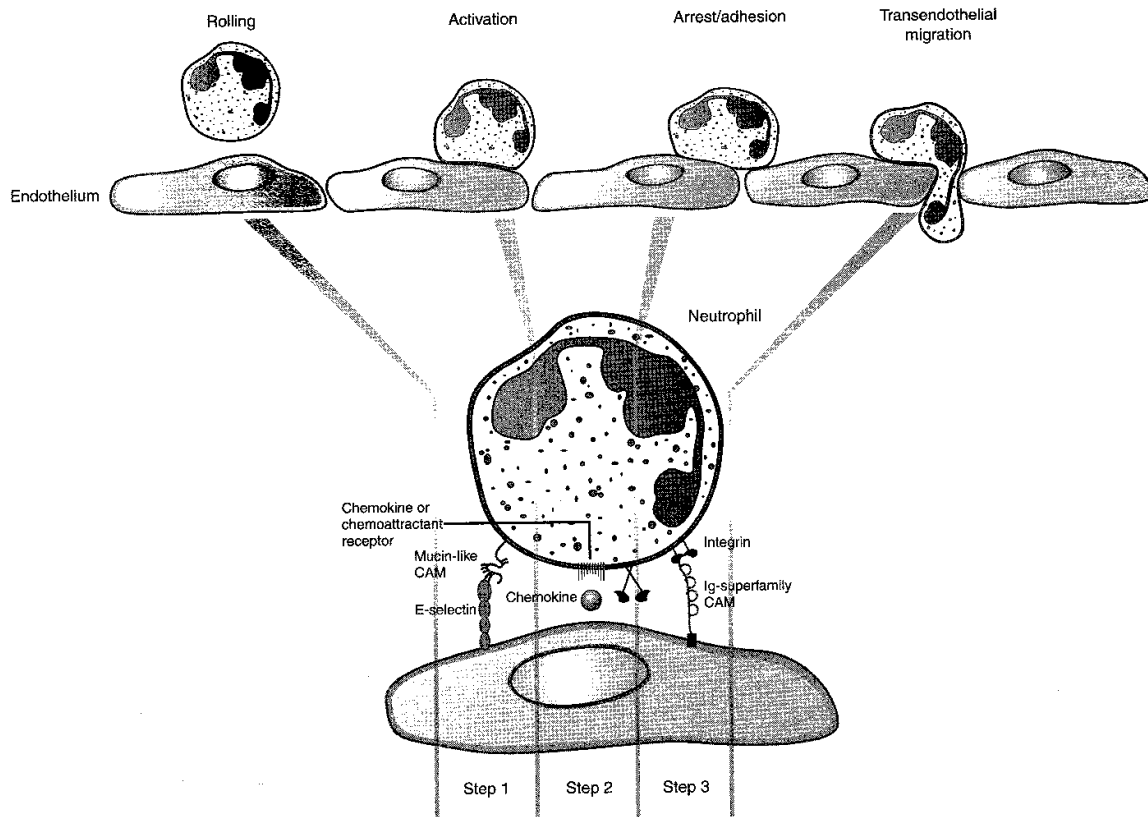


Figure II-5-1. The Steps of Phagocyte Extravasation

Clinical Correlate

Leukocyte adhesion deficiency is a rare autosomal recessive disease in which there is an absence of CD18, which is the common β_2 chain of a number of integrin molecules. A key element in the migration of leukocytes is integrin-mediated cell adhesion, and these patients suffer from an inability of their leukocytes to undergo adhesion-dependent migration into sites of inflammation. The first indication of this defect is often omphalitis, a swelling and reddening around the stalk of the umbilical cord. These patients are no more susceptible to virus infection than are normal controls, but they suffer recurrent, chronic bacterial infections. These patients frequently have abnormally high numbers of granulocytes in their circulation, but migrations into sites of infection is not possible, so **abscess and pus formation do not occur**.

One method of diagnosing LAD involves evaluating expression (or lack thereof) of the β chain (CD18) of the integrin by flow cytometry.

Bacterial infections in these patients can be treated with antibiotics, but they recur. If a suitable bone marrow donor can be found, the hematopoietic system of the patient is destroyed with cytotoxic chemicals and a bone marrow transplant is performed.

In A Nutshell

Substances chemoattractive to neutrophils include IL-8, C5a, fibrinopeptides, leukotriene B₄, and formyl methionyl peptides.

Once in the tissues, neutrophils express increased levels of receptors for chemoattractants and exhibit **chemotaxis** migrating up a concentration gradient toward the attractant. Neutrophils release chemoattractive factors that call in other phagocytes.

Table II-5-1. Chemoattractive Molecules

Chemoattractive Molecule	Origin
Chemokines (IL-8)	Tissue mast cells, platelets, neutrophils, monocytes, macrophages, eosinophils, basophils, lymphocytes
Complement split product C5a	Endothelial damage → activation Hageman factor → plasmin activation
Fibrinopeptides	Endothelial damage → activation Hageman factor → thrombin → fibrin clot degradation
Leukotriene B ₄	Membrane phospholipids of macrophages, monocytes, neutrophils, mast cells → arachidonic acid cascade → lipoxygenase pathway
Formyl methionyl peptides	Released from microorganisms

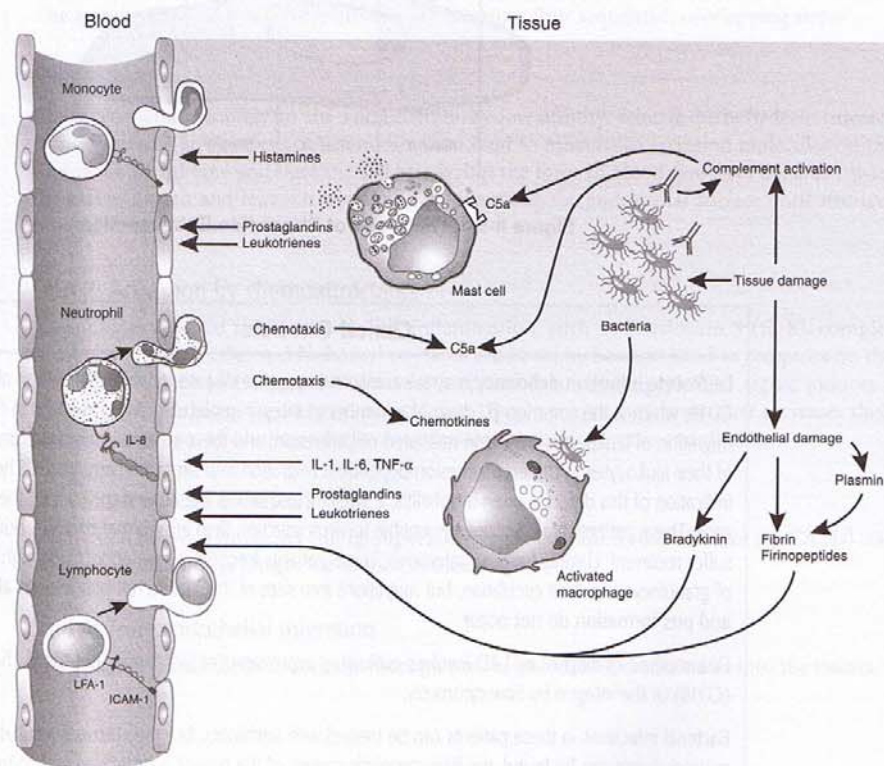


Figure II-5-2. The Acute Inflammatory Response

Once chemotaxis of phagocytic cells into the area of antigen entry is accomplished, these cells ingest and digest particulate debris, such as microorganisms, host cellular debris, and activated clotting factors. This process, called **phagocytosis**, involves:

- Extension of pseudopodia to engulf attached material
- Fusion of the pseudopodia to trap the material in a phagosome
- Fusion of the phagosome with a lysosome to create a phagolysosome
- Digestion
- Exocytosis of digested contents

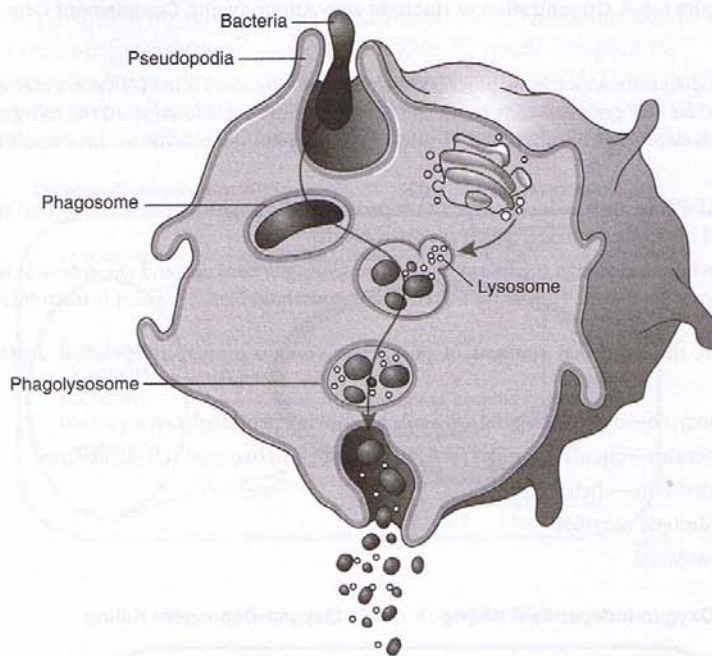


Figure II-5-3. Phagocytosis

Both macrophages and neutrophils have membrane receptors for certain types of antibody (IgG) and certain complement components (C3b). If an antigen is coated with either of these materials, adherence and phagocytosis may be enhanced by up to 4,000-fold. Thus, antibody and complement are called **opsonins**, and the means by which they enhance phagocytosis is called **opsonization**.

In A Nutshell

Phagocytosis involves:

- Extension of pseudopodia
- Formation of phagosome
- Fusion with lysosome to form phagolysosome
- Digestion
- Exocytosis

In A Nutshell

Opsonization is enhancement of phagocytosis with:

- IgG
- C3b

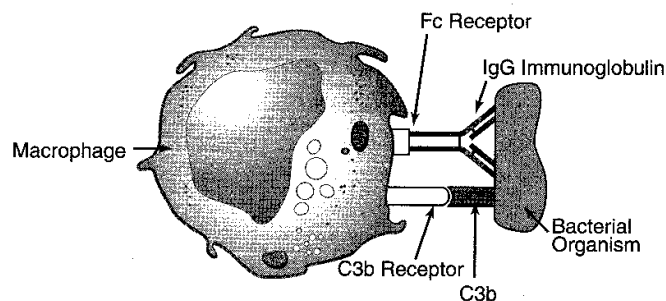


Figure II-5-4. Opsonization of Bacteria with Antibody and Complement C3b

In A Nutshell

Intracellular killing mechanisms include:

- Toxic oxygen metabolites
- Toxic halide radicals
- Lysosomal contents:
 - Lysozyme
 - Defensins
 - Lactoferrin
 - Hydrolytic enzymes

During phagocytosis, a metabolic process known as the respiratory burst activates a membrane-bound oxidase that generates oxygen metabolites, which are toxic to ingested microorganisms. Two oxygen-dependent mechanisms of intracellular digestion are activated as a result of this process.

- **NADPH oxidase** reduces oxygen to superoxide anion, which generates hydroxyl radical and hydrogen peroxide, which are microbicidal.
- **Myeloperoxidase** in the lysosomes acts on hydrogen peroxide and chloride ions to produce hypochlorite (the active ingredient in household bleach), which is microbicidal.

In addition, the lysosomal contents of phagocytes contain oxygen-independent degradative materials:

- Lysozyme—digests bacterial cell walls by cleaving peptidoglycan
- Defensins—circular peptides that form channels in bacterial cell membranes
- Lactoferrin—chelates iron
- Hydrolytic enzymes

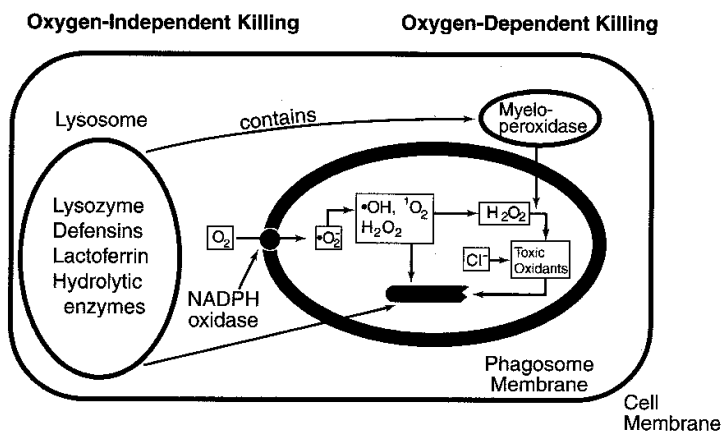


Figure II-5-5. Metabolic Stimulation and Killing Within the Phagocyte

Clinical Correlate

Chronic Granulomatous Disease

When defects occur in the ability of phagocytes to perform their critical functions as first responders and intracellular destroyers of invading antigens, clinically important pathologic processes ensue. Such defects tend to make the patient susceptible to severe infections with **extracellular bacteria and fungi**.

Chronic granulomatous disease (CGD) is an inherited deficiency in the production of one of several subunits of **NADPH oxidase**. This defect eliminates the phagocyte's ability to produce many critical oxygen-dependent intracellular metabolites ($\cdot\text{O}_2^-$, $\cdot\text{OH}$, $^1\text{O}_2$, and H_2O_2). The two other intracellular killing mechanisms remain intact (myeloperoxidase + $\text{H}_2\text{O}_2 \rightarrow \text{HOCl}$ and lysosomal contents). If the patient is infected with a catalase-negative organism, the H_2O_2 waste product produced by the bacterium can be used as a substrate for myeloperoxidase, and the bacterium is killed. If, however, the person is infected with a catalase-positive organism (e.g., *Staphylococcus*, *Klebsiella*, *Serratia*, *Aspergillus*), the myeloperoxidase system lacks its substrate (because these organisms destroy H_2O_2), and the patient is left with the oxygen-independent lysosomal mechanisms that prove inadequate to control rampant infections. Thus, CGD patients suffer from chronic, recurrent infections with **catalase-positive** organisms.

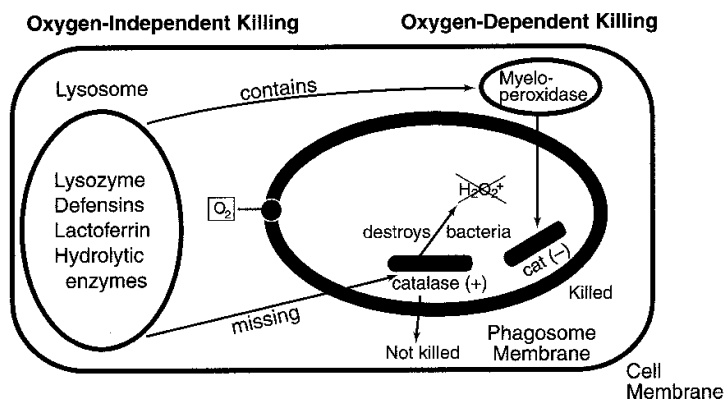


Figure II-5-6. Intracellular Killing in CGD

Failures of phagocytic cells to generate oxygen radicals are easily detected by the nitroblue tetrazolium (NBT) reduction test or neutrophil oxidative index (NOI; a flow cytometric assay).

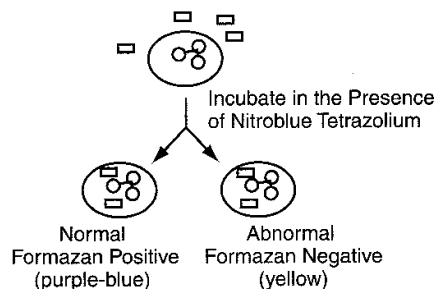


Figure II-5-7. Nitroblue Tetrazolium Reduction

Chapter Summary

- An antigen or immunogen is a substance capable of activating and generating a response from any committed lymphocyte.
- To be immunogenic, a substance must be recognized as foreign and have chemical complexity and sufficient size.
- The portion of an antigen that fits into the idiotype of an antigen receptor is called the epitope or antigenic determinant.
- Haptens are single antigenic determinants that can only generate immune responses if they are linked to larger carrier proteins. This is the mechanism of elicitation of many drug allergies.
- After antigen is introduced across an anatomic barrier, phagocytes (first neutrophils and then monocytes) will extravasate into the area of injury by 1) rolling, 2) activation with chemoattractants, 3) arrest/adhesion, and 4) transendothelial migration.
- Important chemoattractants include IL-8, C5a, leukotriene B₄, and N-formyl peptides.
- Integrins on the phagocyte membrane bind to Ig-CAMs on endothelia to mediate adhesion.
- Once through the endothelium, phagocytes are attracted to the area of injury by chemokines of the IL-8 family, complement split products, fibrinopeptides, leukotriene B₄, and formyl methionyl peptides.
- Phagocytosis involves: the formation of pseudopodia that trap particulate material in a phagosome; addition of lysosomes to cause intracellular digestion within the phagolysosome; and exocytosis of the digested materials.
- Opsonization is the coating of particles with IgG or C3b (or both) to enhance phagocytosis.
- Ingestion is accompanied by a respiratory burst, which generates the toxic metabolites necessary to destroy intracellular materials.
- NADPH oxidase produces toxic oxygen radicals, and myeloperoxidase generates hypochlorite.
- Lysosomal contents (lysozyme, defensins, lactoferrin, and hydrolytic enzymes) are also toxic to ingested material.
- Patients with chronic granulomatous disease (CGD; an inherited deficiency of NADPH oxidase) are susceptible to chronic, recurrent infections with catalase-positive organisms.

Review Questions

1. A rabbit hunter in Arkansas is diagnosed with ulceroglandular tularemia and treated with streptomycin. Within a week, he returns to the hospital. The tularemic papule, lymphadenopathy, and bacteremia have resolved, but he has now developed a raised, itching skin rash and a fever. The drug was discontinued, and the symptoms subsided. What was the role of streptomycin in this case?
 - (A) It acted as a B-cell mitogen
 - (B) It acted as a hapten
 - (C) It acted as a provider of costimulatory signals
 - (D) It acted as a superantigen
 - (E) It acted as an immunogen

2. During World War II, when quinine was used as a prophylactic against malaria infections in U.S. personnel on long-term assignment to the South Pacific, a small proportion of soldiers developed blackwater fever (chronic kidney damage from the autoimmune effects of complement-mediated hemolysis of quinized red blood cells). In this case, the quinine played the role of
 - (A) autoantigen
 - (B) carrier
 - (C) hapten
 - (D) immunogen
 - (E) reagin

3. A two-year-old child who has suffered recurrent bacterial infections is evaluated for immunologic deficiency. The child has age-normal numbers of CD19+ and CD3+ cells in the peripheral blood and an extreme neutrophilia. The nitroblue tetrazolium dye reduction test is normal. What is the most likely defect in this child?
 - (A) Absence of CCR4
 - (B) Absence of CD18
 - (C) Absence of interleukin-1
 - (D) Absence of interleukin-4
 - (E) Absence of tumor necrosis factor- α

4. A 2-year-old boy is admitted to the hospital for workup of a possible immunologic disorder. His history is remarkable for the occurrence of multiple skin infections involving *Staphylococcus*, *Pseudomonas*, and *Candida*. On examination the child has cervical lymphadenopathy and mild hepatosplenomegaly. Blood tests reveal an elevated erythrocyte sedimentation rate and neutrophilia. The nitroblue tetrazolium dye reduction test and neutrophil oxidative index are negative. What is the most likely defect in this child?
- (A) C3 deficiency
 - (B) Deficiency of CD18
 - (C) Deficiency of myeloperoxidase
 - (D) NADPH oxidase deficiency
 - (E) Phagocyte granule structural defect
5. It has been learned in animal experiments that there are advantages to eliciting nonspecific inflammation at the site of inoculation of antigen toward the ultimate development of a protective immune response to that immunogen. Which of the following substances, if introduced with a vaccine, would serve the purpose of attracting a neutrophilic infiltrate into the area?
- (A) Complement component 3b
 - (B) Immunoglobulin G
 - (C) Interleukin-8
 - (D) Myeloperoxidase
 - (E) Tumor necrosis factor- α

Answers and Explanations

1. **The correct answer is B.** Many drug allergies, such as the one described here, are hapten-carrier immune responses. The drug is not large enough by itself to elicit an immune response (it is a hapten), but when it becomes covalently coupled to the body's own proteins (which act as carriers), the combined molecule becomes immunogenic, and a response against one's own tissues is elicited.

Acting as a B-cell mitogen (**choice A**) is not correct. B-cell mitogens, such as pokeweed mitogen and lipopolysaccharide, cause polyclonal proliferation of B cells and elaboration of IgM antibodies. The drug allergy described here is not a polyclonal response, but a specific anti-altered-self response generated by T and B lymphocytes and production of IgE antibodies.

Acting as a provider of costimulatory signals (**choice C**) is not correct. The costimulatory signals required to activate B and T lymphocytes include CD28/B7 and CD40/CD40L interactions. These are additional interactions (beyond the specific recognition of antigen) required for the activation of B and T lymphocytes. Although these costimulatory signals would be involved in the evolution of this allergic response, the streptomycin does not serve as a costimulatory signal.

Acting as a superantigen (**choice D**) is not correct. Superantigens are materials that crosslink the variable β domain of the T-cell receptor and the α -chain of class II MHC molecules. They induce activation of all T cells that express receptors with a particular V β domain. The resulting T-cell mitogenesis causes overproduction of T-cell and macrophage cytokines and system-wide pathology. Toxic shock syndrome toxin-1 and *Streptococcus pyogenes* erythrogenic exotoxins act as superantigens.

Acting as an immunogen (**choice E**) is not correct because streptomycin is not large enough to be immunogenic. Immunogens must be large enough to have at least two epitopes. It is only through binding to a larger carrier protein (the patient's own tissue proteins) that a hapten such as a drug can become immunogenic.

2. **The correct answer is C.** This is a case of autoimmune hemolytic anemia generated during a hapten-carrier immune response. The drug is not large enough by itself to elicit an immune response (it is a hapten), but when it becomes covalently coupled to the red blood cell proteins (which act as carriers), the combined molecule becomes immunogenic. Antibodies are generated and bind to the drug-RBC complex, complement is activated, and the red blood cells are lysed.

Autoantigen (**choice A**) is not correct because the immune response is being generated against an "altered-self" component. The RBCs of the patient are not themselves the immunogen: It is the complex of drug-RBC protein that elicits the response. The effect is certainly similar to an autoimmune response: The patient destroys his own tissues. However, the eliciting immunogen is altered-self, so this is not a case where failure of self-tolerance causes disease.

Carrier (**choice B**) is not correct because carriers must be protein molecules. The carrier component of a hapten-carrier complex is recognized by T lymphocytes. Because T lymphocytes are only capable of recognizing peptides presented in the groove of major histocompatibility complex antigens, a molecule such as quinine is "invisible" to T cells.

Immunogen (**choice D**) is not correct because quinine is not large enough to be immunogenic. Immunogens must be large enough to have at least two epitopes. It is only through binding to a larger carrier protein (the patient's own tissue proteins) that a hapten such as a drug can become immunogenic.

Reagin (**choice E**) is a word that is used to denote an immunogen that elicits an IgE-antibody response. Although some drug allergies are indeed IgE-mediated immediate hypersensitivity responses, this one is not. The antibodies that activate complement are IgM and IgG. IgE does not activate complement.

3. **The correct answer is B.** This child has leukocyte adhesion deficiency (LAD), which is a genetic deficiency of CD18. CD18 is an essential component of a number of integrins, and absence of these molecules causes the inability of WBCs to migrate into sites of inflammation. Thus in this patient, the blood contained abnormally high numbers of neutrophils, but they were unable to extravasate. CD18 is a component of LFA-1, CR3, and CR4.

Absence of CCR4 (**choice A**) would cause difficulties in extravasation and migration of activated T cells and monocytes. This chemokine receptor is not found on neutrophils and therefore would have no effect on neutrophil migration.

Absence of interleukin-1 (**choice C**) might cause difficulties in producing the acute and chronic inflammatory responses. This cytokine, frequently referred to as the endogenous pyrogen, produces fever, acute phase protein production, and many other results critical to inflammation. However, the actions of IL-1 are extremely redundant with those of IL-6 and tumor necrosis factor- α , so such a condition might have no clinically observable effects.

Absence of interleukin-4 (**choice D**) would result in defects in the ability to mount a normal IgE antibody response. This cytokine also serves as the major stimulus for the development of TH2 cells from naive helper T cells, so its absence would be likely to have profound effects on all aspects of the secondary antibody response.

Absence of tumor necrosis factor- α (**choice E**) might cause difficulties in producing the acute and chronic inflammatory responses. This cytokine has many functions that are redundant with those of IL-1 and IL-6, so such a condition might have no clinically observable effects.

4. **The correct answer is D.** This child has chronic granulomatous disease (CGD). The history indicates he has had recurrent infections with catalase-positive organisms and has a defect in generating oxygen radicals intracellularly in his phagocytic cells (the negative nitroblue tetrazolium test and neutrophil oxidative index). This genetic defect arises from a failure to produce one of the subunits of NADPH oxidase, which makes the individual incapable of producing intracellular oxygen radicals. Redundant intracellular killing mechanisms (myeloperoxidase and lysosomal contents) are still functional in these patients, but when they are infected with catalase-positive organisms, the substrate for myeloperoxidase (hydrogen peroxide) is destroyed, and the only remaining intracellular killing mechanism (lysosomes) is insufficient to protect from infection.

C3 deficiency (**choice A**) would cause increased susceptibility to pyogenic infections because C3b is an important opsonin that enhances phagocytosis of extracellular organisms. All extracellular bacteria would be included in this list, not simply catalase-positive ones, as mentioned here. The NBT and NOI would not be negative in this case.

Deficiency of CD18 (**choice B**) is the cause of leukocyte adhesion deficiency (LAD). Because CD18 is the common β chain of the β_2 integrins, its absence compromises leukocyte function antigen (LFA)-1, as well as complement receptors 3 and 4. Patients with LAD suffer recurrent infections with extracellular pathogens (not just catalase-positive ones) because of defective opsonization, mobilization, adhesion, and chemotaxis. The NBT and NOI would be positive.

Deficiency of myeloperoxidase (**choice C**) results from a deficiency of an important granule enzyme in phagocytic cells. However, because there are so many redundant mechanisms of intracellular killing, these patients generally have mild symptoms or none at all.

A phagocyte granule structural defect (**choice E**) is responsible for the Chediak-Higashi syndrome. These patients have chemotactic and degranulation defects, lack NK activity, and have partial albinism.

5. **The correct answer is C.** The only substance on the list that is chemotactic for neutrophils is IL-8. Other neutrophil chemotactic factors that might have been mentioned would include C5a, fibrinopeptides, leukotriene B₄, and formyl methionyl peptides.

Complement component 3b (**choice A**) is not chemotactic for neutrophils. It acts as an opsonin, enhancing the phagocytosis of coated particles.

Immunoglobulin G (**choice B**) is not chemotactic for neutrophils. It acts as an opsonin, enhancing the phagocytosis of coated particles.

Myeloperoxidase (**choice D**) is not chemotactic for neutrophils. It is an enzyme in the lysosomes of phagocytic cells that generates toxic halide radicals intracellularly when exposed to its substrate, hydrogen peroxide.

Tumor necrosis factor- α (**choice E**) is not chemotactic for neutrophils. It is produced by macrophages and is involved with the production of chronic inflammation and cytotoxicity.

The Processing and Presentation of Antigen

6

What the USMLE Requires You to Know

- How the grooves of class I and II MHC are loaded with peptides
- The three signals required for T-cell activation (TCR binding, costimulatory molecules, and cytokines)
- The clinical results of MHC class II deficiency (bare lymphocyte syndrome)
- How superantigens act
- The subclasses of T helper (TH) cells, their functions, and regulation

Although some small, easily digestible antigens are almost totally degraded and exocytosed by phagocytes, as we saw in the last chapter, the critical first step in the elicitation of the adaptive immune response to a first antigenic challenge is the processing of such antigen for the presentation to naive T lymphocytes. Professional antigen-presenting cells (dendritic cells, macrophages) load partially degraded peptides they have ingested into the groove of the class II MHC molecule, so that this can be presented to T cells with idiotypes complementary to that structure. This is accomplished by the **endosomal (exogenous) pathway** of MHC loading.

When MHC class II molecules are produced in the endoplasmic reticulum of an antigen-presenting cell (APC), in addition to the α and β chains, a third chain called the **invariant chain** is synthesized at the same time. This blocks the peptide-binding groove so **no** normal cellular peptides can accidentally be attracted there. As the molecule is completed, it is transported in a vesicle to the location of endocytic vesicles containing the ingested internalized peptides. As these vesicles fuse, the invariant chain is degraded, and peptides in the vacuole are loaded into the MHC II groove. The MHC class II-peptide complex is then transported to the cell surface where it will be accessible for interaction with any T lymphocyte with a complementary TCR.

In A Nutshell

MHC class II molecules are loaded with peptides by the endosomal (exogenous) pathway.

Note

The invariant chain prevents any normal cellular peptides from being bound.

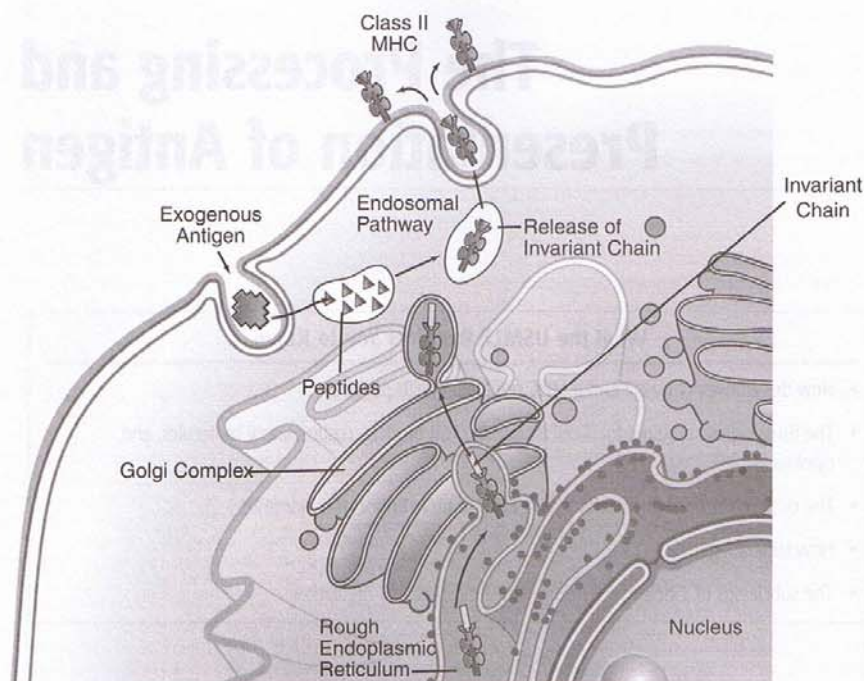


Figure II-6-1. The Exogenous Pathway of Binding Peptides to Class II MHC Molecules

In A Nutshell

- MHC I molecule is loaded with peptides via the endogenous pathway.
- CD8⁺ T cells recognize MHC I/peptide.

As with MHC class II molecules expressed on the surface of APCs and lymphocytes throughout the body, the MHC class I molecule presents small peptides in the groove at its N-terminal end, but the origin of these peptides is not from endocytic processing, but a different, **endogenous** pathway of production. Proteins synthesized in the cell cytosol are routinely degraded in proteasomes, and the peptides from these proteins are transported through a peptide transporter, known as the TAP complex, into the endoplasmic reticulum, where they have the opportunity to bind to freshly synthesized MHC class I proteins. These are then transported to the cell membrane where they may be presented to **CD8⁺ T lymphocytes** (see Chapter 8).

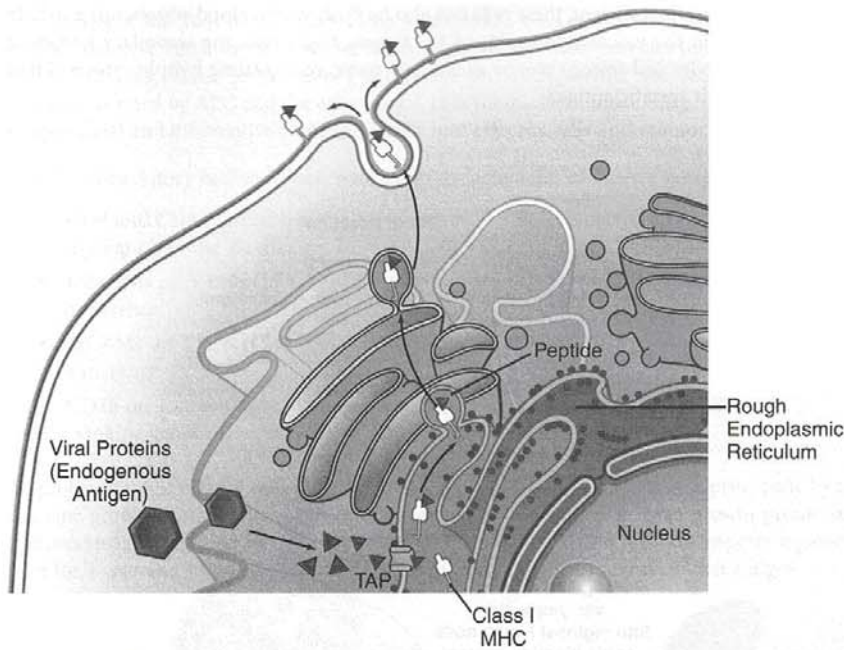


Figure II-6-2. The Endogenous Pathway of Binding Peptides to Class I MHC Molecules

Table II-6-1. Human MHC Summary

	MHC Class I	MHC Class II
Names	HLA-A, -B, and -C	HLA-DP, -DQ, -DR
Tissue distribution	All nucleated cells, platelets	B lymphocytes, monocytes, macrophages, dendritic cells, Langerhans cells, activated T cells, activated endothelial cells
Recognized by	Cytotoxic T cells (CD8+)	TH cells (CD4+)
Peptides bound	Endogenously synthesized	Exogenously processed
Function	Elimination of abnormal (infected) host cells by cytotoxic T cells	Presentation of foreign antigen to TH cells
Invariant chain	No	Yes
β_2 microglobulin	Yes	No

Within a few hours of the initiation of the acute inflammatory response by the breaching of the mucosa or epithelia, the professional APCs that have phagocytosed and processed the invading antigen begin to leave the area via lymphatic vessels. Dendritic cells, with their long, finger-like processes, are probably the most efficient of these cells and retract their membranous processes to round up and begin the journey to the closest lymph node. Thus, phagocytes with MHC class II molecules loaded with peptides digested from the invading antigen enter the lymph node through the afferent lymphatics and become trapped in the meshwork of the organ. If the

In A Nutshell

APCs with MHC class II/peptide molecules travel to the secondary lymphoid organs.

initial tissue damage is sufficient, these cells can also be flushed into blood vessels, ultimately to become trapped in the vascular sinusoids of the spleen. Regardless, the secondary lymphoid organs (lymph nodes and spleen) are the sites where naive, recirculating lymphocytes will first be exposed to their specific antigen.

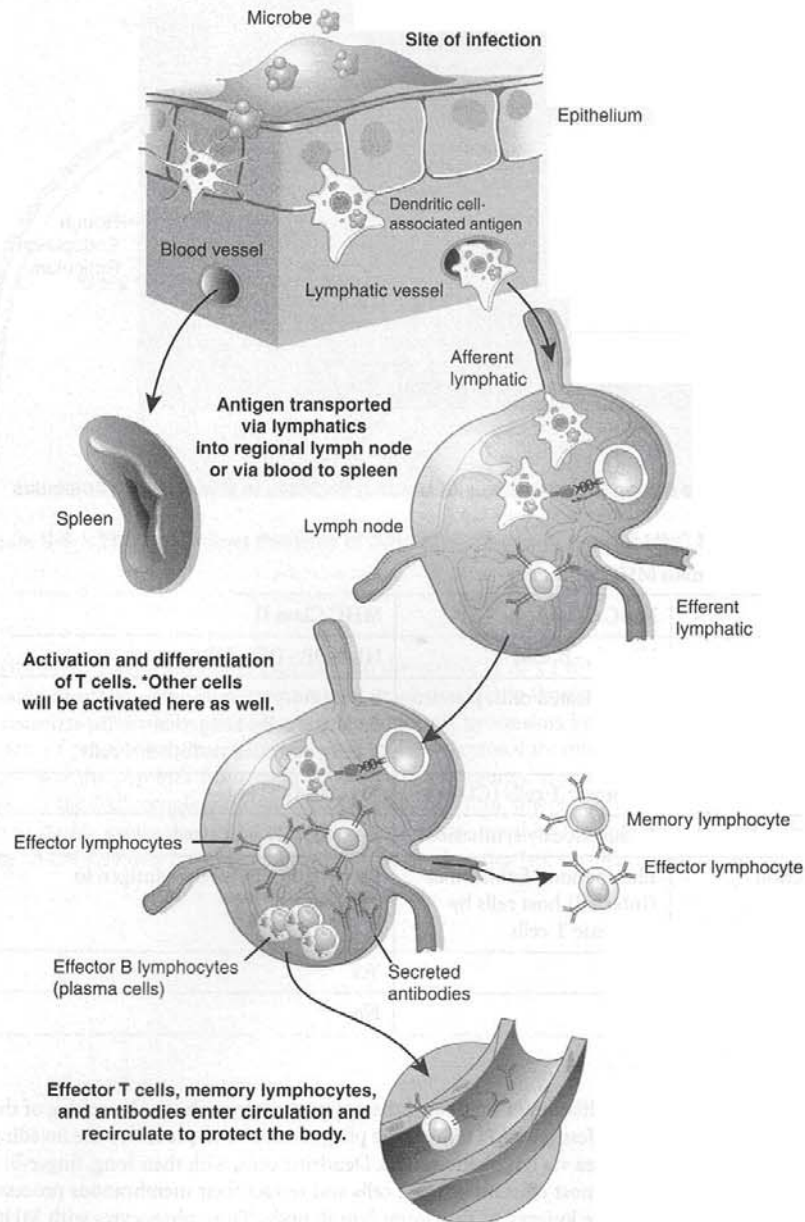


Figure II-6-3. The Transportation of Antigen to the Secondary Lymphoid Organs

The binding of the TCR of the naive T cell to the MHC class II–peptide complex of the APC provides the first signal to the T cell to begin its activation. This provides the antigenic specificity of the response. Costimulatory molecules on APC provide the second signal, and cytokines secreted by APC and the activating T cells themselves induce the proliferation (**clonal expansion**) and differentiation of the T cells into **effector cells** and **memory cells**.

Several costimulatory molecules are involved in the activation of naive T lymphocytes:

- **CD4 and CD8** (coreceptors for MHC classes II and I, respectively) transduce activating signals to the T cells.
- **Integrins** on T cells (LFA-1) bind to IgCAMs on APCs (ICAM-1) to increase cell–cell adherence.
- **IgCAMs** on T cells (CD2) bind to integrins (LFA-3) on APCs to increase cell–cell adherence.
- **CD28** on T cells binds to B7 on APCs and triggers the transcription of several cytokine genes.

The proliferation of naive T cells in response to antigen recognition is mediated principally by an autocrine growth pathway, in which the responding T cell secretes its own growth-promoting cytokines and also expresses receptor molecules for these factors. IL-2 is the most important growth factor for T cells and stimulates the proliferation of clones of T cells specific to that antigen.

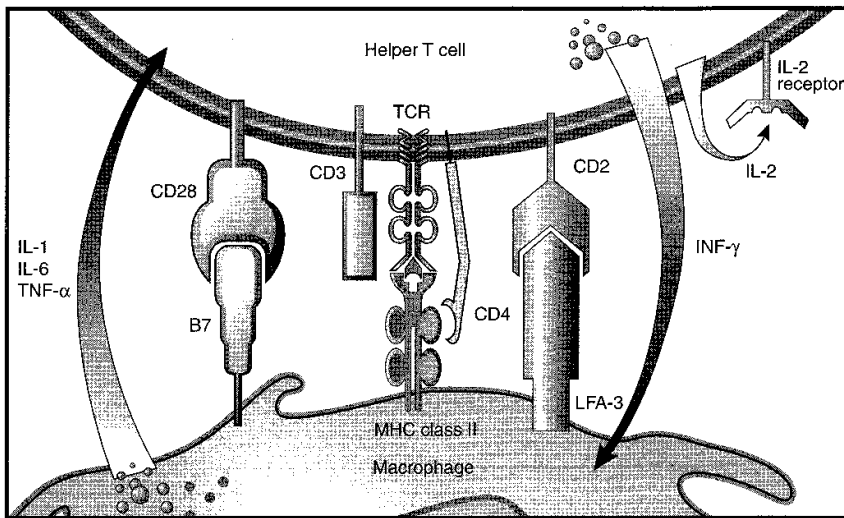


Figure II-6-4. Helper T Cell and Macrophage Adhesion

In A Nutshell

T-cell Activation—First signal = binding TCR to MHC II/peptide complex

In A Nutshell

Second Signal (Costimulatory Molecules)

- CD4 binds MHC II.
- CD8 binds MHC I.
- LFA-1 binds ICAM-1.
- CD2 binds LFA-3.
- CD28 binds B7.

In A Nutshell

Third Signal (Cytokines)

- IL-2
- IL-1
- IL-6
- TNF- α

Clinical Correlate

Superantigens are viral or bacterial proteins that cross-link the variable β domain of a T-cell receptor to an α chain of a class II MHC molecule. This cross-linkage provides an activating signal that induces T-cell activation and proliferation, in the absence of antigen-specific recognition of peptides in the MHC class II groove. Because superantigens bind outside of the antigen-binding cleft, they activate any clones of T cells expressing a particular variable β sequence and thus cause polyclonal activation of T cells, resulting in the overproduction of IFN- γ . This, in turn, activates macrophages, resulting in overexpression of proinflammatory cytokines (IL-1, IL-6 and TNF- α). Excess amounts of these cytokines induce systemic toxicity. Molecules produced during infectious processes and known to act as superantigens include staphylococcal enterotoxins, toxic-shock syndrome toxin-1 (TSST-1), and streptococcal pyrogenic exotoxins.

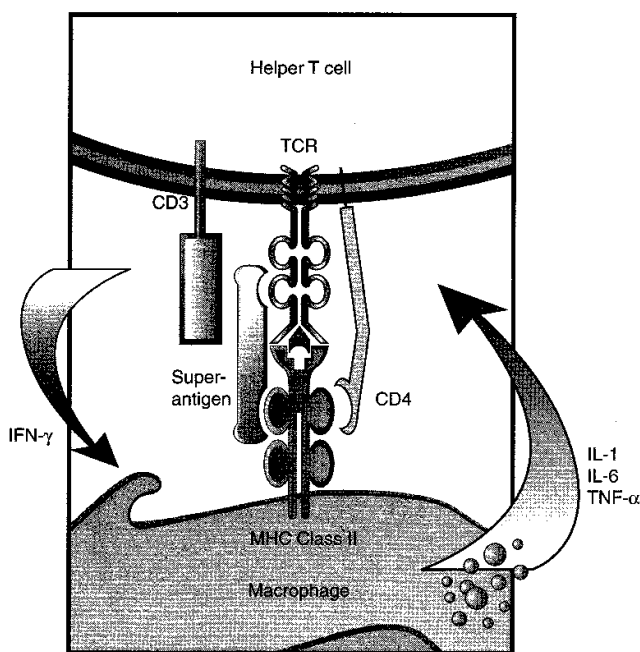


Figure II-6-5. Superantigen Activation of a Macrophage

In A Nutshell

TH cells control effector mechanisms of immunity (antibody, macrophage activation, CTL killing, and NK killing).

The activated CD4+ (helper) T lymphocytes, which have thus been generated in the lymph nodes and spleen following antigen administration, are now ready to serve as the orchestrators of virtually all the possible **effector mechanisms** that will arise to destroy the antigenic invaders. The effector mechanisms that are controlled totally or at least in part by TH cells include antibody synthesis, macrophage activation, cytotoxic T-cell killing, and NK cell killing. The "decision" as to which of these mechanisms should be encouraged is a function of the characteristics of the invading pathogen and is controlled by the differentiation of specialized classes of helper cells.

Clinical Correlate

Bare Lymphocyte Syndrome

Rare cases of MHC class II molecule deficiency (also known as **bare lymphocyte syndrome**) inherited in an autosomal-recessive fashion have been observed in humans. Immune problems tend to appear early in infancy and present as a mild form of severe combined immunodeficiency (SCID; see Chapter 11) with increased susceptibility to pyogenic and opportunistic infections. However, these defects can be distinguished from true SCID in that these patients will have T cells that can respond to nonspecific T-cell mitogens, such as phytohemagglutinin, and also to stimuli with allogeneic cells (cells from genetically nonidentical human beings). They do not develop graft-versus-host disease when given HLA-mismatched bone marrow transplants (see Chapter 14) because they do not express the MHC class II molecules against which such grafted cells can react. Patients with MHC class II deficiency are deficient in CD4⁺ cells due to failure of positive selection in the thymus, and they have moderate to severe hypogammaglobulinemia. This defect results from defects in the transcription factors required to coordinate their expression on the cell surface. Because MHC class I antigens are expressed normally, they do have CD8⁺ cells, although their function is diminished by the absence of TH1 cell cytokines.

There are two major classes of helper T (TH) cells, both of which arise from the same precursor, the naive TH lymphocyte, sometimes called the TH0 cell. The pattern of differentiation is determined by the stimuli present early in the immune response, at the site of antigen introduction. Differentiation of a TH0 cell into a TH1 cell seems to be stimulated by microbes that stimulate a strong initial innate immune response with the resultant production of IL-12 by macrophages or IFN- γ by NK cells. The differentiation of TH1 cells is stimulated by many intracellular bacteria, such as *Listeria* and mycobacteria, and by some parasites such as *Leishmania*—all of which infect macrophages. Differentiation of a TH0 cell into a TH2 cell seems to be encouraged in the absence of such innate immune stimuli, perhaps by default when persistent antigen remains in the system without significant macrophage or NK-cell stimulation. In this way, naive TH0 cells seem to produce IL-4 constitutively, and in the absence of IL-12 stimulation, these cells will upregulate their production of IL-4 to encourage differentiation into TH2 cells. The differentiation into the TH2 subset seems to be favored in infections with helminths and in response to allergens.

These classes of TH cells are distinguished almost entirely on the basis of the cytokines they produce. IFN- γ , TNF- β , and IL-2 are produced by TH1 cells, and IL-2, IL-4, IL-5, IL-6, and IL-10 are the cytokines produced by TH2 cells. These cytokines not only determine the stimulatory pathways that the cells will employ, but they also expand and develop the cells of the respective subset. For example, IFN- γ produced by TH1 cells promotes further TH1 development and inhibits the proliferation of TH2 cells. IL-4 and IL-10 produced by TH2 cells promote TH2 differentiation and inhibit the activation of TH1 cells. Thus, each subset amplifies itself and cross-regulates the other set so that immune responses become increasingly polarized over time, reaching extremes in cases where antigen exposure becomes chronic.

In A Nutshell

- TH1 cells arise from precursor when a strong innate immune response is stimulated.
- TH2 cells arise in the absence of such responses.

In A Nutshell

- TH1 cells produce IFN- γ , TNF- β , and IL-2.
- TH2 cells produce IL-2, IL-4, IL-5, IL-6, and IL-10.
- IFN- γ produced by TH1 inhibits TH2.
- IL-4 and IL-10 produced by TH2 inhibit TH1.

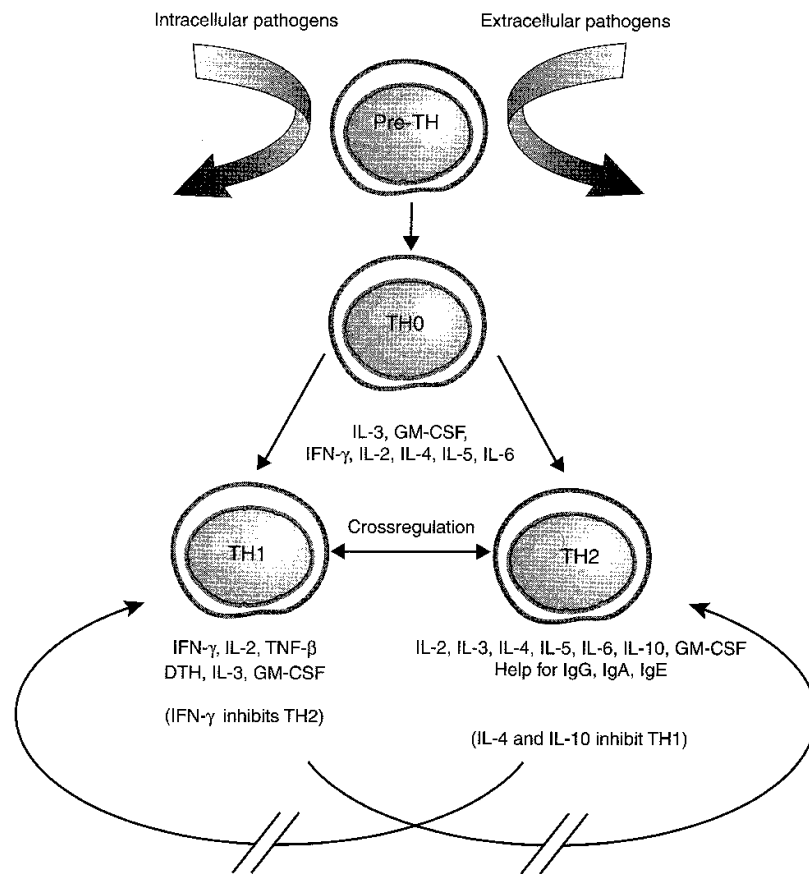


Figure II-6-6. Subsets of Helper T Cells

The effector mechanisms stimulated by the different classes of TH cells are specialized to optimally destroy different classes of invading antigens and are roughly (and somewhat artificially) divided into **cell-mediated** or **humoral** effector mechanisms.

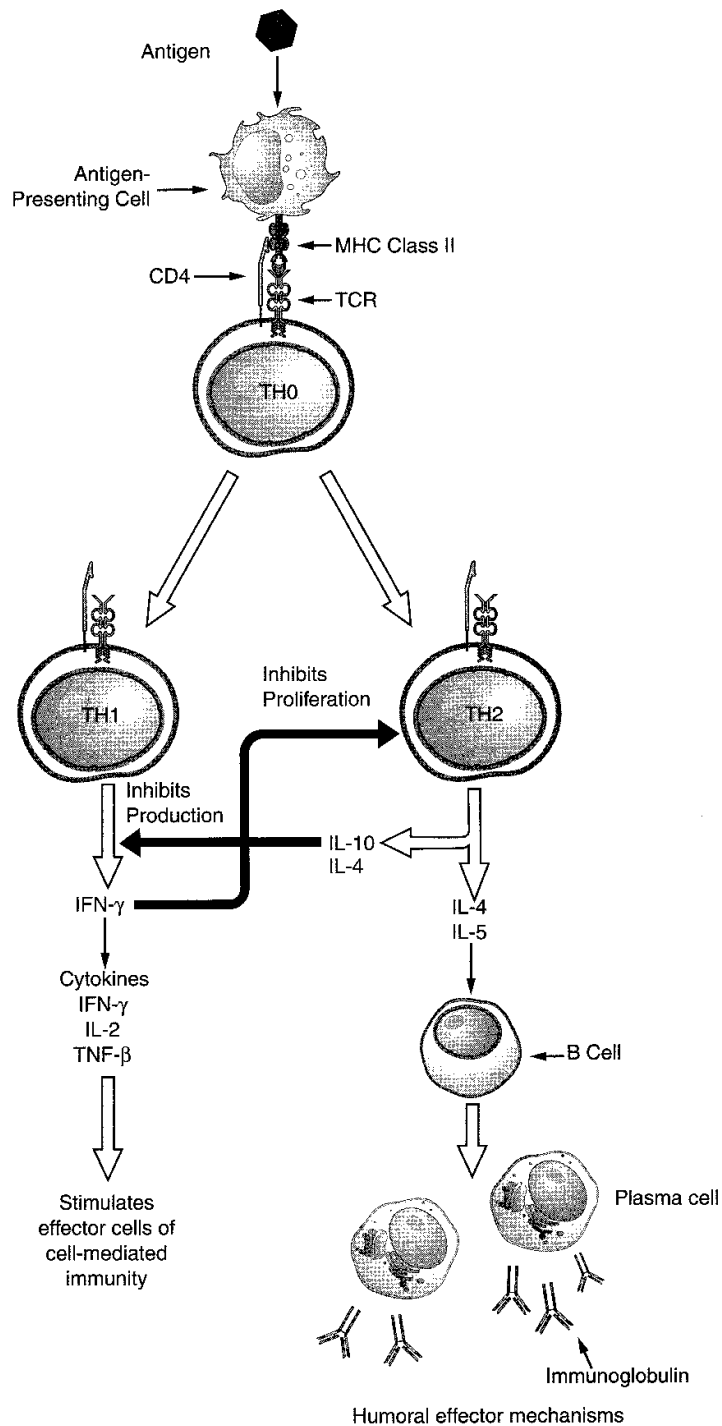


Figure II-6-7. Overview of T-Helper Cell Regulation of the Adaptive Immune Response

Clinical Correlate

Tuberculoid Versus Lepromatous Leprosy

The progression of disease with *Mycobacterium leprae* in humans is a well-documented example of the crucial balance between TH1 and TH2 subsets. Leprosy is not a single clinical entity, but presents as a spectrum of diseases, with tuberculoid and lepromatous forms being at the far poles. In **tuberculoid** leprosy, the patient has a **strong TH1** response, which eradicates the intracellular pathogens by granuloma formation. There is some damage to skin and peripheral nerves, but the disease progresses slowly, if at all, and the patient survives. In **lepromatous** leprosy, the **TH2 response is turned on**, and because of reciprocal inhibition, the cell-mediated response is depressed. These patients develop antibodies to the pathogen that are not protective, and the mycobacteria multiply inside macrophages, sometimes reaching levels of 10^{10} per gram of tissue. Hypergammaglobulinemia may occur, and these cases frequently progress to disseminated and disfiguring infections.

Chapter Summary

- Partially digested peptides are loaded into the groove of class II MHC molecules on antigen-presenting cells by the endosomal (exogenous) pathway.
- APCs migrate to the secondary lymphoid organs, where they present this processed antigen to recirculating naive lymphocytes.
- The binding of the TCR to the peptide/MHC class II complex provides the first signal in T-cell activation.
- Costimulatory molecule interactions (e.g., CD28 binds to B7, CD4 binds to MHC II, CD8 binds to MHC I, LFA-1 binds to ICAM-1, and CD2 binds to LFA-3) serve as the second signal in T-cell activation.
- Cytokines (IL-2, IL-1, IL-6, and TNF- α) serve as the final signal in T-cell activation.
- Superantigens are viral or bacterial proteins that cross-link the variable β domain of a T-cell receptor to an α chain of a class II MHC molecule and thereby cause polyclonal activation of T cells, overproduction of cytokines, and systemic toxicity.
- Activated TH cells act as the orchestrators of the effector mechanisms of the immune response (antibody synthesis, macrophage activation, cytotoxic T cell killing, and NK cell killing).
- Naive TH cells (TH0) differentiate into TH1 cells when IL-12 from macrophages or IFN- γ from NK cells is present. TH1 cells secrete IFN- γ , IL-2, and TNF- β .
- Naive TH0 cells differentiate into TH2 cells when there is extracellular attack. TH2 cells secrete IL-2, IL-4, IL-5, IL-6, and IL-10.
- The cytokines produced by TH subsets are cross-regulatory: IFN- γ produced by TH1 cells inhibits TH2 cells, and IL-4 and IL-10 produced by TH2 cells inhibit TH1 cells.

Review Questions

1. Human infections with *Mycobacterium leprae* express a spectrum of clinical presentations depending on the extent and expression of their immune response to the intracellular organism. On one end of the spectrum, patients with tuberculoid leprosy produce an effective cell-mediated immune response, which is successful at killing the intracellular organisms and, unfortunately, produces tissue damage. Patients with tuberculoid leprosy have granulomas that have elevated amounts of IL-2, IFN- γ , and TNF- β . The immune cell responsible for this pattern of cytokine production is the
 - (A) Cytotoxic T lymphocyte
 - (B) Epithelioid cell
 - (C) Macrophage
 - (D) TH1 cell
 - (E) TH2 cell

2. There is evidence that the immunologic pathway that distinguishes the selection between the two polar forms of leprosy depends on the initial means of antigen presentation, as well as individual human differences in response. If early events of antigen recognition elicit production of IL-4, IL-5, IL-6, and IL-10, lepromatous leprosy is more likely to result, with the outcome of failure to mount a protective delayed-type hypersensitivity response. What differential characteristic of the lepromatous form is predicted based on the fact of overproduction of IL-4, IL-5, IL-6, and IL-10 in lepromatous lesions?
 - (A) Autoimmunity
 - (B) Granuloma formation
 - (C) Hypergammaglobulinemia
 - (D) Immediate hypersensitivity
 - (E) Inflammation

3. A 10-month-old infant girl is admitted to the hospital with signs of *Pneumocystis carinii* pneumonia. Studies of her peripheral blood demonstrate age-normal counts of CD19+ cells, but CD3+ and CD4+ cell numbers are depressed. Immunoelectrophoresis of her serum reveals a moderate hypogammaglobulinemia. Her peripheral blood lymphocytes proliferate normally in response to phytohemagglutinin and MHC class I mismatched allogeneic cells. In a one-way mixed lymphocyte reaction using her cells as the stimulator cells, allogeneic T lymphocytes did not proliferate. Which of the following best describes the molecule most likely lacking from her lymphocytes?
 - (A) It is designed to bind endogenously produced peptides
 - (B) It is designed to bind exogenously processed peptides
 - (C) It possesses β_2 microglobulin
 - (D) It possesses two chains of unequal length
 - (E) It should be present on all nucleated cells in the body

4. An elderly man with diabetes develops a blister on the heel of his foot, which becomes infected. Although nursing staff in the home where he is a resident clean and treat the wound with topical antibiotic ointment, he develops a fever and hypotension, and a desquamating rash spreads from the site of the original blister. How does the toxin responsible for his symptoms cause these signs?
- (A) It acts as an IL-1 homologue
 - (B) It activates B lymphocytes polyclonally
 - (C) It activates complement
 - (D) It cross-links MHC class II molecules to TCRs polyclonally
 - (E) It stimulates neutrophils
5. It has been learned in several experimental systems that proliferation and differentiation of T lymphocytes in response to tumor cells is low because tumor cells lack the necessary costimulatory molecules for lymphocyte activation. If melanoma cells from a patient were induced to express these costimulatory molecules by transfection, production of an effective antitumor response might occur. Which of the following molecules would be the best candidate for transfection of tumor cells to achieve this end?
- (A) B7
 - (B) CD2
 - (C) CD4
 - (D) CD28
 - (E) LFA-1

Answers and Explanations

1. **The correct answer is D.** IL-2, IFN- γ , and TNF- β are all elaborated by the TH1 cell. TNF- β can also be made by NK cells. In tuberculoid leprosy, the TH1 arm of the immune response is most active, resulting in a protective (but also damaging) cell-mediated response and a dampening of the antibody response. In lepromatous leprosy, the patient has an overabundance of TH2 responses, causing the production of a nonprotective antibody response.

Cytotoxic T lymphocytes (**choice A**) are an effector cell in the cell-mediated immune response. They do not elaborate many cytokines but produce cytotoxic molecules, which cause the destruction of specific target cells.

Epithelioid cells (**choice B**) are modified macrophages. They are extremely secretory and may produce IL-1, IL-6, TNF- α , IFN- γ , and GM-CSF. They are prominent in granulomas, and their cytokines would be elevated in a patient with tuberculoid leprosy, but that was not the question.

Macrophages (**choice C**), once activated, may produce IL-1, IL-6, TNF- α , IFN- γ , and GM-CSF. They are prominent in granulomas, and their cytokines would be elevated in a patient with tuberculoid leprosy, but again, that was not the question.

TH2 cells (**choice E**) would be elevated during lepromatous leprosy. The cytokines they secrete include GM-CSF, IL-2, IL-3, IL-4, IL-5, IL-6, and IL-10. These cells are stimulators of the humoral immune response.

2. **The correct answer is C.** In lepromatous leprosy, the activation of the TH2 arm of the immune response results in elicitation of those cytokines that stimulate production of antibody (IL-2, IL-4, IL-5, IL-6, and IL-10) and those that inhibit the development of the protective cell-mediated immune response (IL-4 and IL-10). Therefore, hypergammaglobulinemia is a frequent finding in lepromatous leprosy.

Autoimmunity (**choice A**) may develop following infectious processes, but there is no evidence that stimulation of TH2 cells, by itself, causes autoimmune disease.

Granuloma formation (**choice B**) would be decreased after exposure to these cytokines. Granulomas are an expression of the delayed-type hypersensitivity response, which is a function of TH1 cells. IL-10 and IL-4 would depress the TH1 response.

Immediate hypersensitivity (**choice D**) requires sensitized mast cells and IgE antibodies. Although this result could occur in persons predisposed to atopic allergy, it is not the most likely result of stimulation with TH2 cytokines.

Inflammation (**choice E**) is primarily mediated by substances released during tissue injury (leukotrienes, histamine, etc.) and the cytokines of activated macrophages (IL-1, IL-6, and TNF- α). It is not enhanced by TH2 cytokines.

3. **The correct answer is B.** This child has bare lymphocyte syndrome, a rare autosomal-recessive disease in which there is absence of MHC class II molecules on cells. Thus, her cells can recognize other cells as foreign and proliferate to T-cell mitogens, but they cannot be recognized by allogeneic lymphocytes because they do not express class II MHC antigens on their surface. The phrase which best describes the MHC class II molecule on this list is that it is designed to bind exogenously processed peptides. Other descriptions that could apply would be that it has two chains of similar length, is produced with an invariant chain, and is designed to present foreign peptides to TH cells.

It is designed to bind endogenously produced peptides (**choice A**) is a description that fits the class I MHC molecule. If this were a case of class I MHC deficiency, she would not have made a normal proliferative response to mismatched allogeneic cells.

It possesses β_2 microglobulin (**choice C**) is a description that fits the class I MHC molecule.

It possesses two chains of unequal length (**choice D**) is a description of the class I MHC molecule. It has an α chain with three domains, and a smaller chain, β_2 microglobulin, becomes associated with the α chain.

It should be present on all nucleated cells in the body (**choice E**) describes the class I MHC molecule. Class II MHC will be found on all antigen-presenting cells in the body.

4. **The correct answer is D.** This patient is showing signs of toxic shock syndrome, caused by infection of the blister with *Staphylococcus aureus* and the resultant elaboration of the exotoxin TSST-1. This toxin acts as a superantigen, cross-linking the variable β region of the TCR to the α chain of the class II MHC molecule. This binds TH cells and APC together without the specificity of antigen recognition, and so clonal proliferation of T cells and production of IFN- γ leads to activation of macrophages. As a result, the macrophages overproduce the cytokines IL-1, IL-6, and TNF- α , which are toxic at high levels.

It acts as an IL-1 homologue (**choice A**) is not true. IL-1 is produced by macrophages as a result of T-cell activation, but TSST-1 does not itself act as an IL-1 homologue.

It activates B lymphocytes polyclonally (**choice B**) is not true. TSST-1 acts on TH cells to stimulate macrophage cytokines. It does not have a direct effect on B-cell proliferation.

It activates complement (**choice C**) is not correct. TSST-1 does not have an effect on complement.

It stimulates neutrophils (**choice E**) is not correct. Although neutrophils are stimulated during *Staphylococcus aureus* infection and produce IL-1, which causes fever, the mechanism of action of TSST-1 and other superantigens is not through neutrophil activation.

5. **The correct answer is A.** The B7 molecule on antigen-presenting cells binds to the CD28 molecule on T lymphocytes and serves as a costimulatory signal for their activation. If the tumor cells could be induced to express this costimulatory molecule, they would provide the important activating signal to the T cells.

CD2 (**choice B**) is the molecule on T lymphocytes that binds to LFA-3 on antigen-presenting cells. If the tumor cell were induced to express CD2, it would bind to the complementary structure on macrophages and not activate the T cells.

CD4 (**choice C**) is the molecule on T lymphocytes that stabilizes the interaction of MHC class II and the TCR. If the tumor cell were induced to express CD4, it would not increase the tumor-specific response.

CD28 (**choice D**) is the molecule on T cells that binds to B7. If the tumor cell were induced to express CD28, it would bind to the complementary structure on macrophages and not activate the T cells.

LFA-1 (**choice E**) is the molecule on T cells that binds ICAM-1 on the antigen-presenting cells. If the tumor cells were induced to express LFA-1, it would bind to the complementary structure on macrophages and not activate the T cells.

The Generation of Humoral Effector Mechanisms

7

What the USMLE Requires You to Know

- The humoral responses to thymus-independent and -dependent **antigens**
- The cell surface molecules that contribute to production of T- and B-cell **conjugates**
- The basic structures and effector functions of the five **antibody isotypes**
- The meaning of valence, affinity, and avidity
- The role of somatic hypermutation in affinity maturation
- How the alternative and classical complement cascades are initiated
- The biologic functions of complement components

Humoral immunity is mediated by antibodies synthesized by B lymphocytes and secreted by their fully differentiated end cell, the **plasma cell**. This arm of the immune response is directed toward the defense against **extracellular** microbes or toxins and may culminate in the extracellular degradation of such materials or the enhancement of their destruction via phagocytosis.

As mature naive B lymphocytes leave the bone marrow following successful rearrangement of their membrane immunoglobulin receptor genes, they recirculate throughout the body, attracted to **follicular areas** of the lymph nodes and spleen. If antigen entering these secondary lymphoid organs binds to and cross-links the idiotypes of these membrane receptors, this provides the first signal for the activation of the B lymphocyte.

Most antigens introduced into the body fall into the category of **thymus-dependent antigens**. Response to such molecules requires the direct contact of B cells with TH cells and their **cytokines**. After the cross-linking of receptors on the B-cell surface with antigen, the material is endocytosed and processed via the endocytic (exogenous) pathway to generate MHC class II/peptide conjugates, which are then inserted in the membrane, just as we have seen during phagocytosis and processing in the professional phagocyte lineages. Simultaneously, expression of costimulatory molecules such as B7 is upregulated on the B lymphocyte, making them effective presenters of antigen to TH cells in the area. Once a TH cell recognizes a processed antigenic peptide displayed with MHC class II molecules on the membrane of the B cell, the two cells form a **conjugate**, and the TH cell is activated and induced to become a TH2 cell. TH2 cells in conjugates rearrange their Golgi apparatus toward the junction with the B cell leading to the directional release of cytokines toward the B cell. In addition, expression of a molecule known as CD40L on the surface of the TH2 cell is upregulated, and this molecule interacts with CD40 on the B cell to provide the second signal for B-cell activation. The B cells are encouraged to proliferate after this interaction. The final signal delivered by the TH2 cell is the release of cytokines, which will induce the differentiation of B cells into fully differentiated, antibody-secreting cells and memory cells (see Chapter 9) and induce **class switching**.

In A Nutshell

- Humoral immunity = antibodies
- Defends against extracellular agents
- B lymphocytes are attracted to follicular areas of secondary lymphoid organs.
- Most pathogens are thymus-dependent antigens.

In A Nutshell

B-cell contact with TH cells requires:

- MHC II/peptide presentation
- Costimulatory molecules (B7)
- CD40/CD40L binding

In A Nutshell

TH2 cytokines induce B-cell:

- Differentiation
- Memory
- Class switching

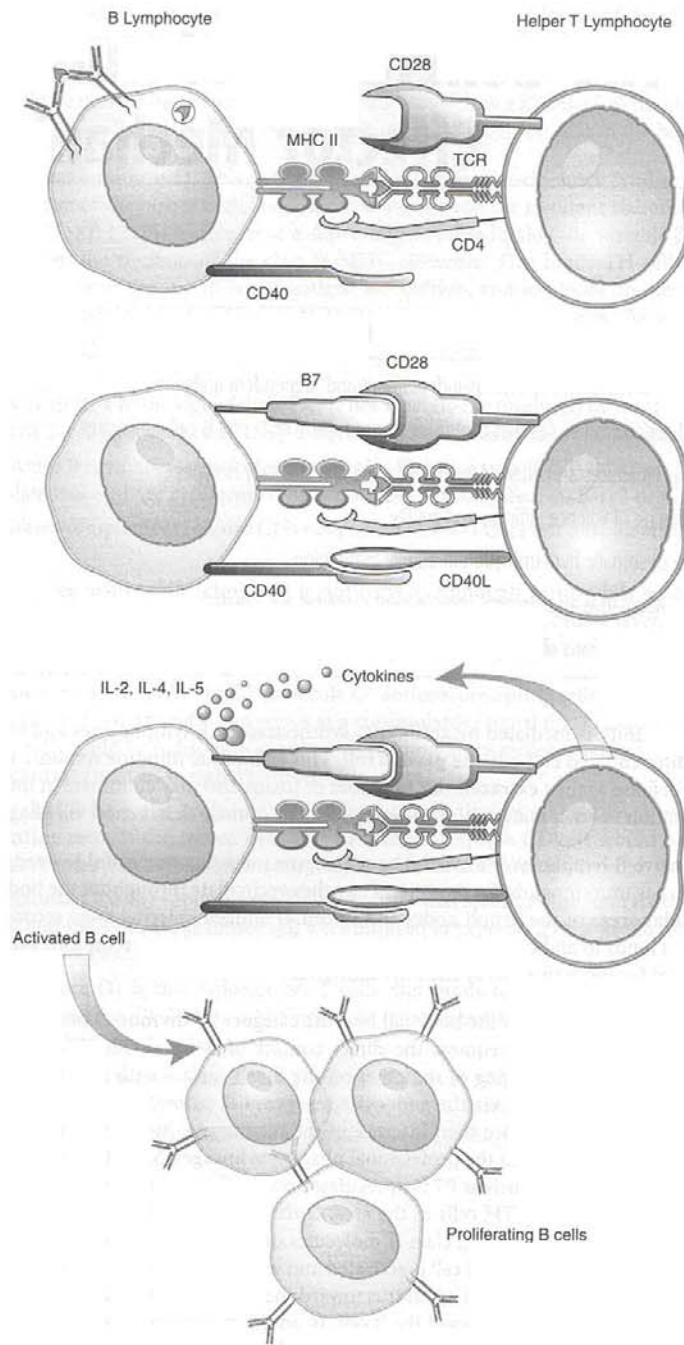


Figure II-7-1. The Formation of T- and B-Cell Conjugates

T cells recognize peptides bound to MHC molecules. Therefore, antigens that possess no peptide structure cannot be recognized by T cells. These antigens are called **thymus-independent antigens** and include lipopolysaccharide from the cell envelope of gram-negative bacteria and polysaccharide capsular antigens. These antigens may directly stimulate B cells to cause proliferation and secretion of antibody, or they may act as B-cell **mitogens**, directly activating these cells regardless of their antigenic specificity. The response to thymus-independent antigens is generally **weaker** than the response to other classes of antigens, resulting in the **secretion of IgM antibodies only** and the **absence of immunologic memory**.

Although all of the antibody molecules secreted by a clone of B lymphocytes will have identical idiotypes (see Chapter 2), the B cell is induced to make new classes, or **isotypes**, of immunoglobulin in response to cytokine-directed instruction from the TH2 cell. The progression of new antibody isotypes produced by B cells is defined by the sequence of constant domain coding in the B-lymphocyte DNA, and each isotype of immunoglobulin is designed with a different effector function in mind. Just as the three-dimensional structure of the **idiotype defines antigen specificity**, the sequence of amino acids in the constant domains of the immunoglobulin molecule (**isotype**) **dictates the effector functions** that will be expressed.

The biologic function of segments of the antibody molecule was first elucidated by digestion of these molecules with proteolytic enzymes. If an antibody molecule is digested with **papain**, cleavage occurs above the disulfide bonds that hold the heavy chains together. This generates three separate fragments, two of which are called Fab (fragment antigen binding), and one is called Fc (fragment crystallizable). Cleavage of the antibody molecule with **pepsin** generates one large fragment called $F(ab')_2$ and a digested Fc fragment. The **bridging of antigens** by antibody molecules is required for **agglutination of particulate antigens** or the **precipitation of soluble antigens**.

In A Nutshell

Thymus-independent antigens:

- contain no peptides.
- stimulate only IgM.
- create no memory.

Note

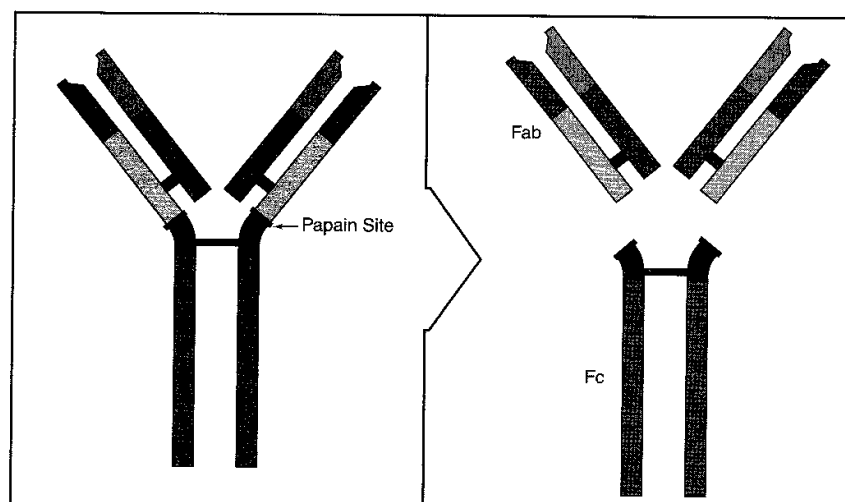
Mitogens activate many clones of B cells and are used clinically to assess lymphocyte function.

In A Nutshell

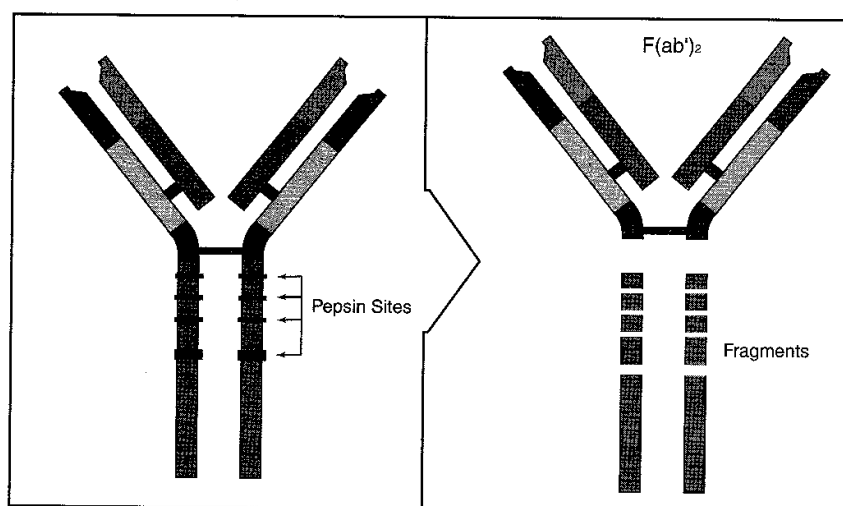
- Isotype switching is directed by TH2 cells.
- Isotypes dictate effector function of the antibody molecule.

In A Nutshell

- Papain generates 2 Fab + 1 Fc.
 - Fab (monovalent)—capable of binding
- Pepsin generates 1 $F(ab')_2$.
 - $F(ab')_2$ (divalent)—capable of binding and bridging



Proteolytic Cleavage with Papain



Proteolytic Cleavage with Pepsin

In A Nutshell

- IgM is the first isotype produced.
- Plasma IgM exists as a pentamer.
- J chain joins the monomer units.

Figure II-7-2. Proteolytic Cleavage of Immunoglobulin by Papain/Pepsin

The first isotype of immunoglobulin that can be produced by a B cell with or without T-cell help is IgM. This is because coding for the constant domains of the heavy chain of IgM (μ chains) are the first sequences downstream from the coding for the idiotype of the molecule. The IgM molecule on the surface of the B cell is a monomer, but the secreted form of this molecule is a **pentamer**, held together in an extremely compact form by a **J chain** synthesized by the cell.

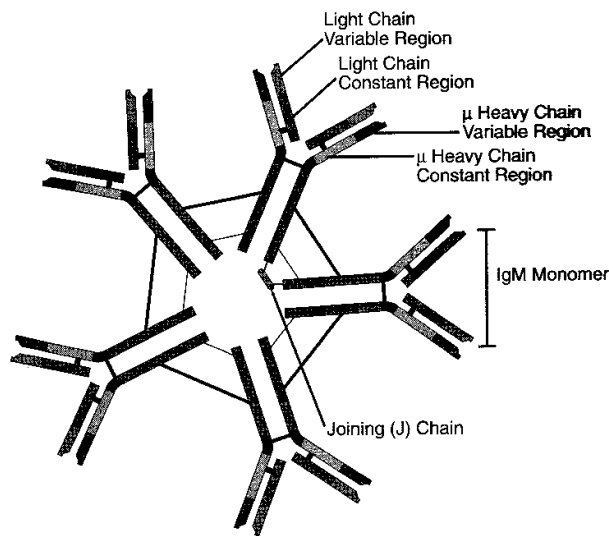


Figure II-7-3. The IgM Pentamer

The design of the IgM pentamer maximizes the effector functions critical to the body early during antigenic challenge. Because of its multimeric structure (5 of the Y-shaped monomers joined into one unit), plasma IgM has five times the capacity for binding antigenic epitopes as any monomeric immunoglobulin unit. The valence of the molecule is therefore 10: In other words, 10 identical epitopes can be simultaneously bound, as compared with 2 for the monomeric structure. This makes IgM the most effective immunoglobulin isotype at "sponging" the free antigen out of the tissues and proves critical, as the humoral response evolves, in trapping antigen so that it can be presented to the lymphocytes, which will ultimately refine the choice of effector mechanism. Although the binding strength (affinity) of the idiotype for the epitope may not be strong early in the immune response, the IgM molecule possesses the highest avidity (number of combining sites available to bind epitopes) of any immunoglobulin molecule produced in the body.

In A Nutshell

IgM

- Plasma IgM valence = 10
- Functions in trapping free antigen
- Affinity (binding strength) may be low
- Avidity (multipoint binding) highest of all isotypes

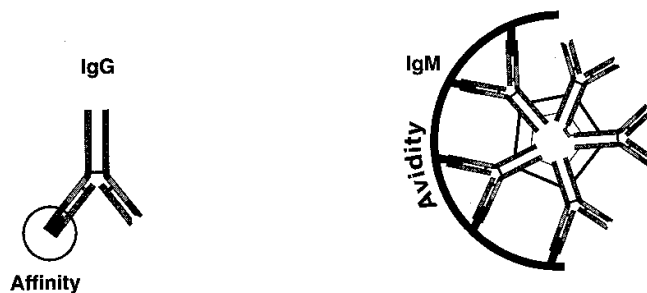


Figure II-7-4. Affinity and Avidity

In A Nutshell

- IgM is most effective isotype at activating complement.
- It is not an opsonin.
- It does not mediate ADCC.

Note

Isotype switching is induced by TH2 cells.

The multimeric structure of IgM also makes it the most effective antibody at activating complement, a set of serum proteases important in mediating inflammation and antigen removal. Serum IgM is incapable of binding to cellular Fc receptors and thus cannot act as an opsonin (see Chapter 5) or a mediator of antibody-dependent cell-mediated cytotoxicity (ADCC) (see Chapter 8).

As the B lymphocyte receives cytokine signals from the activated TH2 cells in the secondary lymphoid organs, it is induced to undergo **isotype switching**, changing the heavy-chain constant domains to classes of antibodies with new and different effector functions. It does this by rearranging the DNA encoding the constant region of the heavy chain by activating switch regions that cause the intervening DNA to be looped out, excised, and degraded. The idiotype is then joined to a new constant region domain coding, and an antibody molecule with identical antigenic specificity but a new effector function is produced. This isotype switch is **one-way**: Because the excised DNA is degraded, a cell that has begun to produce an isotype downstream from IgM coding can never produce IgM again.

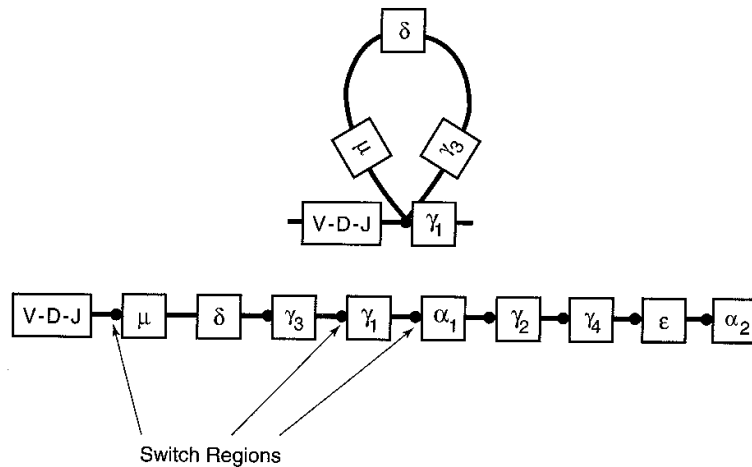


Figure II-7-5. Immunoglobulin Heavy Chain Switching

Note

IgM is the main immunoglobulin of the primary immune response.

Clinical Correlate

- IgM is used as a measure of a primary response (acute infection).
- Convalescent serum will have mostly IgG with subthreshold levels of IgM.

This is why IgM is the principal immunoglobulin of the **primary immune response** when antigen is first encountered, and it is replaced in later responses by antibodies of different isotypes. Although IgM antibodies are occasionally produced at low levels during secondary and later immunologic responses, they are always produced by cells seeing that antigen for the first time; namely, naive cells newly emerging from the bone marrow (see Chapter 9).

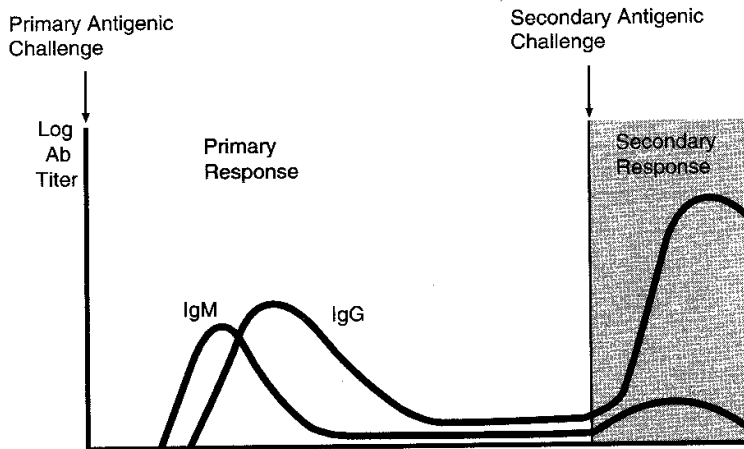


Figure II-7-6. Primary and Secondary Antibody Responses

During the activation of B lymphocytes by TH2 cells, intense proliferation of the B cells results in the formation of **germinal centers** in the follicles of the lymph nodes and spleen. These are clones of proliferating, antigen-specific cells. During the intense proliferative response of the B cell, random mutations in the coding of the variable domain region may occur. This is called **somatic hypermutation** and creates single point mutations in the antibody idiotype. If these slightly altered idiotypes have increased affinity for the antigen, then the cell expressing them will be at a selective advantage in competing to bind antigen. Because binding antigen serves as the first signal for proliferation, over time, clones of cells with higher receptor affinity will begin to predominate in the germinal center. This **clonal selection** results in the predominance of clones capable of producing antibodies with increasing affinity for the antigen, a process known as **affinity maturation**. This means that although isotype switching will necessarily **decrease the avidity** of the preponderance of antibody molecules as the immune response evolves, this will be substituted by an **increase in antibody affinity** over time.

The choice of activation of particular switch regions in the B-cell DNA sequence is apparently dictated by the release of specific cytokines by activated TH2 cells.

In A Nutshell

- Germinal centers are clones of proliferating antigen-specific B cells.
- Somatic hypermutation may cause minor idiotype changes.
- Clonal selection by competition for antigen causes affinity maturation.

In A Nutshell

TH2 cytokines dictate switch region activation.

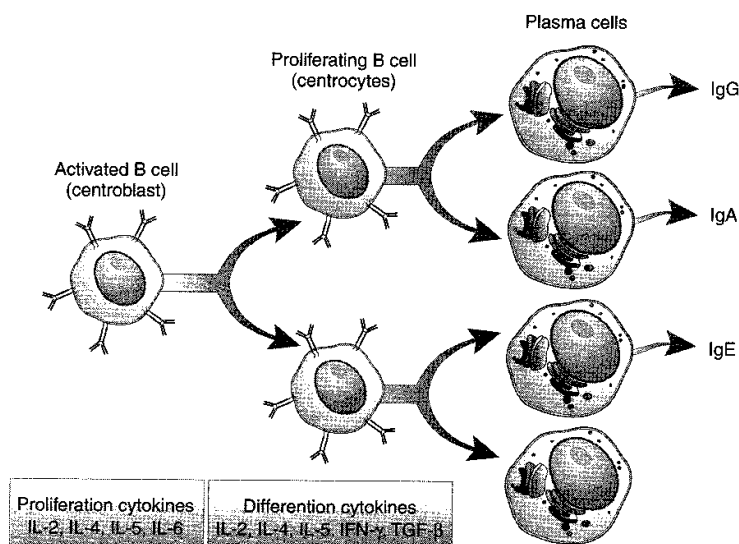


Figure II-7-7. TH-Cell Direction of B-Lymphocyte Proliferation and Differentiation

Clinical Correlate

X-Linked Hyper-IgM Syndrome is characterized by a deficiency of IgG, IgA, and IgE and elevated levels of IgM. IgM levels can reach 10 mg/mL (normal is 1.5 mg/mL). It is most commonly inherited as an X-linked recessive disorder, but some forms seem to be acquired and can be seen in both sexes. The peripheral blood of infected individuals has high numbers of IgM-secreting plasma cells, as well as autoantibodies to neutrophils, platelets, and red blood cells. These patients fail to make germinal centers during a humoral immune response. Children with this condition suffer recurrent respiratory infections, especially those caused by *Pneumocystis carinii*.

The defect in this syndrome is in the gene encoding the CD40 ligand, which maps to the X chromosome. Therefore, TH cells from these patients will fail to express functional CD40L on their membrane and will thereby fail to give the costimulatory signal necessary for the B-cell response to T-dependent antigens, so only IgM antibodies are produced. The B-cell response to T-independent antigens is unaffected.

In A Nutshell

- IgG is the major antibody produced after IgM.
- IgG exists in four subisotypes.
- IgG activates complement, opsonizes, and mediates ADCC.
- IgG is actively transported across the placenta.

The preponderant isotype of immunoglobulin that begins to be produced after IgM during the primary immune response is IgG. IgG is a monomeric molecule with a γ **heavy chain** and a new set of effector functions. IgG exists in four different **subisotypes** (subclasses) in humans—IgG1, -2, -3 and -4, each of which exhibits slightly different capacity in effector functions. But in general, IgG activates complement, acts as an opsonin, and mediates ADCC. It is also actively transported across the placenta by receptor-mediated transport and thus plays a crucial role in protection of the fetus during gestation.

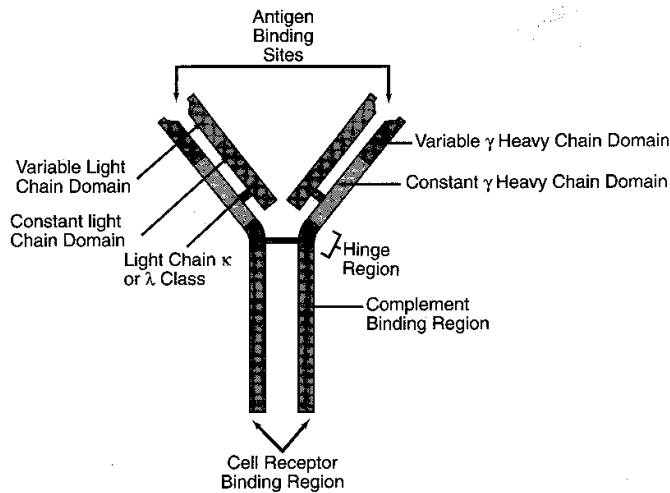


Figure II-7-8. The Basic Structure of IgG

Another isotype of antibody that can be produced following class switching is IgA, although this isotype is much more commonly produced in the submucosa than in the lymph nodes and spleen (see Chapter 9). IgA generally exists as a **dimer**, held together by a J chain similar to that produced with IgM, and serves as a major protective **defense of the mucosal surfaces of the body**. Its sole function appears to be the inhibition of binding of toxins or adhesive microbial components to the mucosa of the digestive, respiratory, and urogenital systems, and it does not activate complement or act as an opsonin.

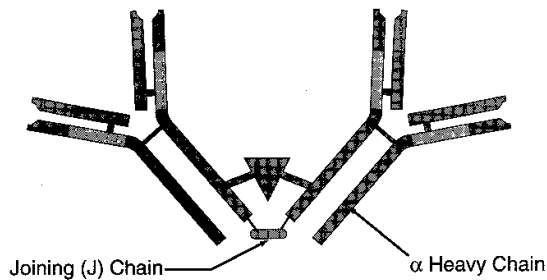


Figure II-7-9. The IgA Dimer

IgE is the so-called **homocytotropic antibody** because it binds directly to Fcε receptors present on mast cells and basophils (without binding antigen) and is involved in elicitation of protective immune responses against helminth parasites and many allergic responses (see Chapter 13). It does not activate complement or act as an opsonin. Its heavy chain is called an ε chain.

In A Nutshell

- Most IgA is produced in the submucosa.
- IgA is a dimer with a J chain.
- IgA inhibits binding of adhesive substances to mucosal surfaces.
- IgA is an important component in breast milk.

In A Nutshell

- IgE is bound to mast cells and basophils.
- IgE mediates immediate type I allergic reactions.
- IgE protects against parasites.

Clinical Correlate

Immunodeficiencies Involving B Lymphocytes

Patients with B-cell deficiencies usually present with recurrent pyogenic infections with extracellular pathogens. The absence of immunoglobulins for opsonization and complement activation is a major problem (see Chapter 11). The T-cell immune system is intact, and T-cell activities against intracellular pathogens, delayed-type hypersensitivity, and tumor rejection are normal (see Chapter 8).

In A Nutshell

The Complement System

- Has two pathways of activation
- Enhances inflammation
- Enhances phagocytosis
- Causes lysis

Table II-7-1. Summary of the Biologic Functions of the Antibody Isotypes

	IgM	IgG	IgA	IgD	IgE
Heavy chain	μ	γ	α	δ	ϵ
Adult serum levels (in mg/dL)	40–345	650–1,500	75–390	Trace	Trace
Functions					
Complement activation, classic pathway	+	+	–	–	–
Opsonization	–	+	–	–	–
Antibody-dependent cell-mediated cytotoxicity (ADCC)	–	+	–	–	–
Placental transport	–	+	–	–	–
Naive B-cell antigen receptor	+	–	–	+	–
Memory B-cell antigen receptor (one only)	–	+	+	–	+
Trigger mast cell granule release	–	–	–	–	+

Complement

The **complement system** is a set of interacting proteins released into the blood after production in the liver. The components act together as zymogens, activating one another in cascade fashion after initiation from a variety of stimuli. Two different pathways of activation occur in the body and culminate similarly in the production of important split products that mediate **inflammation**, enhance phagocytosis by **opsonization**, and cause **lysis** of particles by membrane pore formation.

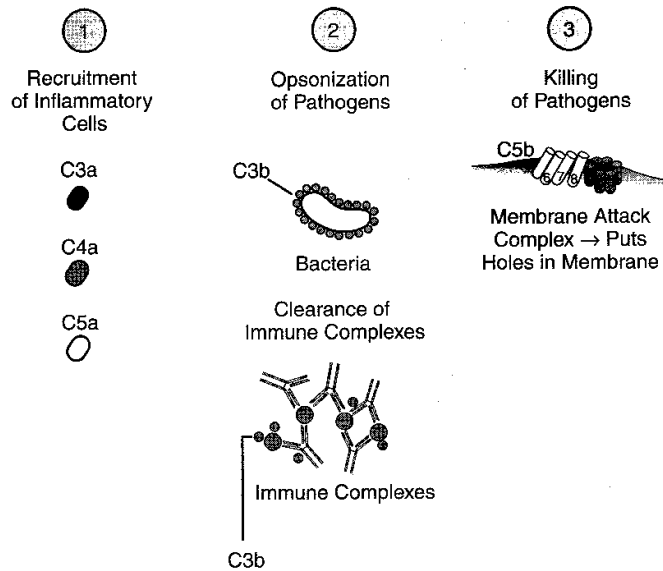


Figure II-7-10. Three Functions of the Complement System

The **alternative pathway** of complement activation is probably the more primitive of the two pathways because it is initiated by simple attraction of the early factors to the surfaces of microbes. Bacterial polysaccharides and the lipopolysaccharide of the cell envelope of gram-negative bacteria both serve as potent, initiating stimuli.

In A Nutshell

The alternative pathway is initiated by surfaces of pathogens.

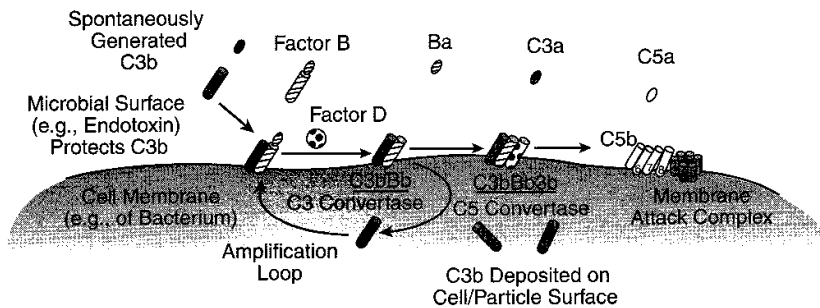


Figure II-7-11. The Alternative Complement Pathway

The **classical pathway** is activated by antigen-antibody complexes and is probably the more phylogenetically advanced system of activation. Both IgG and IgM can activate the system by this pathway, although IgM is the most efficient.

Note

The classical pathway is activated by Ag/Ab complexes.

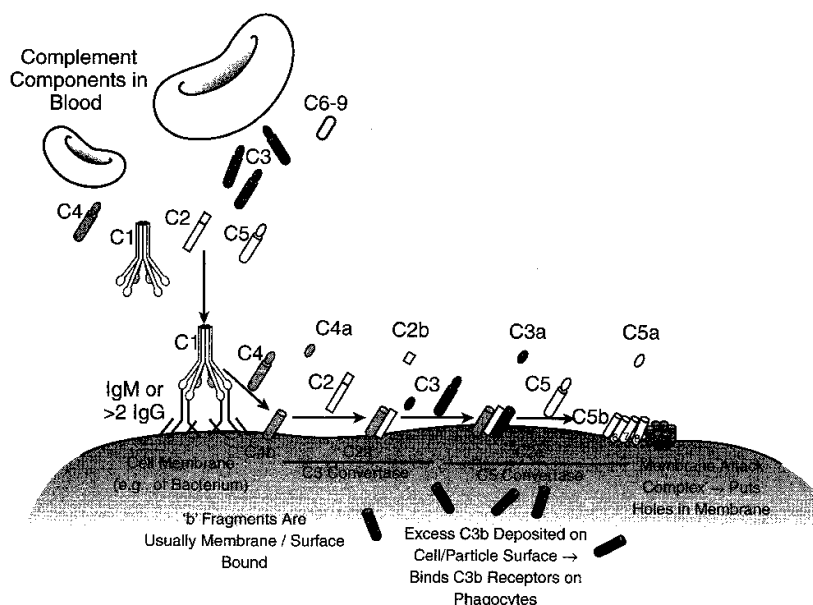


Figure II-7-12. The Classical Complement Pathway

Although the complement cascade is considered a constituent of the innate immune response, its overlapping stimulation of effector functions of cells of the adaptive immune response, as well as its role in enhancement of inflammation, make it a critical effector system for removal of extracellular invaders and concentration of antigens into the secondary lymphoid organs, where the adaptive immune responses are elicited.

When uncontrolled activation of complement occurs in certain disease states (see Chapters 11 and 13), damage to host tissues can ensue. Physiologic controls on complement activation occur at the level of C1 (classical pathway) and C3 and C5 (both pathways).

In A Nutshell

Physiologic controls on complement activation act at level of C1, C3, and C5.

Clinical Correlate

Complement Deficiencies

Genetic deficiencies have been described for each of the components of complement and their regulatory proteins (see Chapter 11). These deficiencies highlight the critical role of the early components of complement in generating C3b and the essential role of C3b for clearance of immune complexes from the body. Furthermore, even though gram-positive bacteria may be resistant to the membrane attack complex of complement, the early components of the cascade mediate localized inflammation and opsonize the bacteria. In a great number of cases then, it is the complement cascade that converts immunoglobulins into powerful effectors of bacterial destruction.

When deficiencies of complement regulatory components occur, then the uncontrolled activation of the complement cascade can have dangerous results in the body's own innocent bystanders. In hereditary angioedema, uncontrolled complement activation at the mucosal surfaces causes edema and pain. In paroxysmal nocturnal hemoglobinuria, the absence of regulatory proteins causes paroxysms of hemolysis of RBCs and the resultant hemoglobinuria.

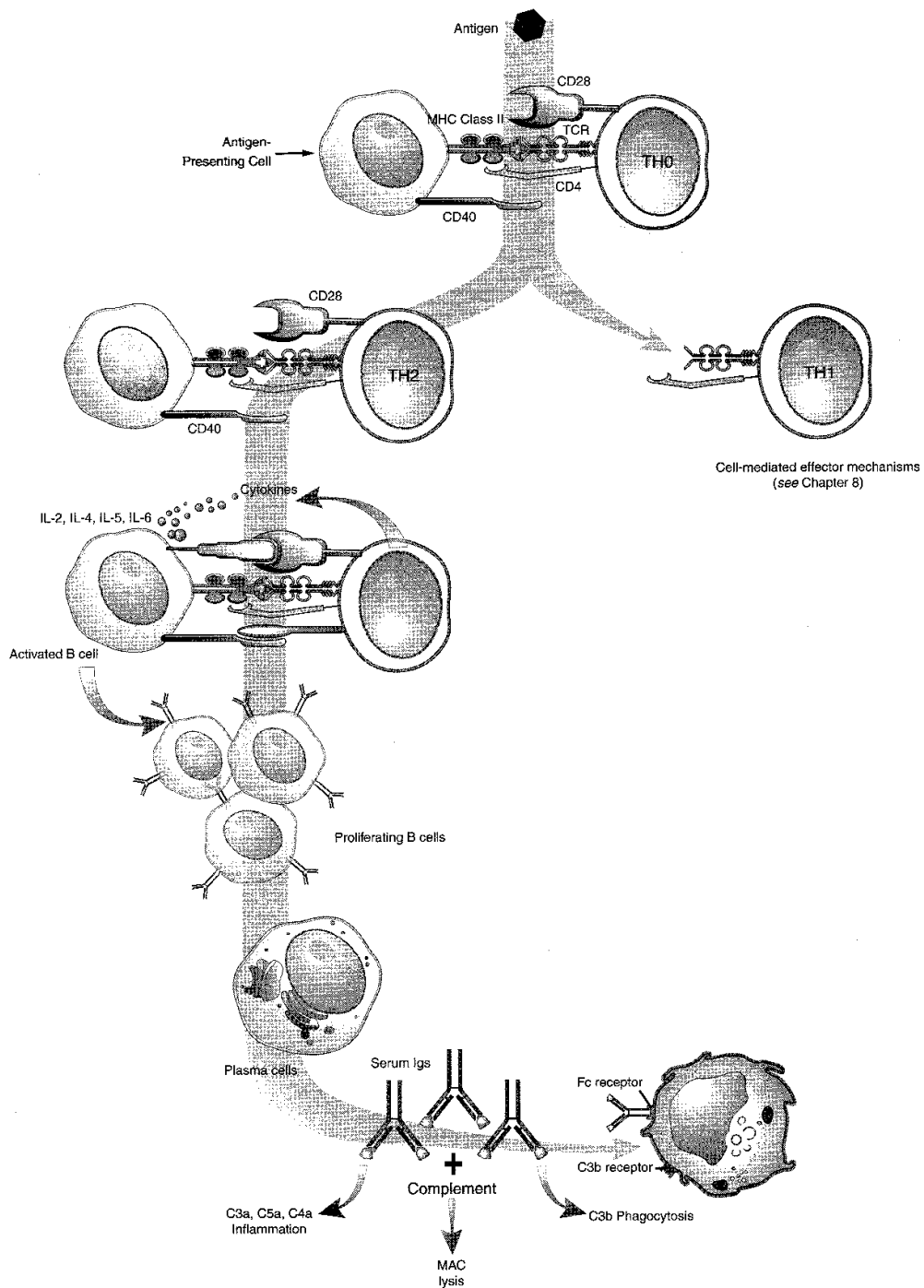


Figure II-7-13. Overview of the Generation of Humoral Effector Mechanisms

Chapter Summary

- Humoral immunity is mediated by antibodies synthesized by B cells and secreted by plasma cells.
- Humoral immunity is the major defense mechanism against extracellular microbes and toxins.
- Most naturally occurring antigens are thymus-dependent: They require collaboration of TH and B cells.
- Contact between specific B and TH cells involves MHC class II/peptide presentation, costimulatory molecules (B7/CD28), CD40/CD40L binding, and cytokine production (IL-2, IL-4, IL-5, and IL-6).
- TH2 cells direct isotype switching by B cells, which changes the effector function of the antibody produced.
- Thymus-independent antigens, such as bacterial lipopolysaccharide, cross-link the receptors of B lymphocytes and cause them to proliferate and secrete IgM antibodies. These antigens *do not* create "immunologic memory."
- IgM is the first isotype of antibody that can be produced. It exists in serum as a pentamer held together by a joining (J) chain.
- The functions of IgM are as a monomer (receptor on B cells), antigen capture in the secondary lymphoid organs and as a pentamer in plasma, and activation of complement.
- TH2 activation of B lymphocytes causes intense proliferation in the germinal centers, and somatic hypermutation may cause slight variation in the shape of the idiotype. Clonal selection of the idiotype with the highest affinity for antigen results in "affinity maturation": a general improvement in the "goodness-of-fit" for the antigen as the immune response progresses.
- IgG is the major isotype produced after IgM. It exists in four subisotypes. It activates complement, opsonizes, mediates ADCC, and is actively transported across the placenta.
- IgA is the major isotype produced in the submucosa, colostrum, and breast milk. It is a dimer with a J chain holding it together. It functions in inhibiting the binding of substances to cells or mucosal surfaces. It does not activate complement or mediate opsonization.
- IgE is the antibody that binds to mast cells and is responsible for antihelminthic and allergic responses.
- Complement is a set of interacting serum proteins that enhance inflammation (C3a, C4a, C5a) and opsonization (C3b) and cause lysis of particulate substances (C5b-9).
- The alternative pathway of complement is activated by interaction with microbial surfaces.
- The classical pathway is activated by antigen-antibody complexes.
- Inappropriate activation of the complement cascade is controlled at the level of C1, C3, and C5.

Review Questions

1. An antibody preparation is being used in a laboratory protocol to study B lymphocytes. The preparation does not activate the cells or cause capping. It **does not** cause precipitation of its purified ligand, and it does not cause agglutination of latex beads covalently coupled to its ligand. Which of the following is the most likely antibody preparation?
 - (A) Monoclonal anti-CD19 IgG
 - (B) Monoclonal anti-CD56 IgG
 - (C) Papain-treated anti-CD19 IgG
 - (D) Papain-treated anti-CD56 IgG
 - (E) Pepsin-treated anti-CD19 IgG
 - (F) Pepsin-treated anti-CD56 IgG
2. IgM isohemagglutinins from an individual of blood group A are treated with pepsin. When the product of this reaction is added to group B erythrocytes, they will be
 - (A) agglutinated
 - (B) lysed
 - (C) phagocytized
 - (D) precipitated
 - (E) unaffected
3. A 26-year-old obstetric patient becomes ill during the first trimester of pregnancy with fever and lymphadenopathy. She is found to have a rising titer of anti-*Toxoplasma gondii* antibodies. She delivers a full-term baby with no apparent signs of in utero infection. The best test to diagnose acute infection in the neonate would be a parasite-specific ELISA for which isotype of immunoglobulin?
 - (A) IgA
 - (B) IgD
 - (C) IgE
 - (D) IgG
 - (E) IgM
4. A 4-year-old boy is evaluated for a possible immunologic deficiency. He has suffered repeated infections of mucosal-surface pathogens and has shown delayed development of protective responses to the standard childhood vaccinations. Immunoelectrophoresis of his serum demonstrates absence of a macroglobulin peak, and his sputum is devoid of secretory IgA. Normal numbers of B lymphocytes bearing monomeric IgM are found by flow cytometry, and serum levels of monomeric IgA, IgE, and each of the four subisotypes of IgG are normal. Which of the following deficiencies could account for these findings?
 - (A) Absence of CD40
 - (B) Absence of J chains
 - (C) Absence of IL-4
 - (D) Absence of Tdt
 - (E) Absence of TH2 cells

5. A 56-year-old homeless, alcoholic, and febrile man is brought to the emergency department after a difficult night during which his coughing kept everyone at the shelter awake. On arrival his pulse is rapid, and his breathing is labored with diffuse rales. Endotracheal aspirates produce a mucopurulent discharge containing numerous gram-positive cocci in chains. His serum contains high titers of IgM antibodies specific for the polysaccharide capsule of *Streptococcus pneumoniae*. The effector mechanism most likely to act in concert with this early IgM production to clear infection is
- (A) ADCC
 - (B) complement-mediated opsonization
 - (C) cytotoxic T lymphocytes
 - (D) LAK cells
 - (E) NK cells
6. A 3-year-old boy has had several bouts with pneumonia. *Streptococcus pneumoniae* was isolated and identified in each of the cases. The child was treated with penicillin each time, and the condition resolved. He is now being evaluated for a potential immunologic deficiency. Serum electrophoresis shows age-normal values for all isotypes of immunoglobulin, but serum levels of some components of complement are depressed. Which of the following deficiencies could explain his problem?
- (A) C1
 - (B) C2
 - (C) C3
 - (D) C4
 - (E) C5

Answers and Explanations

1. **The correct answer is C.** The cell surface marker which is typically used to identify B lymphocytes is CD19. This is a component of the B-cell/signal transduction complex and thus will be found on all B cells. Treatment of IgG with papain yields two monovalent antigen binding (Fab) fragments and destroys the function of the Fc portion of the molecule. Immunoglobulin molecules that are disrupted this way lose their ability to cross-link the receptors on cells, to promote precipitation or agglutination, and to activate cells by providing a first stimulatory signal.

Monoclonal anti-CD19 IgG (**choice A**) is a divalent antibody molecule that recognizes the signal transduction complex on B cells. Monoclonal antibodies can cross-link cell-surface receptors and cause capping, cell activation, and precipitation. Agglutination is usually accomplished using IgM because a very large molecule is needed to overcome the zeta potential (repulsive charge) of erythrocytes. If IgG is used, a second developing antibody must be added.

Monoclonal anti-CD56 IgG (**choice B**) is a divalent antibody molecule that recognizes a molecule found on NK cells. Because both arms of the molecule are intact, it is capable of causing capping, cell activation, precipitation, and agglutination if a developing antiserum is added.

Papain-treated anti-CD56 IgG (**choice D**) would not be used for the study of B lymphocytes because CD56 is a marker for NK cells.

Pepsin-treated anti-CD19 IgG (**choice E**) is a divalent molecule possessing two Fab fragments joined together ($F(ab')_2$), and a fragmented Fc region. The $F(ab')_2$ portion of the antibody is capable of causing capping, cell activation, precipitation, and, with a developing antiserum, agglutination.

Pepsin-treated anti-CD56 IgG (**choice F**) is a divalent molecule possessing two Fab fragments joined together ($F(ab')_2$) and a fragmented Fc region. Its specificity is for NK cells. Additionally, the $F(ab')_2$ portion of the antibody is capable of causing capping, cell activation, precipitation, and, with a developing antiserum, agglutination.

2. **The correct answer is A.** Isohemagglutinins are IgM antibodies that will agglutinate the RBCs of individuals with another blood type. They are believed to be made due to exposure to cross-reactive antigens found on the surface of normal gut flora organisms. Thus, a person of blood group A will produce isohemagglutinins that will agglutinate type B cells. If these antibodies are pretreated with pepsin, a divalent $F(ab')_2$ fragment and destruction of the Fc will result. A divalent fragment is capable of causing agglutination.

Lysed (**choice B**) is not correct because it would require the integrity of the complement-binding regions of the IgM, which are found in the Fc, and the question stem does not provide complement in the mix.

Phagocytized (**choice C**) is not correct because it would require an intact cell-binding region in the Fc, and the question stem does not provide phagocytic cells in the mix.

Precipitated (**choice D**) would be the correct answer if the antigen in question were a soluble protein. Proteins precipitate when treated with specific antibodies, particles agglutinate. The two particles used in laboratory medicine are latex beads and erythrocytes. If neither of these is mentioned, then the student can assume that treatment would result in precipitation, not agglutination. Precipitation has exactly the same requirements as agglutination: a divalent antigen-binding molecule.

Unaffected (**choice E**) would be the correct answer if papain had been used to treat the isohemagglutinins. Because papain produces two monovalent Fab fragments, these are incapable of cross-linking antigen (whether soluble protein or particle), so neither agglutination nor precipitation would be possible.

3. The correct answer is E. The only way to identify a neonatal infection serologically is by detection of pathogen-specific IgM antibodies. This is because the fetus receives IgG antibodies from the mother by active transport across the placenta. Because you cannot identify the source of the antibodies, IgG detection in the child can simply reflect this natural passive type of protection. Because IgM does not cross the placenta, any IgM detected in the neonate is being produced in the child and is reflective of a response to infection. In this way, all children born to HIV-infected mothers will be seropositive by both ELISA and Western blot, but only 20% will actually be infected in utero, even in the absence of antiviral therapy.

IgA (choice A) does not usually begin to be produced by a child until one to two years after birth. At the end of the first year, most children have no more than 20% of adult values, so it would not be a useful diagnostic in the neonate. Additionally, because *Toxoplasma gondii* is an intracellular parasite, IgA would not be the most effective immune response in any individual.

IgD (choice B) will be produced by an infected neonate along with IgM because of alternative RNA splicing, but this is not a useful diagnostic. IgD rarely reaches levels easily detected by serology, and the immunoglobulin has the shortest half-life of all the immunoglobulins. The function of secreted IgD, if any, is not clear, so it is not a useful serologic test.

IgE (choice C) does not usually begin to be produced by a child until well into the second year after birth. Additionally, because *Toxoplasma gondii* is an intracellular parasite, IgE would not be the most effective immune response in any individual.

IgG (choice D) is not a useful serologic test in a neonate because it is impossible to determine the origin of such molecules. Children infected in utero will begin to produce IgG due to isotype switching late in gestation, but because the placenta is actively transporting all maternal IgG into the fetus, it is not possible to distinguish whether the child is actually infected or simply passively protected using this technique.

4. The correct answer is B. IgM and secretory IgA are similar in that they are held together by a J chain synthesized by the B cell or plasma cell. Without the presence of the J chain, IgM would exist only in monomeric form, and the macroglobulin peak would be absent on electrophoresis. Because pentameric IgM is important for capturing newly introduced foreign antigen and thus beginning the immune response, the child is delayed in his development of protective responses to vaccination. Because secretory IgA is a dimer that protects the mucosal surfaces, such a child would be especially susceptible to infectious agents crossing the mucosal surfaces.

Absence of CD40 (choice A) would affect the production of IgG, IgA, and IgE, but would not prevent macroglobulin synthesis. Indeed, most patients with this defect have hypermacroglobulinemia because the CD40/CD40L interaction is necessary for isotype switching.

Absence of IL-4 (choice C) would cause problems with the ability to produce IgG, IgA, and IgE. This cytokine, produced by TH2 cells, is necessary for the differentiation and development of most antibody responses other than IgM. Thus, IgM levels either would not be affected or would be increased in a compensatory fashion.

Absence of Tdt (choice D) would cause problems with the patient's ability to perform the genetic rearrangements necessary to form the idiotype of the antibody molecule. They would not affect the isotype of antibody produced.

Absence of TH2 cells (choice E) would affect the production of IgG, IgA, and IgE, but would not affect IgM production.

5. **The correct answer is B.** One of the most effective protective responses to infections with extracellular, encapsulated bacteria, such as *Streptococcus pneumoniae*, is complement-mediated opsonization. Because IgM is the most effective antibody at activating complement, generation of C3b fragments during this process coats the bacteria and makes them more susceptible to ingestion and intracellular killing by cells of the phagocytic system.

ADCC (**choice A**), or antibody-dependent cell-mediated cytotoxicity, is a mechanism by which NK cells, neutrophils, macrophages, and eosinophils can use their Fc receptor to bind specific antibody and target an agent for lysis. No cells have Fc receptors for IgM, so this is not a mechanism that could act in concert with early IgM production.

Cytotoxic T lymphocytes (**choice C**) identify altered-self/MHC class I molecule conjugates on the surfaces of cells that are malignantly transformed or infected with intracellular pathogens. They are not a protective mechanism that acts in concert with any antibody molecule.

LAK cells (**choice D**), or lymphokine-activated killer cells, are NK cells that have been stimulated in vitro with cytokines that enhance their killing activity. These cells have a function in early surveillance against altered-self cells, but are not believed to play a role in protection against extracellular pathogens, such as this one.

NK cells (**choice E**) are members of the innate immune system and are believed to play a role in surveillance against tumor cells and other altered-self cells that fail to express MHC class I antigens on their surfaces. They would not act in concert with IgM production, and they would not be effective against an extracellular pathogen, such as this one.

6. **The correct answer is C.** The component of complement that is most important in clearance of extracellular pathogens such as *Streptococcus pneumoniae* is C3b. This fragment acts as an opsonin and enhances the ingestion and intracellular killing of the bacteria by phagocytic cells.

C1 (**choice A**) is the first component of the complement cascade activated in the classic pathway. Although it is critical to initiating those events that can culminate in the production of the membrane attack complex, it is not the most important component for the clearance of infections such as this one.

C2 (**choice B**) is the third component of the complement cascade activated in the classic pathway. Although it is critical to initiating those events that can culminate in the production of the membrane attack complex, it is not the most important component for the clearance of infections such as this one.

C4 (**choice D**) is the second component of the complement cascade activated during the classic pathway. Although it is critical to initiating those events that can culminate in the production of the membrane attack complex, it is not the most important component for the clearance of infections such as this one.

C5 (**choice E**) is the fifth component of the complement cascade activated during the classic pathway and the first step in the formation of the membrane attack complex (C5b-9). It is not the most important component for the clearance of infections such as this one.

The Generation of Cell-Mediated Effector Mechanisms

8

What the USMLE Requires You to Know

- The biologic function of cell-mediated immunity (CMI)
- The effector cells of CMI, their targets, and mechanisms of killing
- The means of regulation of CMI responses

The cell-mediated arm of the immune response (CMI) is designed to identify and eradicate antigenic stimuli that arise from **inside** the cells of the body. This occurs when cells of the host become infected with intracellular pathogens, such as viruses, some parasites, and bacteria, or when malignant transformation causes cells to express aberrant surface molecules. In such cases, TH1 cells primed in the lymph nodes and spleen serve to provide the cytokine stimuli to activate the three potential effector cells to destroy the infected or altered cells: **cytotoxic CD8+ T lymphocytes (CTLs), macrophages, and NK cells.**

One example of a cell-mediated effector mechanism that is enhanced by the action of TH1 cells is macrophage killing. This is a critical protective mechanism in the defense against organisms invading macrophages and attempting to live there (mycobacteria, *Leishmania*) or in the case where phagocytosed microbes have protective mechanisms that make them resistant to intracellular digestion (*Listeria*). In CMI against phagocytosed microbes, the specificity of the response arises from T cells, but the actual effector function is mediated by the phagocytes. This provides an important link between the adaptive and innate immune responses, and in essence, converts phagocytes into agents of the adaptive immune response. The most important cytokine elaborated by TH1 cells and CD8+ T lymphocytes to enhance the microbicidal capabilities of phagocytes is IFN- γ . In addition, production of TNF- α and TNF- β by T cells enhances inflammation and provides other stimuli that activate phagocytic cells. Macrophages and other phagocytes kill microbes intracellularly (as discussed in Chapter 6) in contrast to the mechanism observed with CTLs and NK cells.

When TH1 cytokines activate macrophages and cause tissue damage, the result is **delayed-type hypersensitivity (DTH)** (see Chapter 13). Assay of DTH by skin testing is often used as a measure of the patient's ability to mount a CMI response (e.g., Mantoux test, Lepromin test).

The CTL recognizes the cell it will ultimately kill by interaction between its TCR and MHC class I antigens on the surface of the target cell. If the cell in question is performing normal functions and therefore producing normal "self" peptides, there should be no CD8+ T cells that have a complementary TCR structure. If the cell is infected with an intracellular parasite or is expressing neoantigens reflective of tumor transformation, however, some small proportion of those CD8+ cells generated from the thymus should be capable of binding their TCRs to this MHC class I/non-self peptide combination. Unfortunately, because of the extreme polymorphism of the HLA system in humans, when tissues are transplanted between nonidentical individuals,

In A Nutshell

- CMI protects against intracellular pathogens.
- TH1 cells stimulate:
 - Macrophages
 - CTLs
 - NK cells

In A Nutshell

- Macrophages kill intracellularly.
- Killing is enhanced by IFN- γ , TNF- α , and TNF- β .
- The DTH skin test measures TH1 function.

cells of the transplant are often targeted by CTLs as abnormal. In spite of the fact that they may only be presenting normal cellular peptides, in these cases the HLA molecules themselves are different enough to turn on the system (see Chapter 14).

Clinical Correlate

MHC Class I Deficiency

A recessively inherited deficiency in the production of MHC class I molecules has been described in rare individuals. Some of these cases result from the failure of TAP molecules to transport MHC I molecules to the surface of the cell, and others are due to the production of aberrant or nonfunctional MHC I molecules themselves. These patients, as anticipated, suffer from profound deficiencies of CD8+ T cells, although numbers of CD4+ T cells are normal. This is because MHC class I expression in the thymus is essential to the development of committed CD8+ cells. These individuals are susceptible to multiple, recurrent viral infections, but interestingly, not all viral infections appear to be involved. It may be that they are able to compensate in the case of some specific viral infections, by using $\gamma\delta$ T cells (which are not dependent on thymic education and the expression of MHC) or NK cells to control those infections, whereas other viruses require killing by CD8+ cells alone. Patients with this defect will possess a normal ability to mount CD4+ cell responses such as DTH and antibody production, and indeed their antibody responses may be higher than usual, presumably because of the absence of inhibitory feedback signals (IFN- γ) from CD8+ cells directed toward TH2 responses. Thus, they are not very efficient at terminating antibody responses and tend to overproduce them as a result.

In A Nutshell

CTL stimulation requires:

- Non-self peptide/class I MHC
- CD28-B7
- IL-2

In A Nutshell

CTLs kill by:

- Perforins
- Cytokines
- Granzymes
- Fas/FasL

CTLs are capable of differentiation and cloning by themselves in the presence of the appropriate non-self peptide/class I MHC antigen stimulus, but are much more effective in so doing if they are assisted by signals from APCs or TH1 cells. CD8+ T cells express CD28 molecules, which can be bound by the B7 complementary molecule on the surface of APCs, which simultaneously express the appropriate MHC class I/peptide combination. This provides a costimulatory signal that stimulates CTL differentiation and cloning. The TH1 cell secretes IL-2 similarly acting on CD8+ cells to enhance their differentiation and cloning.

CTLs kill their target by the delivery of toxic granule contents that induce the apoptosis of the cell to which they attach. This process occurs in four phases:

- **Attachment** to the target (mediated by TCR, CD8, and LFA-1 integrin)
- **Activation** (cytoskeletal rearrangement to concentrate granules against attached target)
- **Exocytosis** of granule contents (perforin and granzymes)
- **Detachment** from the target

The actual production of apoptosis in the target may be mediated in distinct fashions. First, **perforin** present in the CTL granules creates pores in the membrane of the target cell through which **granzymes** (serine proteases) enter the target, inducing the activation of caspases, which in turn induces apoptosis. Second, cytokines such as IFN- γ with TNF- α or TNF- β can also induce apoptosis. Furthermore, activated CTLs express a membrane protein called **Fas ligand** (FasL), which may bind to its complementary structure on the target, Fas. When this occurs, caspases are induced and apoptosis results.

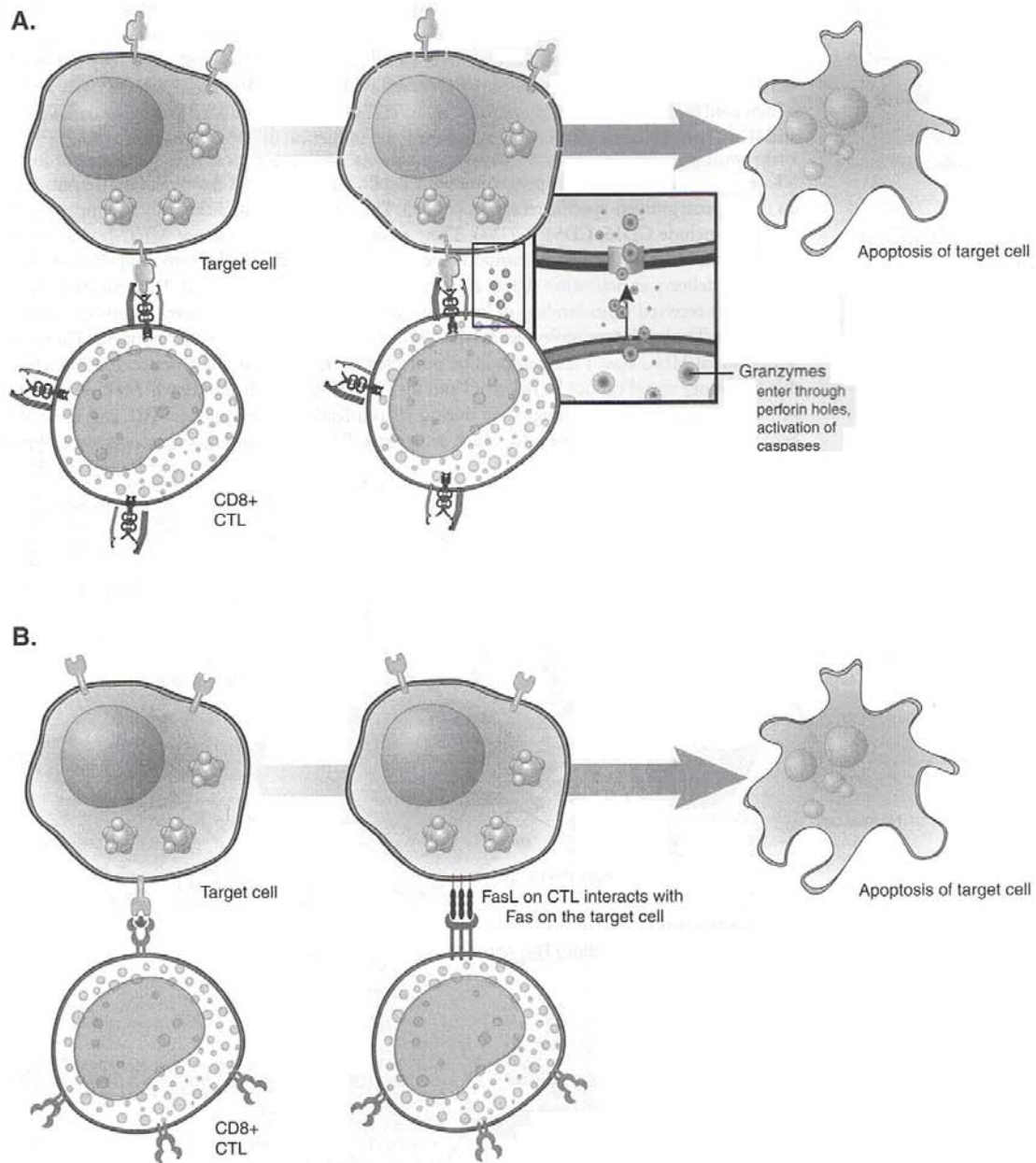


Figure II-8-1. Mechanisms of Cytotoxic T-Cell Killing. *A*: Perforin and Granzymes, and *B*: Fas/Fas Ligand Interaction

In A Nutshell

NK Cells

- Kill tumor cells and virus-infected cells
- Kill by granzymes and perforin
- Enhanced by IFN- α , IFN- β , and IL-12
- Inhibited by MHC class I
- Counted with CD16 and CD56

Another cell-mediated effector mechanism enhanced by the action of TH1 cells is NK cell killing. NK cells are the only lymphocyte members of the innate branch of the immune response. They exhibit the capacity to kill cells infected with some viruses and tumor cells, and they kill via the same mechanisms of inducing apoptosis observed with CTLs (granzymes, perforin). NK activity is increased in the presence of interferons (IFNs) α and β (IFNs stimulated during viral infections) and IL-12 (produced by phagocytic cells during the induction of TH1 responses).

NK cells share a common early progenitor with T cells, but they do not develop in the thymus. They do not express antigen-specific receptors or CD3. The markers are used clinically to enumerate NK cells and include CD16, CD56 (CD3-). Their recognition of targets is not MHC-restricted, and their activity does not generate immunologic memory. NK cells employ two categories of receptors: One delivers an activation signal, and one delivers an inhibitory signal. The activation signals seem to be received from binding of lectins possibly conserved among many groups of common pathogens. The inhibitory molecules on the NK cell seem to bind MHC class I antigens: Thus, a cell with normal MHC class I antigens will be protected from killing. In the absence of MHC class I, this inhibitory signal cannot be delivered and the NK cell will kill the target cell. MHC class I antigen expression may be downregulated during virus infections (see Appendix IV), and these antigens may be lost among tumor cells, which are genetically unstable and may delete portions of their genome (see Chapter 15).

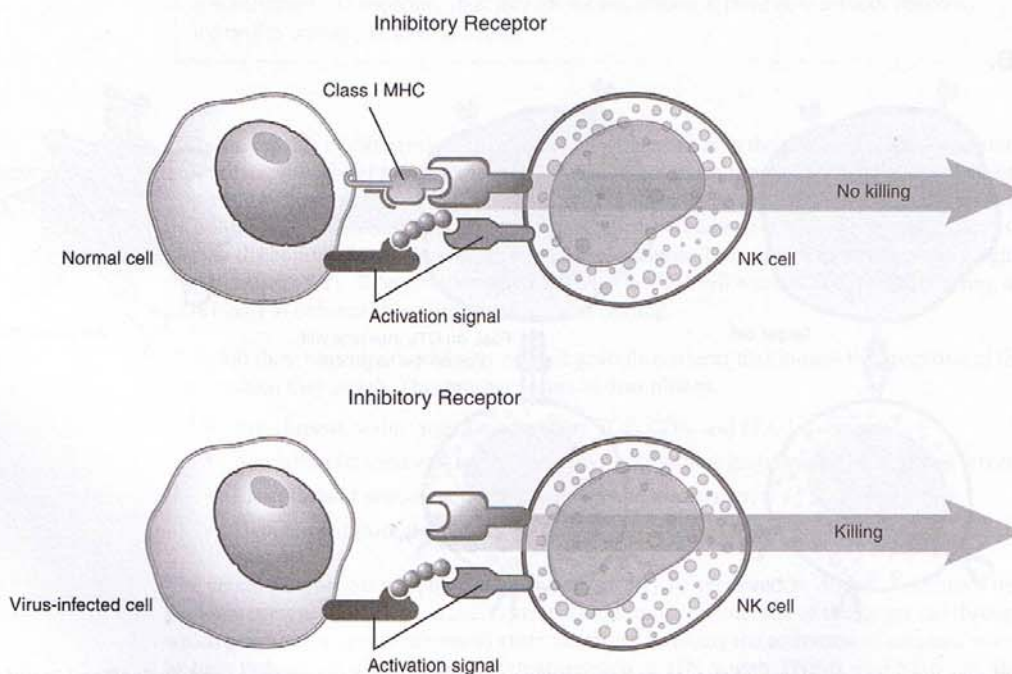


Figure II-8-2. Activation of NK Cells

A final mechanism of cell-mediated cytotoxicity that bridges humoral and cell-mediated effector systems in the body is **antibody-dependent cell-mediated cytotoxicity (ADCC)**. A number of cells with cytotoxic potential (NK cells, macrophages, monocytes, neutrophils, and eosinophils) have membrane receptors for the Fc region of IgG. When IgG is specifically bound to a target cell, the cytotoxic cells can bind to the free Fc "tail" and subsequently cause lysis of the target cell. Although these effectors are not specific for antigen, the specificity of the idiotype of the antibody directs their cytotoxicity. The mechanism of target cell killing in these cases may involve

- Lytic enzymes
- Tumor necrosis factor
- Perforin

In A Nutshell

ADCC

- NK cells
- Macrophages and monocytes
- Neutrophils
- Eosinophils
- Target recognition via IgG
- Killing by lytic enzymes, TNF, and perforin

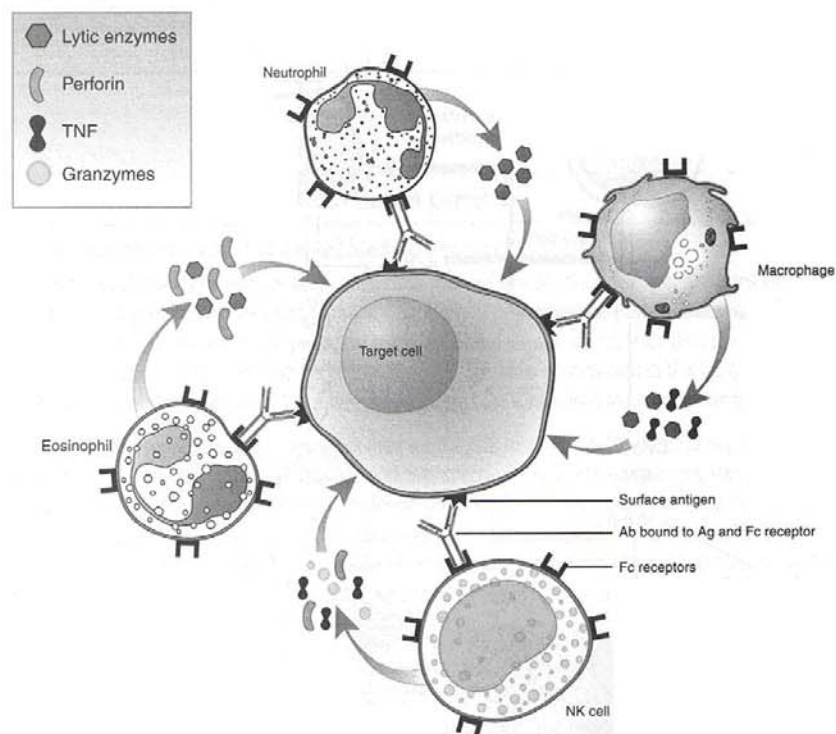


Figure II-8-3. Antibody-Dependent Cell-Mediated Cytotoxicity

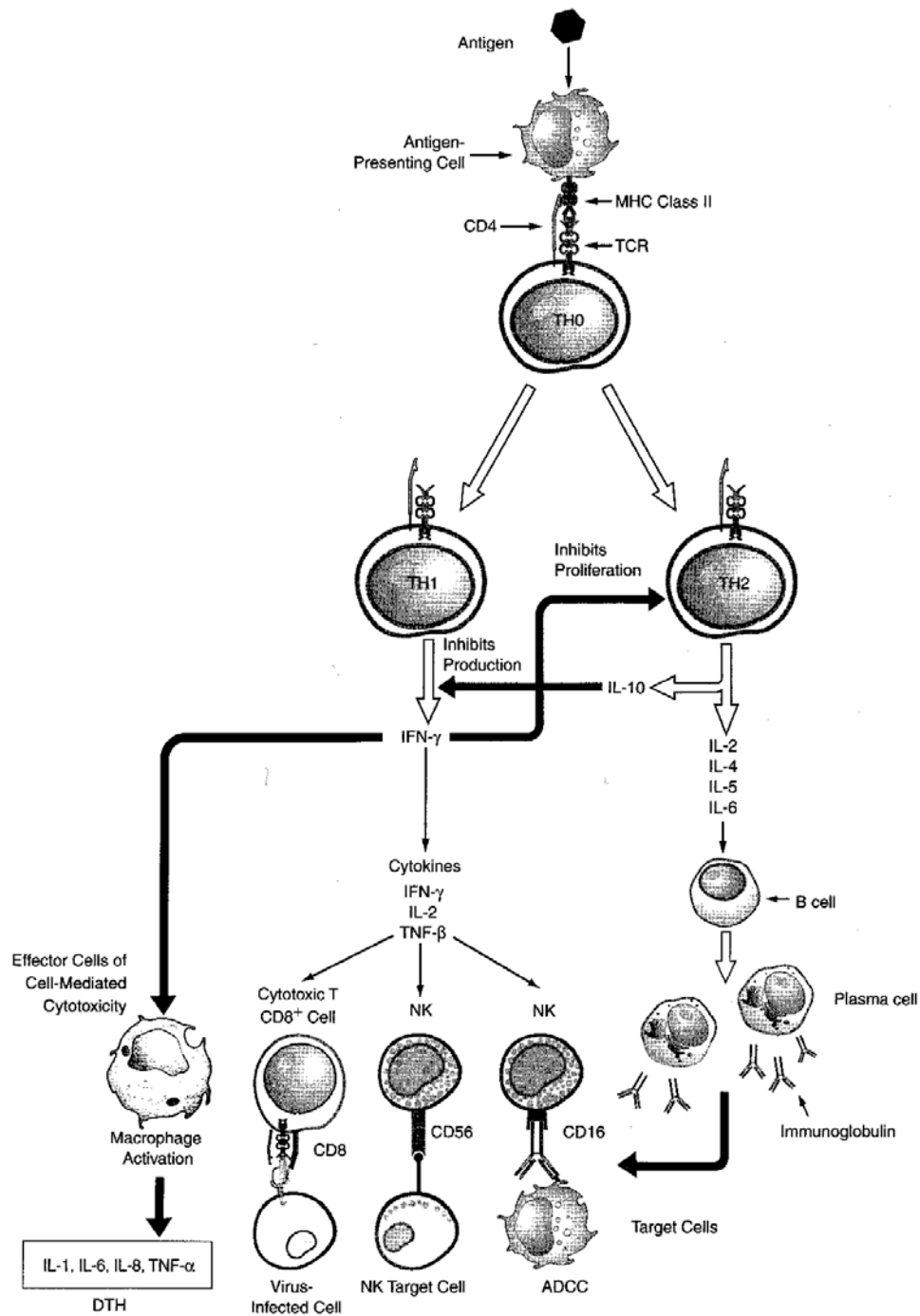


Figure II-8-4. Overview of Cell-Mediated Immunity

Table II-8-1. Effector Cells in Cell-Mediated Immunity

Effector Cell	CD Markers	Antigen Recognition	MHC Recognition Required for Killing	Effector Molecules
CTL	TCR, CD3, CD8, CD2	Specific, TCR	Yes, class I	Perforin, granzymes, cytokines (TNF- β , IFN- γ)
NK cell	CD16, CD56, CD2	ADCC: specific by IgG; otherwise, recognizes lectins	No, MHC I recognition inhibits	Perforin, granzymes, cytokines (TNF- β , IFN- γ)
Macrophage	CD14	Nonspecific	No	TNF- α , enzymes, NO, oxygen radicals

Clinical Correlate

Viral Strategies for Evasion of the Cell-Mediated Immune Response

In the struggle for survival inside the human host, viruses have developed several strategies for evasion of protective cell-mediated immune responses. In viruses that replicate quickly, as do most RNA viruses, CTLs are the major protective response. These viruses will replicate until the TH1 response turns on CTLs and destroys all infected cells. The virus is eradicated in this host, but the rate of viral replication has ensured that a new nonimmune host is now infected to continue the cycle.

In viruses that replicate slowly and cannot out-race the CTL response, a different strategy is necessary. Many such viruses block host cell protein synthesis, specifically MHC class I synthesis, transport, and expression (see Appendix IV). With this molecule downregulated, infected cells become "immune" to CTL recognition and killing. The downregulation of class I MHC, however, makes most of these viruses susceptible to the second cell-mediated immune mechanism, NK killing. Therefore, in most cases, the conjunction of CTL and NK mechanisms is sufficient to eradicate viral infections.

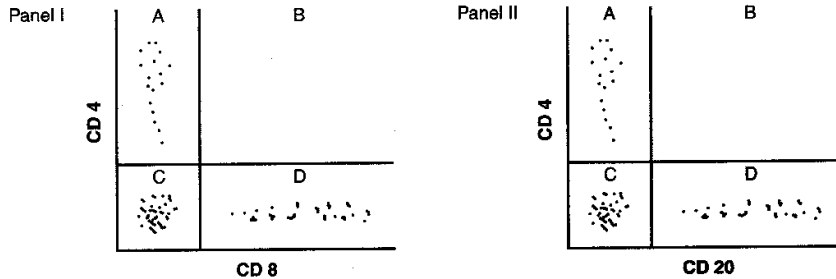
The exception to this rule is cytomegalovirus (CMV). CMV downregulates the class I MHC molecule like many other viruses, but also produces a "decoy" MHC class I-like molecule. This virally encoded decoy molecule is too different from the host's to be recognized by CTLs, but it is sufficient to fool the NK cell, so CMV escapes both of these killing mechanisms. The reason humans are not all overwhelmed by this pathogen is that CMV can be successfully killed by the third cell-mediated immune response, ADCC. IgG molecules with specificity for the surface-expressed CMV antigens, such as the decoy molecule, will bind to virus-infected cells and be recognized by Fc receptor-bearing effector cells, like NK cells, which will then kill the target.

Chapter Summary

- The cell-mediated immune response protects against intracellular pathogens.
- TH1 cells activate macrophages, CTLs, and NK cells.
- Macrophages kill intracellularly in response to TNF- α , TNF- β , and IFN- γ activation.
- The DTH skin test measures TH1 function.
- CTLs kill targets wearing MHC class I/ altered-self peptides, using perforin, granzymes, and Fas ligand.
- CTLs are stimulated by CD28/B7 interaction and IL-2 from TH1 cells.
- NK cells kill tumor and virus-infected cells using granzymes and perforin.
- NK cells are stimulated by IFN- α , IFN- β , and IL-12, and kill targets lacking MHC I.

Review Questions

1. An experimental treatment for melanoma involves in vitro stimulation of tumor-specific killer cells with tumor cells transfected with a gene for production of altered-self MHC class I molecules. As a first step, peripheral blood leukocytes from the patient are incubated with fluorescent-labeled antibodies against CD4, CD8, and CD20. The cells are then subjected to flow cytometry and separated into different populations based on their expression of these cell surface markers. In which quadrant would you find the cell sub-population most likely to produce a beneficial anti-tumor response in this protocol?



- (A) Panel I, quadrant A
 - (B) Panel I, quadrant B
 - (C) Panel I, quadrant C
 - (D) Panel I, quadrant D
 - (E) Panel II, quadrant A
 - (F) Panel II, quadrant B
 - (G) Panel II, quadrant C
 - (H) Panel II, quadrant D
2. A 62-year-old accountant develops a solid tumor that is unresponsive to chemotherapy. He elects to participate in an experimental treatment protocol to stimulate his own immune effector cells to recognize and kill the malignant cells. The tumor cells are found to have no expression of MHC class I antigens. Which of the following in vitro treatments of his tumor cells is likely to stimulate the most effective immune response when reinfused into the patient?
- (A) IFN- γ
 - (B) IL-2
 - (C) IL-8
 - (D) IL-10
 - (E) TNF- β

3. *Toxoplasma gondii* is an intracellular parasite that lives inside phagocytic and nonphagocytic cells by generating its own intracellular vesicle. This may allow it to avoid recognition and killing by CD8⁺ lymphocytes, which require the presentation of foreign peptides transported into the endoplasmic reticulum and loaded onto MHC molecules that have
 - (A) a β_2 domain instead of a β_2 microglobulin
 - (B) invariant chains
 - (C) a peptide-binding groove
 - (D) a single transmembrane domain
 - (E) two similar chains

4. Before 1960, children with enlarged thymus glands were frequently irradiated to functionally ablate this organ, whose role was not yet known. Over the lifetime of such individuals, which of the following conditions was likely to develop?
 - (A) Depressed immune surveillance of tumors
 - (B) Depressed oxygen-dependent killing by neutrophils
 - (C) Depressed primary response to soluble antigens
 - (D) Increased cellularity of lymph node paracortical areas
 - (E) Increased tendency toward atopy

5. A 42-year-old Nigerian man who is in the United States visiting with his brother comes into the hospital clinic. He complains of several months of weight loss, night sweats, mild sputum production, and the spitting up of blood. You run a PPD skin test and the results are positive. What can you conclude from this result?
 - (A) A cell-mediated immune response has occurred
 - (B) A humoral immune response has occurred
 - (C) The B-cell system is functional
 - (D) The B- and T-cell systems are functional
 - (E) The neutrophilic phagocyte system is functional

Answers and Explanations

1. **The correct answer is D.** To generate tumor-specific killer cells in vitro that would kill tumor cells transfected with an altered-self MHC class I gene, one would need to start with potential killer cells that use MHC I as a stimulatory signal. The only cytotoxic cell in the body that meets these criteria is the cytotoxic T lymphocyte (CTL). To determine which quadrant of the two figures would contain CTLs, the student needs to know how to interpret the data generated by the flow cytometer. In each case, the instrument plots the fluorescence of a cell passing through it as an individual dot on the histogram. So in panel I, increasing levels of fluorescence with antibody to CD8 are plotted as one moves to the right, and increasing levels of fluorescence with antibody to CD4 are plotted as one moves upward. Thus, the cells most strongly positive with CD8 are found the farthest to the right in quadrant D of panel I.

Panel I, quadrant A (**choice A**) would contain cells that are CD4+ and CD8-. These would be helper cells, and they would not be cytotoxic to transfected tumor cells.

Panel I, quadrant B (**choice B**) would contain cells that are double-positive for CD4 and CD8. These cells would be found as immature thymocytes in the thymus and not in the blood; thus, there are no double-labeled cells shown in this quadrant.

Panel I, quadrant C (**choice C**) contains the cells that have only background levels of fluorescence with antibodies to CD4 and CD8. These would be nonhelper, noncytotoxic cells, so they could be B lymphocytes, NK cells, or any other peripheral blood leukocyte.

Panel II, quadrant A (**choice E**) would contain cells which are strongly CD4+ and CD20-. These are helper T lymphocytes.

Panel II, quadrant B (**choice F**) would contain cells positive for CD4 and CD20. Because CD4 is a TH cell marker, and CD20 is a B-cell marker, such cells do not exist and thus this quadrant is empty.

Panel II, quadrant C (**choice G**) contains the cells that have only background levels of fluorescence with antibodies to CD4 and CD20. These would be nonhelper, non-B cells, so they could be cytotoxic T lymphocytes, NK cells, or any other peripheral blood leukocyte. Although this quadrant clearly contains some of the cytotoxic cells that this question asks about, there are other cells present, so this is not the best answer.

Panel II, quadrant D (**choice H**) contains the cells strongly positive for CD20 and negative for CD4. These are B lymphocytes.

2. **The correct answer is A.** The killer cells cytotoxic to targets lacking MHC class I antigens are NK cells. These cells are members of the innate immune response, and as such their response is not enhanced over time. The most specific, inducible cytotoxic cells in the body are cytotoxic T lymphocytes (CTLs), which depend on MHC class I recognition of their target. Because this question asks how the tumor cells can be altered to make them better stimulators of an immune response, one approach would be to increase their expression of MHC class I molecules. This can be accomplished by treatment of the tumor cells with interferon (IFN)- γ . IFN- γ increases expression of both classes I and II MHC products on cells.

IL-2 (**choice B**) is a product of TH1 lymphocytes and induces proliferation of antigen-primed TH and cytotoxic T cells. It also supports their long-term growth. It would not have an effect on this patient's tumor cells.

IL-8 (**choice C**) is a product of macrophages and endothelial cells and acts on neutrophils to cause their chemotaxis and extravasation into tissues. It would not have an effect on this patient's tumor cells.

IL-10 (**choice D**) is a product of TH2 cells and acts on macrophages to suppress their cytokine production. It therefore indirectly reduces cytokine production by TH1 cells and

dampens the activation of the cell-mediated arm of the immune response. It would not have an effect on this patient's tumor cells.

TNF- β (**choice E**) is a product of macrophages and NK cells and acts on tumor cells to cause direct cytotoxicity. It acts on inflammatory cells to induce cytokine secretion and causes the cachexia associated with chronic inflammation. It would not have an effect on this patient's tumor cells.

3. **The correct answer is D.** CD8+ lymphocytes, or cytotoxic T lymphocytes recognize their target cells by binding to MHC class I molecules containing altered-self peptides. The class I molecule is a two-chain structure, with one long α chain that passes through the cellular membrane and a shorter chain called β_2 microglobulin that becomes associated with the α chain.

A β_2 domain instead of a β_2 microglobulin (**choice A**) describes the class II MHC molecule. It is loaded with peptides by the endosomal (exogenous) pathway and is recognized by CD4+ T cells.

Invariant chains (**choice B**) are found blocking the peptide-binding groove of the class II MHC molecule immediately after synthesis. These chains are digested away when the class II MHC is exposed to the contents of the phagocytic vesicles of macrophages, and the groove is loaded with peptides from the ingested particle.

A peptide-binding groove (**choice C**) would be found in both class I and II MHC molecules and is therefore not the best answer.

Two similar chains (**choice E**) would be found in the class II MHC molecule. It is composed of an α and a β chain of similar lengths, both of which have transmembrane domains. The class II MHC molecule is loaded with peptides by the endosomal (exogenous) pathway and is recognized by CD4+ T cells.

4. **The correct answer is A.** Although the ablation of the thymus in early childhood will ultimately have far-reaching consequences in the development of many immune responses, the immune surveillance of tumors is performed only by cytotoxic T cells and NK cells, and thus would be profoundly affected by this treatment. Other parameters that could be depressed include immune responses to intracellular pathogens and secondary antibody responses.

Depressed oxygen-dependent killing by neutrophils (**choice B**) would not be expected in this case because neutrophils are components of the innate immune response and function in the absence of T-cell help.

Depressed primary response to soluble antigens (**choice C**) would not be expected in this case because the IgM response to many antigens is T-cell independent. It is class switching that would be impossible without T-cell help.

Increased cellularity of lymph node paracortical areas (**choice D**) would not be expected in this case because the paracortex of lymph nodes is a T-cell area. Therefore, following thymic irradiation, decreased cellularity of these regions would occur.

Increased tendency toward atopy (**choice E**) would not be expected in this case because atopic allergies are those that involve IgE antibodies and mast cells. IgE cannot be produced without T-cell help, so athymic individuals will have decreased tendency toward atopy.

5. **The correct answer is A.** The Mantoux test, or tuberculin test (or simply the TB skin test), is the classic clinical demonstration of the function of the delayed-type hypersensitivity response. This is a cell-mediated reaction caused by sensitization of TH1 cells and demonstrated by the influx and activation of macrophages in response to the cytokines that they elaborate.

That a humoral immune response has occurred (**choice B**) is not true. Antibodies are not involved in the production of a DTH response, and they are not important products during infections with most intracellular pathogens.

That the B-cell system is functional (**choice C**) is not true. B cells do not play a role in the DTH response, and they do not play a major role in defense during infections with most intracellular pathogens.

That the B- and T-cell systems are functional (**choice D**) is not true. The DTH response certainly demonstrates that the TH1 response is functional, but it says nothing about the function of B cells.

That the neutrophilic phagocyte system is functional (**choice E**) is not true. Neutrophils do not play a role in the elicitation of the DTH response.

The Generation of Immunologic Memory

9

What the USMLE Requires You to Know

- The characteristics of memory B and T lymphocytes
- The recirculation patterns of memory B and T lymphocytes
- The characteristics and function of secretory IgA
- The comparative attributes of the primary and secondary immune responses

As long as foreign antigen is present in the system, the activation, proliferation, and differentiation of lymphocytes into effector cells will continue in the secondary lymphoid organs. As the effector mechanisms generated are successful in destroying or causing clearance of the invader, however, the system will slowly return to its baseline quiescent state, and **immunologic memory** will be generated. This is important because it avoids expending energy on the generation of cells and molecules that are no longer needed and may be potentially harmful in the absence of the invading stimulus. It also “resets” the baseline homeostatic function of the immunologic organs so that they can efficiently respond to new and emerging challenges.

Both B- and T-lymphocyte populations will respond to a primary antigenic challenge by the production of long-lived memory cells. In B lymphocytes, this is primarily accomplished by the fact that their differentiation into plasma cells is antigen-dependent, and as that antigen disappears, the stimulus for that differentiation is removed. Plasma cells, which function only as factories for immunoglobulin synthesis, are relatively short-lived (two weeks) and as they die and are not replaced from the differentiating B-cell pool, the response wanes. Memory B lymphocytes differ from naive B cells in that they have undergone isotype switching; will bear membrane immunoglobulin of IgG, IgA, or IgE isotype; and enter a resting stage of the cell cycle.

The body’s mechanism of dampening T-cell activity after the primary immune response is more active, presumably because the cytokines they produce can have harmful effects if they are generated unnecessarily. These cells no longer benefit the host and can be potentially harmful, so apoptosis is induced in all but a few, which will become quiescent in a resting stage of the cell cycle. This is called **activation-induced cell death (AICD)** and is mediated through the Fas pathway. In this way, trimerization of the Fas molecule expressed on the surface of activated T cells with the Fas ligand molecule on neighboring cells initiates a signal-transduction cascade that leads to apoptosis of the Fas-bearing cell.

Although the memory cells generated during the waning stages of the primary immune response are small and relatively quiescent, they exhibit high-level expression of adhesion molecules, which will help them to recirculate throughout the body and home to areas of new antigen introduction. In this way, a response that was initiated in a single draining lymph node will become generalized throughout the body and available to initiate a rapid and powerful secondary response if that challenge is reintroduced.

In A Nutshell

As pathogens are eliminated, immunologic memory is generated.

In A Nutshell

Memory B cells have surface IgG, IgA, or IgE.

In A Nutshell

AICD removes activated T cells after the primary response.

In A Nutshell

Memory cells home to inflamed tissues.

Table II-9-1. Characteristics of Naive, Effector, and Memory Lymphocytes

	Naive Lymphocytes	Activated or Effector Lymphocytes	Memory Lymphocytes
T lymphocytes			
Migration	To peripheral lymph nodes	To inflamed tissues	To inflamed tissues or mucosa
Effector functions	None	Cytokine secretion, cytotoxicity	None
Cell cycling	Absent	Present	+/-
IL-2 receptor	Low	High	Low
B lymphocytes			
Isotype of membrane Ig	IgM and IgD	IgG, IgA, or IgE	IgG, IgA, or IgE
Affinity of Ig	Low	Increasing	High
Effector function	None	Antibody secretion	None
Morphology	Small, little cytoplasm	Large, more cytoplasm	Small

In A Nutshell

Memory cells return to the tissue where they first encountered antigen.

Clinical Correlate

Dissemination of Immunologic Memory

The vaccine used by the military against adenovirus types 4 and 7 is an enteric-coated, live, nonattenuated virus preparation. This vaccine produces an asymptomatic intestinal infection and thereby induces mucosal IgA memory cells. These cells then populate the mucosal immune system throughout the body. Vaccine recipients are thus protected against adenovirus acquired by aerosol, which could otherwise produce pneumonia.

The trafficking patterns of memory lymphocytes are different than those of either naive or effector lymphocytes. As discussed in Chapter 4, naive cells tend to home to regions of the secondary lymphoid organs specific for their cell type (T cells to paracortical areas, etc.). Effector cells tend to home to areas of active inflammation because of their expression of cell adhesion molecules such as LFA-1. Memory cells tend to home in a tissue-specific fashion, presumably returning to the type of tissue in which they first encountered antigen. In this way, some memory cells express adhesion molecules that direct them to a protective location along the digestive tract, whereas others express different adhesion molecules that tend to direct them to protective sites in the dermis of the skin.

When an antigen is introduced into the system a second time, the response of lymphocytes is accelerated and the result amplified over that of the primary immune response. The increased speed of response is due to the presence of the memory-cell progeny of the first response throughout the body, and the increased amplitude of effector production is due the fact that activation and cloning now begin from a much larger pool of respondents.

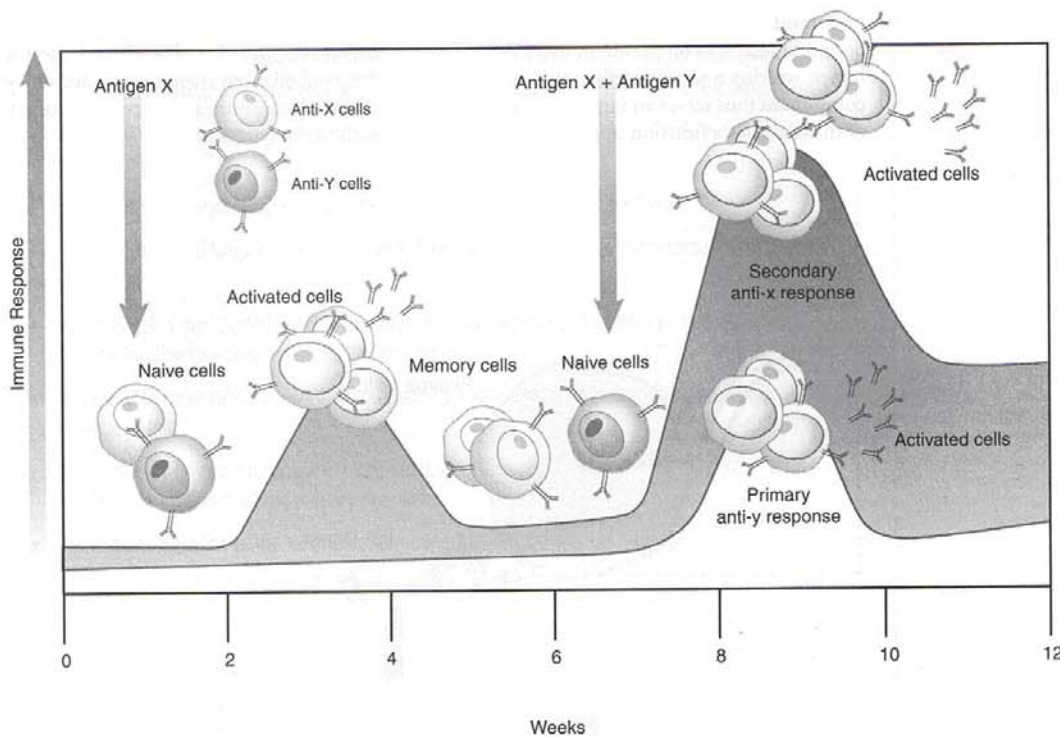


Figure II-9-1. The Primary and Secondary Immune Responses

Table II-9-2. Comparison of the Primary and Secondary Immune Responses

Feature	Primary Response	Secondary Response
Time lag after immunization	5–10 days	1–3 days
Peak response	Small	Large
Antibody isotype	IgM, then IgG	Increasing IgG, IgA, or IgE
Antibody affinity	Variable to low	High (affinity maturation)
Inducing agent	All immunogens	Protein antigens
Immunization protocol	High dose of antigen (often with adjuvant)	Low dose of antigen (often without adjuvant)

The homing of specific memory cells to epithelial and mucosal surfaces leads to the production of specialized lymphoid aggregations along these barriers. Collectively referred to as **mucosal-associated lymphoid tissues**, or **MALT**, they include the tonsils and Peyer patches, as well as numerous less well-organized lymphoid accumulations in the lamina propria. TH2 cells in these sites are dedicated to providing help for class switching to IgA. Most IgA-secreting B lymphocytes and plasma cells in the body will be found in these locations. **Secretory IgA** (that which is released across the mucosa of the respiratory, digestive, and urogenital tracts) differs from serum IgA in an

In A Nutshell

- MALT contains TH2 cells assisting IgA production.
- Secretory IgA is a dimer with secretory component.
- Secretory component:
 - Transepithelial transport
 - Protection from proteolytic cleavage

important fashion. As the IgA dimer is produced by plasma cells and B lymphocytes, it becomes bound to receptors on the abluminal side of the epithelia, is endocytosed, and is released into the lumen wearing a secretory piece that is the residue of the epithelial-cell receptor. The **secretory component** thus serves an important function in transepithelial transport, and once in the lumen of the tract, has a function in protecting the molecule from proteolytic cleavage.

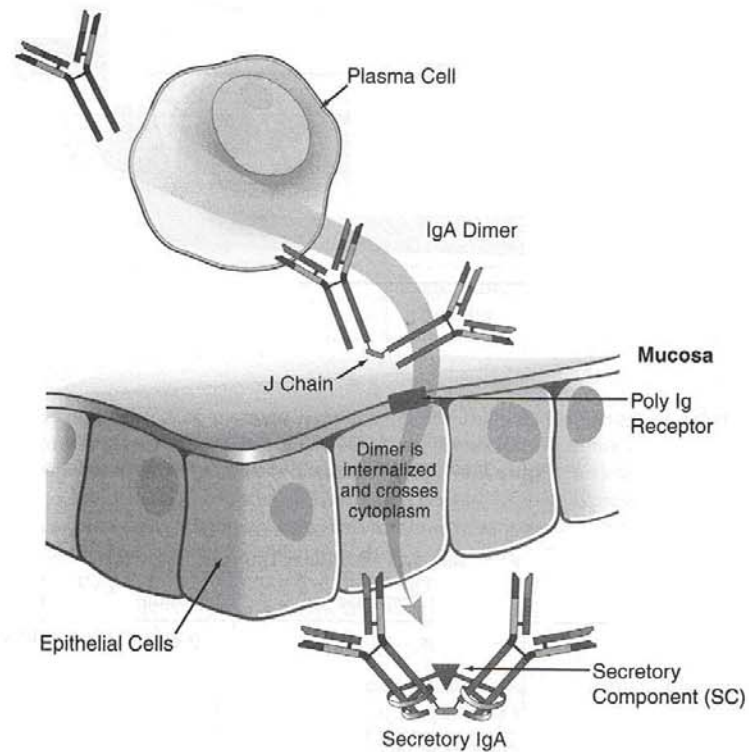


Figure II-9-2. Secretory IgA

Chapter Summary

- Immunologic memory is generated as pathogens are eliminated by the immune response.
- Long-lived B and T memory cells are generated and recirculate through the body in surveillance for the antigen.
- Memory B lymphocytes express IgG, IgA, or IgE molecules as antigen receptors.
- Mucosal-associated lymphoid tissues (tonsils, Peyer patches) are the primary sites of TH2 help for IgA production.
- Secretory IgA is transported into the lumen of the gastrointestinal, respiratory, or genitourinary tracts by binding to a polyimmunoglobulin receptor.
- This receptor (now called a secretory component) is retained as protection from proteolytic cleavage of IgA.
- The secondary immune response is more rapid, is larger, has higher affinity, and requires less antigenic stimulation than the primary response.
- The immunoglobulins of the secondary immune response are IgG, IgA, and IgE.

Review Questions

1. A patient is suffering from lymphadenopathy and splenomegaly. He has greatly increased numbers of lymphocytes, reduced numbers of platelets, and autoimmune anemia. When his peripheral blood leukocytes are exposed to T-cell mitogens, they proliferate wildly, even for weeks after the mitogenic stimulus is removed. Which of the following is most likely to be the genetic defect?
 - (A) Absence of complement
 - (B) Absence of Fas
 - (C) Absence of interferon- γ
 - (D) Absence of perforin
 - (E) Absence of TNF

2. Up until the 1970s, tonsillectomies were routinely performed on children with swollen tonsils. This procedure has lost its widespread appeal as we have learned the important role of mucosal-associated lymphoid tissue (MALT) in the protective immune response. What is the major immunoglobulin produced by the MALT?
 - (A) A dimeric immunoglobulin with secretory component
 - (B) A monomeric immunoglobulin that crosses the placenta
 - (C) A monomeric immunoglobulin bound by mast cells
 - (D) A monomeric immunoglobulin that opsonizes
 - (E) A pentameric immunoglobulin that activates complement

3. In a lifetime, a person may receive a dozen or more tetanus toxoid inoculations. When boosters are administered at 10-year intervals, which of the following would be true of the B lymphocytes that respond?
 - (A) Their receptors would have high avidity
 - (B) They would be large and highly metabolic
 - (C) They would have low levels of adhesion molecules
 - (D) They would have surface IgG, IgA, or IgE
 - (E) They would have surface IgM

4. A 64-year-old man undergoes surgery to excise 18 inches of bowel with adenocarcinoma. When the tissue and draining mesenteric lymph nodes are sent for pathologist's examination, the Peyer patches are noted to be hyperplastic with IgA-secreting plasma cells, but there is no secretory IgA found in the lumen of the colon. Which of the following changes in the bowel epithelium could explain this finding?
- (A) Failure of isotype switching
 - (B) Failure of variable domain gene-segment rearrangement
 - (C) Loss of J chain synthesis
 - (D) Loss of the polyimmunoglobulin receptor
 - (E) Loss of TH2 cells
5. An immunologic laboratory is studying the migration patterns of different lymphocyte subpopulations. One population of small, nondividing lymphocytes, which are CD3+ and CD4+ and express low levels of IL-2 receptors but high levels of LFA-1, are labeled with a radioactive marker and traced as they migrate through the body. What type of cell has been labeled in this case?
- (A) A blast cell
 - (B) A memory T cell
 - (C) An activated T cell
 - (D) A naive T cell
 - (E) An effector T cell

Answers and Explanations

1. **The correct answer is B.** This patient has Canale-Smith syndrome, a condition in which a mutant Fas protein is produced. This mutant protein competes with the normal proteins essential for the Fas-mediated death pathway. For this reason, these patients develop a progressively increasing and ultimately unsustainable number of lymphocytes of specific clones and the inability to mount any new protective response.

Absence of complement (**choice A**) is not a condition that would result in a failure of homeostasis in immune cells. Complement is activated by invasion of foreign pathogens or by the complexes of antigen and antibody. Neither of these is believed to play a role in the dampening of specific immune responses when they are no longer necessary.

Absence of interferon- γ (**choice C**) would tend to promote immune deviation toward the TH2 arm of the immune response and promote antibody synthesis. It would not cause uncontrolled cell proliferation or an inability to remove unnecessary cells. This cytokine is a product of TH1 cells, which then inhibits the function of TH2 cells.

Absence of perforin (**choice D**) would inhibit the ability of cytotoxic T lymphocytes or NK cells to kill their specific targets. It is not a major mechanism involved in homeostasis of lymphocyte populations in the body.

Absence of tumor necrosis factor (TNF) (**choice E**) would cause decreased killing of some tumor cell targets and decreased phagocytosis. Neither TNF- α nor TNF- β is believed to be involved in the homeostasis of lymphocyte populations in the body.

2. **The correct answer is A.** The mucosal-associated lymphoid tissues (MALT) are the major sites of synthesis of IgA. IgA is a dimeric molecule held together by a J chain similar to that used in IgM. As IgA is transported across the epithelial surface, it acquires the secretory component, which functions both in transepithelial transport and protection from proteolytic cleavage.

A monomeric immunoglobulin that crosses the placenta (**choice B**) describes IgG. IgG is the major immunoglobulin of the blood and is produced in lymph nodes and spleen, but less commonly in the MALT.

A monomeric immunoglobulin bound by mast cells (**choice C**) describes IgE. IgE is the immunoglobulin that causes immediate hypersensitivity by virtue of its attraction to the Fc receptors of mast cells. It is not the major immunoglobulin produced in the MALT, although it may be produced there.

A monomeric immunoglobulin that opsonizes (**choice D**) describes IgG. IgG is the major immunoglobulin of the blood and is produced in lymph nodes and spleen, but less commonly in the MALT.

A pentameric immunoglobulin that activates complement (**choice E**) describes IgM. IgM is the major immunoglobulin of the primary immune response and is produced in lymph nodes and spleen, but less commonly in the MALT.

3. **The correct answer is D.** The protective response to the tetanus toxoid depends on production of antibodies that prevent the binding of the toxin. After repeated immunizations, the population of memory B cells is stimulated, which is the goal of such prophylaxis. Memory B cells may have IgG, IgA, or occasionally IgE on their surfaces serving as antigen receptors.

That their receptors would have high avidity (**choice A**) is not true because avidity decreases with repeated booster inoculations. This is because IgM, which is the immunoglobulin of the primary immune response and is the receptor on mature naive B lymphocytes, is replaced in secondary and subsequent responses by isotype switching to other isotypes such as IgG or IgA or IgE. All of these molecules have less avidity than IgM

because they have fewer combining sites than IgM. The secondary and subsequent responses should have increased affinity (goodness-of-fit of idiotype for epitope), but decreased avidity.

That they would be large and highly metabolic (**choice B**) is not true because memory lymphocytes are usually small and in a resting phase of the cell cycle. Activated lymphocytes are large and highly metabolic.

That they would have low levels of adhesion molecules (**choice C**) is not true because memory lymphocytes express high levels of adhesion molecules. This allows them to migrate to areas of active inflammation where they can have maximum benefit in protection of the host.

That they would have surface IgM (**choice E**) is not true because this would describe mature, naive B lymphocytes that have not met their antigen before. As soon as the primary response begins, isotype switching to other classes of immunoglobulin is directed by TH cells.

4. **The correct answer is D.** The transport of IgA dimers from the abluminal side of the mucosa to the lumen is mediated via attachment to polyimmunoglobulin receptors on mucosal cells. This allows endocytosis of IgA into the mucosal cell and secretion onto the other side. Secretory IgA be found in the lumen of the bowel retains a residue of this receptor, secretory component, which further protects it from proteolytic cleavage inside the intestine. If this receptor were lacking, transport of IgA across the mucosa would not be possible, and the IgA dimers would be trapped on the abluminal side of the mucosa.

Failure of isotype switching (**choice A**) is not a potential cause of such a condition because isotype switching occurs in secondary lymphoid organs and not in epithelial cells. Because the IgA dimers were present, isotype switching had been successful, but transepithelial transport was not occurring.

Failure of variable domain gene segment rearrangement (**choice B**) is not a potential cause of such a condition because variable domain gene-segment rearrangement occurs in the primary lymphoid organs and not in epithelial cells. Because immunoglobulin was being produced, these gene segment rearrangements had occurred successfully, but transepithelial transport was not occurring.

Loss of J chain synthesis (**choice C**) would result in the inability of an individual to join dimers of IgA and pentamers of IgM. Because the question states that the individual was making IgA dimers, J chain is clearly being made successfully by the B cell.

Loss of TH2 cells (**choice E**) would cause the patient to be unable to switch isotypes. These persons could make only IgM, and this patient clearly has successfully produced IgA.

5. **The correct answer is B.** This is a population of memory T lymphocytes (small, nonmitotic cells rich in adhesion molecules).

A blast cell (**choice A**) is a cell undergoing blastogenesis (rapid proliferation or cloning). These cells would be large and highly mitotic and possess high levels of adhesion molecules.

An activated T cell (**choice C**) would be large and highly mitotic and possess high levels of adhesion molecules.

Naive T cells (**choice D**) would be small with scant cytoplasm. They would be nonmitotic and express low levels of adhesion molecules.

An effector T cell (**choice E**) would be large and highly mitotic and express high levels of adhesion molecules on its surface.

Vaccination and Immunotherapy

10

What the USMLE Requires You to Know

- The clinical applications of active, passive, natural, and artificial immunization
- The immunologic rationales for standard vaccination protocols
- The ontogeny of the immune response in children as it relates to vaccination, diagnosis of prenatal infection, and detection of immunodeficiency diseases
- The role of adjuvants in vaccination

Immunity to infectious organisms can be achieved by **active** or **passive immunization**. The goal of passive immunization is transient protection or alleviation of an existing condition, whereas the goal of active immunization is the elicitation of protective immunity and immunologic memory. Active and passive immunization can be achieved by both **natural** and **artificial** means.

In A Nutshell

Immunization

- Active
- Passive
- Natural
- Artificial

Table II-10-1. Types of Immunity

Type of Immunity	Acquired Through	Examples
Natural	Passive means	Placental IgG transport, colostrum
Natural	Active means	Recovery from infection
Artificial	Passive means	Horse antivenin against black widow spider bite, snake bite Horse antitoxin against botulism, diphtheria Pooled human immune globulin versus hepatitis A and B, measles, rabies, or tetanus "Humanized" monoclonal antibodies versus RSV*
Artificial	Active means	Hepatitis B component vaccine Diphtheria, tetanus, pertussis toxoid vaccine <i>Haemophilus</i> capsular vaccine Polio live or inactivated vaccine Measles, mumps, rubella attenuated vaccine Varicella attenuated vaccine

*Monoclonal antibodies prepared in mice but spliced to the constant regions of human IgG

In A Nutshell

Passive immunotherapy can cause:

- IgE production
- Type III hypersensitivity
- Anti-allotype antibodies

Passive immunotherapy may be associated with several risks:

- Introduction of antibodies from other species can generate IgE antibodies, which may cause systemic anaphylaxis (*see* Chapter 13).
- Introduction of antibodies from other species can generate IgG or IgM anti-isotype antibodies, which form complement-activating immune complexes, which can lead to type III hypersensitivity reactions (*see* Chapter 13).
- Introduction of antibodies from humans can elicit responses against minor immunoglobulin polymorphisms or **allotypes**.

Persons with selective IgA deficiency (1:700 in population, *see* Chapter 11) are at risk to develop reactions against infused IgA (a molecule they have not seen before).

Active immunization (**vaccination**) has played an important role in the reduction of morbidity and mortality from various infectious diseases, especially among children, and is currently recommended on the following schedule:

Recommended Childhood and Adolescent Immunization Schedule -- United States, 2004

Vaccine ▼	Age ►	Range of recommended ages				Catch-up vaccination				Preadolescent assessment			
		Birth	1 mo	2 mos	4 mos	6 mos	12 mos	15 mos	18 mos	24 mos	4-6 yrs	11-12 yrs	13-18 yrs
Hepatitis B		HepB #1	only if mother HBsAg (-)										
				HepB #2									
Diphtheria, Tetanus, Pertussis				DTaP	DTaP	DTaP		DTaP			DTaP		Td
Haemophilus influenzae Type b				Hib	Hib	Hib		Hib					
Inactivated Polio				IPV	IPV						IPV		
Measles, Mumps, Rubella								MMR #1			MMR #2		MMR #2
Varicella								Varicella				Varicella	
Pneumococcal				PCV	PCV	PCV		PCV			PCV	PPV	
Vaccines below this line are for selected populations													
Hepatitis A												Hepatitis A series	
Influenza													Influenza (yearly)

Figure II-10-1

Note

Live viral vaccines are never safe for immunocompromised patients.

Many of these vaccinations require multiple booster inoculations in children to achieve maximal protection. Live attenuated virus vaccines are only given after 12 months of age because residual maternal antibodies would inhibit replication and the vaccine would fail. In cases where children are at exceptionally high risk for exposure to a pathogen, this rule is sometimes broken, but administration of vaccines earlier than 6 to 9 months of age is almost always associated with the need for repeated booster inoculations.

Live viral vaccines should never be given to immunocompromised patients because even attenuated live viruses could cause significant pathology in these individuals.

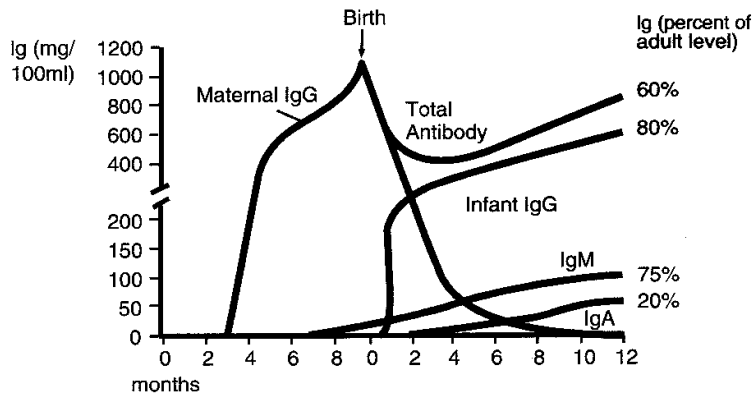


Figure II-10-2. Immunoglobulins in the Serum of the Fetus and Newborn Child

One of the key factors for consideration in the design of vaccines is the arm of the immune response that needs to be stimulated to produce a protective response. Thus, although humoral responses are usually adequate to neutralize bacterial toxins or block virus binding, if TH cells are not elicited during this process, the immune response generated will display no immunologic memory. For this reason, the vaccine for *Haemophilus influenzae* type B in children was engineered in such a way that the capsular polysaccharide of the organism was covalently coupled to a protein carrier (either *Neisseria meningitidis* outer-membrane proteins or the diphtheria toxoid), so that T-cell recognition of the protein carrier would serve to provide the activated TH cells necessary for the generation of IgA and IgG antibodies and immunologic memory. Two other polysaccharide capsular vaccines, against *Streptococcus pneumoniae* and *Neisseria meningitidis*, are not administered with a protein component, but may elicit protective immunity by stimulating previously existing reactive cells formed in response to previous infection or normal flora cross-reactivities.

As a general rule, viral vaccines that are living strains elicit both cell-mediated and humoral immunity, whereas killed viral vaccines elicit predominantly an antibody response.

Recombinant Antigen Vaccines

The hepatitis B vaccine is produced by recombinant DNA technology. In this way, the gene coding for the HBsAg is inserted into yeast cells, which then release this molecule into the culture medium. The molecule is then purified and used as the immunogen in the vaccine.

Recombinant Vector Vaccines

Attenuated viral or bacterial organisms can be used as vectors for the introduction of antigens from harmful pathogens. Vaccinia virus has been used as a vector in this way by inserting gene products from several sources, which will then be produced by the modified vector when inoculated into a host. If the foreign gene product is a viral envelope protein, it is inserted into the membrane of the infected host cell and induces the development of both cell-mediated and humoral immunity. This strategy is being used in clinical trials of HIV antigens introduced into canarypox vectors.

Clinical Correlate

- Persistence of maternal Ab affects vaccinations.
- Infant has 20% of adult IgA at 12 months, so colostrum is important.
- IgM is the only isotype useful in diagnosing infections in neonate.
- Normal infants have few infections during first few months because of maternal IgG.
- Children with immune deficiencies don't become ill until maternal IgG is low.

In A Nutshell

- Hib vaccine is a T-cell-dependent vaccine.
- Living viral vaccines elicit CMI and HMI.
- Killed viral vaccines elicit antibodies.

In A Nutshell

Hepatitis B vaccine is produced by recombinant DNA.

In A Nutshell

Experimental Vaccine Strategies

- Recombinant vector
- DNA vaccines

In A Nutshell

Adjuvants increase immunogenicity nonspecifically.

DNA Vaccines

In this vaccination strategy, plasmid DNA encoding antigenic proteins is injected directly into the muscle of the recipient. The DNA is taken up by muscle cells and dendritic cells in the area, and the protein is expressed in its natural form, without denaturation or modification. These vaccines elicit both humoral and cell-mediated responses, and the prolonged expression of the molecule generates significant immunologic memory.

Adjuvants

Adjuvants are substances that increase the immunogenicity of an antigen when administered with it. They may exert one or more of the following effects:

- Prolonging antigen persistence (aluminum potassium sulfate)
- Enhancing costimulatory signals (muramyl dipeptide)
- Inducing granuloma formation (alum)
- Inducing nonspecific lymphocyte proliferation (lipopolysaccharide and synthetic polyribonucleotides)

Chapter Summary

- Active immunization occurs when an individual is exposed (naturally or artificially) to a pathogen.
- Passive immunization occurs when an individual receives preformed immune products (antibodies, cells) against a pathogen (naturally or artificially).
- Passive immunotherapy is useful in postexposure prophylaxis but runs the risk of eliciting adverse immune responses (hypersensitivity).
- Childhood vaccination protocols must take into account risk of exposure, presence of maternal antibodies, and the type of protective immune response needed.
- Only killed or component vaccines are safe for use in immunocompromised patients.
- Live viral vaccines elicit both cellular and humoral responses, whereas killed viral vaccines elicit primarily antibody responses.
- The hepatitis B vaccine is a component vaccine produced by recombinant DNA technology.
- Recombinant vector vaccines and DNA vaccines are in experimental preparation.
- Adjuvants increase immunogenicity nonspecifically.

Review Questions

1. A 10-year-old child was bitten by a stray dog. The child is started on a course of inoculations of pooled human antirabies immunoglobulin. What is repeated inoculation of this antirabies immunoglobulin preparation likely to induce?
 - (A) Anti-allotype antibodies
 - (B) Anti-epitope antibodies
 - (C) Anti-idiotypic antibodies
 - (D) Anti-isotype antibodies
 - (E) Anti-rabies antibodies

2. All residents of a Chicago nursing home are inoculated intramuscularly with an H3N2 influenza A preparation. The goal of this protocol is to stimulate which of the following types of immunity?
 - (A) Adaptive
 - (B) Artificial active
 - (C) Artificial passive
 - (D) Natural active
 - (E) Natural passive

3. A city sanitation worker is struck by a car and his leg is crushed against his sanitation truck. The extreme trauma to the leg necessitates amputation above the knee. Although the patient's health records reflect a tetanus booster 6 years ago, the man is revaccinated; human, pooled antitetanus immunoglobulin is injected around the macerated tissue. Administration of immunoglobulin is an example of which of the following forms of immunization?
 - (A) Adaptive
 - (B) Artificial active
 - (C) Artificial passive
 - (D) Natural active
 - (E) Natural passive

4. A 28-year-old man was brought into court for nonpayment of child support. A 20-year-old woman insists that he is the father of her child. The court suggests before hearing the paternity case that various genetic tests be performed on the man, woman, and child. One of the sets of tests was for genetic immunoglobulin identification. Which immunoglobulin marker would be useful in this case?
- (A) Allotype
 - (B) Idiotype
 - (C) IgA2
 - (D) IgM
 - (E) Isotype
5. In 1988 a new childhood vaccine was developed to protect against epidemic meningitis by mixing *Haemophilus influenzae* type B capsular polysaccharide with whole, killed *Bordetella pertussis* bacteria. The function of the whole, killed bacteria in this vaccine is as a(n)
- (A) carrier
 - (B) hapten
 - (A) mitogen
 - (D) adjuvant
 - (E) immunogen

Answers and Explanations

1. **The correct answer is A.** Because rabies antitoxin is a pooled, human immunoglobulin product, repeated inoculation will cause a patient to produce anti-allotype antibodies. Allotypes are minor amino-acid sequence variations in the constant domains of heavy and light immunoglobulin chains. Their expression is genetically determined, and repeated exposure to molecules of foreign allotype can cause the production of antibodies, which recognize these sequence variations.

Anti-epitope antibodies (**choice B**) would be produced by repeated inoculation of an immunogen. The epitope of the antigen has a three-dimensional complementarity with the idiotype of the antibody molecule. In this case, anti-epitope antibodies would be generated by rabies vaccination, but the question asks what the result of repeated exposure to immunoglobulins would be.

Anti-idiotype antibodies (**choice C**) would be generated in a human if a monoclonal antibody preparation were repeatedly inoculated into another human. The idiotype of an antibody is the three-dimensional shape of its antigen-combining site. It is unique to the antibodies produced by a clone of cells. Because the material mentioned in this case is a pooled human immunoglobulin, it would contain many different idiotypes and would be unlikely to elicit any one specific anti-idiotype antibody.

Anti-isotype antibodies (**choice D**) are usually raised across species barriers. For example, to produce anti-human IgG, IgG pooled from many humans is repeatedly injected into rabbits, goats, or sheep. These animals will recognize the human determinants in the constant domains of the heavy and light chains (the isotypes) and will produce antibodies that specifically recognize those determinants.

Anti-rabies antibodies (**choice E**) are generated during vaccination. When the killed virus is administered, the patient makes an active, artificial response to the immunogen and produces immunoglobulins, which will protect against virus attachment. In this case, anti-rabies antibodies were inoculated, so there is no possibility that more of the same will be generated.

2. **The correct answer is B.** In this case, high-risk individuals are vaccinated with the serotype of influenza virus that is predicted to be most common in this flu season. This elicits an active immunologic response in the patient and is artificial by definition because it is being administered in a medical setting. This sort of immunization causes the development of memory in the patient that will protect for the whole season, but it requires approximately two weeks for development of protection.

Adaptive (**choice A**) immunity describes all immune responses that have specificity and memory. These immune responses are produced by specific B and T lymphocytes. Although adaptive immunity will be elicited in these patients, this is not the best answer because it is imprecise.

Artificial passive (**choice C**) immunity is achieved when preformed immunologic products (immune cells or antibodies) are given to a patient. These procedures provide passive protection that is rapid but lacks immunologic memory. Because it is administered in a medical setting, it is, by definition, artificial.

Natural active (**choice D**) immunity would result following recovery from an infection.

Natural passive (**choice E**) immunity is acquired across the placenta and in the colostrum and breast milk, from mother to child. The child receives preformed antibodies (IgG across the placenta and IgA in milk) that protect the child until a natural active immune response can be mounted.

3. **The correct answer is C.** In this case, an attempt at postexposure prophylaxis against tetanus is made by inoculating antitetanus immunoglobulin into the patient. When preformed immunologic products (immune cells or antibodies) are given to a patient, the procedure provides passive protection that is rapid but lacks immunologic memory. Because it is being administered in a medical setting, it is by definition artificial.

Adaptive (**choice A**) immunity describes all immune responses that have specificity and memory. These immune responses are produced by specific B and T lymphocytes. Because this patient is being given a product of the adaptive immune response (antibodies), there will be no elicitation of an adaptive immune response in this individual.

Artificial active (**choice B**) immunity is produced during the process of vaccination. The patient is exposed to a modified pathogen or product. As a result, an active immune response to that inoculation is made. This sort of immunization causes the development of memory in the patient.

Natural active (**choice D**) immunity would result after a recovery from an infection.

Natural passive (**choice E**) immunity is acquired across the placenta and in the colostrum and breast milk, from mother to child. The child receives preformed antibodies (IgG across the placenta and IgA in milk), which serve to protect the child until a natural active immune response can be mounted.

4. **The correct answer is A.** Allotypes are minor amino-acid sequence variations in the constant domains of heavy and light immunoglobulin chains. Their expression is genetically determined, and variations can be used as evidence in favor of paternity in some cases. Allotypic markers are most frequently used in studies of population genetics, as certain ethnic groups are likely to have similar allotypic markers on their immunoglobulins. Allotypic markers do not affect the biologic function of the immunoglobulin molecule.

The term "idiotype" (**choice B**) describes the three-dimensional shape of the antigen-combining site of an antibody or T-cell receptor molecule. Because each human is capable of producing many millions of different idiotypic sequences, these would not be useful in paternity cases.

IgA2 (**choice C**) is an isotype of immunoglobulin. Because all normal human beings produce some amount of this immunoglobulin, it would not be useful in paternity cases.

IgM (**choice D**) is an isotype of immunoglobulin. Because all normal human beings produce some amount of this immunoglobulin, it would not be useful in paternity cases.

An isotype (**choice E**) is the heavy- or light-chain constant domain of an immunoglobulin. Thus, there are five heavy-chain isotypes (A, E, G, M, and D) and two light-chain isotypes (κ and λ). Because all human beings produce heavy- and light-chain isotypes, this would not be useful in paternity testing.

5. **The correct answer is D.** Although this vaccine is no longer in use because of the possible side effects of *Bordetella pertussis* inoculation, in this case the whole, killed bacteria served as an adjuvant. They increased local inflammation, thus calling inflammatory cells to the site and prolonging exposure to the immunogen, the capsular polysaccharide of *Haemophilus*.

A carrier (**choice A**) is not correct because a carrier is a protein covalently coupled to a hapten to elicit a response. There is no mention in the question stem here that the polysaccharide is chemically coupled to the bacteria; it is stated that they are only mixed together.

A hapten (**choice B**) is not correct because a hapten is a single antigenic epitope, and a whole, killed bacterium such as *Bordetella* has many epitopes.

A mitogen (**choice C**) is not correct because mitogens are substances that cause the polyclonal activation of immune cells. The mitogens most commonly used in clinical laboratory medicine are lipopolysaccharide, concanavalin A, and pokeweed mitogen.

An immunogen (**choice E**) is not correct because the immunogen in a vaccine is the substance to which the immune response is being made. Because the object of the HIB vaccine is to immunize against *Haemophilus influenzae*, *Bordetella pertussis* bacteria cannot be the immunogen.

Immunodeficiency Diseases

11

What the USMLE Requires You to Know

The molecular defects, signs, and symptoms associated with defects of phagocytic cells, complement, and B and T cells.

If individuals experience defects in the functioning of any of the components of the immune system, clinical manifestations are common. These **immunodeficiency diseases** are favorite topics of USMLE vignettes and are reviewed here in their totality, although many have been discussed previously in the Clinical Correlates spread throughout the chapters.

Defects of Phagocytic Cells

Table II-11-1. Defects of Phagocytic Cells

Disease	Molecular Defect(s)	Symptoms
Chronic granulomatous disease (CGD)	Deficiency of NADPH oxidase (any one of four component proteins); failure to generate superoxide anion, other O ₂ radicals	Recurrent infections with catalase-positive bacteria and fungi
Chediak-Higashi syndrome	Granule structural defect	Recurrent infection with bacteria: chemotactic and degranulation defects; absent NK activity, partial albinism
Glucose-6-phosphate dehydrogenase (G6PD) deficiency	Deficiency of essential enzyme in hexose monophosphate shunt	Same as CGD, with associated anemia
Myeloperoxidase deficiency	Granule enzyme deficiency	Mild or none
Leukocyte adhesion deficiency	Absence of CD18—common β chain of the leukocyte integrins	Recurrent and chronic infections, fail to form pus, and do not reject umbilical cord stump

Defects of Humoral Immunity

Table II-11-2. Defects of Humoral Immunity

Disease	Molecular Defect	Symptoms/Signs	Treatment
Bruton X-linked hypogammaglobulinemia	Deficiency of a tyrosine kinase blocks B-cell maturation	Low immunoglobulin of all classes, no circulating B cells, pre-B cells in bone marrow in normal numbers, normal cell-mediated immunity	Monthly gamma-globulin replacement, antibiotics for infection
Transient hypogammaglobulinemia of infancy	Delayed onset of normal IgG synthesis	Detected in 5th to 6th month of life, resolves by 16–30 months; susceptibility to pyogenic bacteria	Antibiotics and in severe cases, gamma-globulin replacement
Common variable hypogammaglobulinemia	Unknown	Onsets in late teens, early twenties; B cells present in peripheral blood, immunoglobulin levels decrease with time; increased auto-immunity	Antibiotics
Selective IgA deficiency	Deficiency of IgA (most common)	Repeated sinopulmonary and gastrointestinal infections	Antibiotics, not immunoglobulins
X-linked hyper-IgM syndrome	Deficiency of CD40L on activated T cells	High serum titers of IgM without other isotypes, normal B- and T-cell numbers, susceptibility to extracellular bacteria and opportunists	Antibiotics and gammaglobulins

Deficiencies of Complement or Its Regulation

Table II-11-3. Deficiencies of Complement or Its Regulation

Deficiencies in Complement Components	Deficiency	Signs/Diagnosis
Classic pathway	C1q, C1r, C1s, C4, C2	Marked increase in immune complex diseases, increased infections with pyogenic bacteria
Alternative pathway	Factor B, properdin	Increased <i>Neisseria</i> infections
Both pathways	C3	Recurrent bacterial infections, immune complex disease
	C5, C6, C7, or C8	Recurrent meningococcal and gonococcal infections
Deficiencies in complement regulatory proteins	C1-INH (hereditary angioedema)	Overuse of C1, C4, or C2 Edema at mucosal surfaces
	Decay-activating factor or homologous restriction factor	Paroxysmal nocturnal hemoglobinuria

Defects of T Lymphocytes and Severe Combined Immunodeficiencies

Although patients with defects in B lymphocytes can deal with many pathogens adequately, defects in T lymphocytes are observed globally throughout the immune system. Because of the central role of T cells in activation, proliferation, differentiation, and modulation of virtually all naturally occurring immune responses, abnormalities in these cell lines send shock waves throughout the system. It is often a Herculean clinical effort to dissect the cause-and-effect relationships in such inherited diseases, and their diagnosis is often one of trial-and-error, which takes years to unravel. Although in some cases both B- and T-lymphocyte defects may occur, the initial manifestation of these diseases is almost always infection with agents such as **fungi and viruses** that are normally destroyed by T-cell-mediated immunity. The B-cell defect, if any, is usually not detected for the first few months of life because of the passive transfer of immunoglobulins from the mother through the placenta or colostrum. The immune system is so compromised that even attenuated vaccine preparations can cause infection and disease.

Table II-11-4. B- and T-Cell Deficiencies

Category	Disease	Defect	Clinical Manifestations
Selective T-cell deficiency	DiGeorge Syndrome	Failure of formation of 3rd and 4th pharyngeal pouches, thymic aplasia	Facial abnormalities, hypoparathyroidism, cardiac malformations, depression of T-cell numbers, and absence of T-cell responses
	MHC class I deficiency	Failure of TAP 1 molecules to transport peptides to endoplasmic reticulum	CD8+ T cells deficient, CD4+ T cells normal, recurring viral infections, normal DTH, normal Ab production
Combined partial B- and T-cell deficiency	Wiskott-Aldrich Syndrome	Defect in cytoskeletal glycoprotein, X-linked	Defective responses to bacterial polysaccharides and depressed IgM, gradual loss of humoral and cellular responses, thrombocytopenia, and eczema <pre> graph TD A[thrombocytopenia] --> C[eczema] B[immunodeficiency] --> C </pre>
	Ataxia telangiectasia	Defect in kinase involved in the cell cycle	Ataxia (gait abnormalities), telangiectasia (capillary distortions in the eye), deficiency of IgA and IgE production
Complete functional B- and T-cell deficiency	Severe combined immunodeficiency (SCID)	Defects in common γ chain of IL-2 receptor (present in receptors for IL-4, -7, -9, -15), X-linked	Chronic diarrhea; skin, mouth, and throat lesions; opportunistic (fungal) infections; low levels of circulating lymphocytes; cells unresponsive to mitogens
		Adenosine deaminase deficiency (results in toxic metabolic products in cells)	See above
		Defect in signal transduction from T-cell IL-2 receptors	See above
		Bare lymphocyte syndrome/ MHC class II deficiency	T cells present and responsive to nonspecific mitogens, no GVHD, deficient in CD4+ T cells, hypogammaglobulinemia

Chapter Summary

- B-cell, phagocyte, and complement defects predispose to infections with extracellular pathogens.
- T-cell defects predispose to infections with intracellular pathogens.
- Severe combined immunodeficiencies tend to manifest first as T-cell defects.

Review Questions

1. A newborn is evaluated for immunologic function. He has a distortion of the shape of his mouth, low-set and malformed ears, and widely spaced eyes. Radiographically, there is evidence of cardiac malformation and absence of a thymic shadow. Which of the following parameters would be normal in this child?
 - (A) Antibody-dependent cell-mediated cytotoxicity of parasite targets
 - (B) Cellularity of splenic periarteriolar lymphoid sheaths
 - (C) Cytotoxic killing of virus-infected targets
 - (D) Generation of oxygen metabolites in phagocytic cells
 - (E) Proliferative response to concanavalin A

2. A 14-month-old male infant is referred to a specialist for diagnosis of a potential immunologic deficiency. For the past 4 months, the child has suffered repeated episodes of bacterial infections and attempts to induce immunity using the pneumococcal vaccine have failed. Studies of peripheral blood indicate an absence of cells responsive to pokeweed mitogen. Bone marrow aspirates are remarkable for hypercellularity of pro-B cells. What is the most likely diagnosis?
 - (A) Bruton agammaglobulinemia
 - (B) Common variable hypogammaglobulinemia
 - (C) DiGeorge syndrome
 - (D) Selective immunoglobulin deficiency
 - (E) Wiskott-Aldrich syndrome

3. A 31-year-old man is treated for a fourth episode of disseminated *Neisseria gonorrhoeae* infection in the last 5 years. He had no previous history of unusual or recurrent infections. If he has an immunologic defect, which of the following is most likely?
 - (A) Common variable immunodeficiency
 - (B) C8 deficiency
 - (C) DiGeorge syndrome
 - (D) Selective IgA deficiency
 - (E) Severe combined immunodeficiency

4. A patient has been hospitalized three times for painful abdominal edema and is complaining now of swollen lips. What will laboratory findings in this patient most likely include?
- (A) Abnormal superoxide anion production by neutrophils
 - (B) Abnormal T-cell function
 - (C) Abnormal T-cell numbers
 - (D) Defective neutrophil chemotaxis
 - (E) Reduced C4 levels
5. A 4-year-old girl presents with a severe *Staphylococcus aureus* abscess. Her history is significant for a previous infection with *Serratia marcescens*. If she has an enzyme deficiency, which of the following is most likely?
- (A) Adenosine deaminase
 - (B) C1 inhibitor
 - (C) Myeloperoxidase
 - (D) NADPH oxidase
 - (E) Superoxide dismutase
6. A new pediatrician has just opened his office next to the hospital, and one of his first patients is a small, 4-year-old boy. His mother brings him into the office because he has several boil-like lesions on his arm. His mother told the physician that the boy has had these lesions on several different occasions. His other physician had prescribed antibiotics, and the lesions resolved. His records indicate he has had all of his immunizations. The pediatrician orders several different laboratory tests, and the following results are reported:
- | | |
|--|----------|
| Immunoglobulin levels | normal |
| B-cell and T-cell counts | normal |
| Complement levels | normal |
| Calcium and parathyroid hormone levels | normal |
| Nitroblue tetrazolium test | negative |

The mother told the physician that she was not aware of any eczema or bleeding problems. Which disease is indicated by these findings?

- (A) Bruton X-linked agammaglobulinemia
- (B) Chronic granulomatous disease
- (C) DiGeorge syndrome
- (D) Severe combined immunodeficiency disease
- (E) Wiskott-Aldrich syndrome

7. An acutely ill, 2-year-old boy is hospitalized with *Staphylococcus aureus* pneumonia, which is treated appropriately. The patient's history indicates similar bouts of bacterial infections in the past. He had recovered uneventfully from measles 6 months ago. Physical examination discloses scant tonsillar tissue and no palpable lymphadenopathy. Immunoelectrophoresis reveals subnormal levels of gammaglobulins. The nitroblue tetrazolium and chemiluminescence assays indicate normal phagocytic killing. Which of the following disorders is most likely responsible for this child's condition?
- (A) Adenosine deaminase deficiency
 - (B) Defect of the *Btk* gene
 - (C) Defect of the *SAP* gene
 - (D) Defect of the *WAS* gene
 - (E) ICAM-1 deficiency
8. A 2-year-old boy suffering from repeated painful bouts of inflammation of mucosal surfaces, especially affecting the lips, is brought to the pediatrician's office. The mother remembers similar symptoms in previous generations of her family and fears a heritable tendency toward food allergy. What laboratory finding would best support the physician's suspicion?
- (A) Depressed C3
 - (B) Depressed C4
 - (C) Depressed C5
 - (D) Elevated C1
 - (E) Elevated C1, C4, and C2

Answers and Explanations

- The correct answer is D.** This is a case of DiGeorge syndrome, which is a congenital failure in the formation of the third and fourth pharyngeal pouches. As a result, individuals with this defect have aplastic thymus and parathyroids and facial, esophageal, and cardiac malformations. Immunologically, the absence of the thymus will ultimately have global effects on the development of all T-cell-mediated immune responses. At birth, the child will have IgG antibodies that have been transplacentally transferred from the mother, but by 12 months or so after birth, these will be gone and IgM will be the only isotype of immunoglobulin present. Phagocytic killing will be normal until that point, although after all the maternal IgG is gone, opsonization of bacteria will no longer be possible.

Antibody-dependent cell-mediated cytotoxicity of parasite targets (**choice A**) will be depressed in this child because eosinophil-mediated ADCC requires IgE antibodies, and these cannot be produced without T-cell help.

Cellularity of splenic periarteriolar lymphoid sheaths (**choice B**) will be decreased in this child because these are T-cell-dependent areas of the spleen.

Cytotoxic killing of virus-infected targets (**choice C**) will be depressed in this child because cytotoxic T cells will be absent, and only NK cells will be available for antiviral protection.

The proliferative response to concanavalin A (**choice E**) will be depressed in this child because concanavalin A is a T-cell mitogen. If there are no T cells, there will be no proliferation in response to this mitogen.
- The correct answer is A.** This is a case of X-linked agammaglobulinemia, or Bruton agammaglobulinemia. It is caused by a mutation in a thymidine kinase gene, which is important in B-cell maturation. The bone marrow becomes hypercellular with cells that cannot progress from the pro-B to the pre-B stage, while the peripheral blood lacks mature B lymphocytes. There will be no proliferative response to B-cell mitogens (pokeweed mitogen), and CD19+ cells will be absent from the blood. Persons with this condition are unable to mount a normal antibody response; therefore, symptoms appear after the disappearance of maternal antibodies. Susceptibility to extracellular, encapsulated pathogens is profound.

Common variable hypogammaglobulinemia (**choice B**) is a condition that usually appears in the late teens or early twenties. It is believed to be an autoimmune disease and is associated with the disappearance of immunoglobulin isotypes over time.

DiGeorge syndrome (**choice C**), or congenital thymic aplasia, is a condition in which there is failure of formation of the third and fourth pharyngeal pouches. These infants have facial abnormalities, failure of formation of the parathyroids, and cardiac defects, as well as absence of T-lymphocyte development.

Selective immunoglobulin deficiency (**choice D**) would not be manifested by a failure of B-cell development in the bone marrow. Selective IgA deficiency is most common of these and would manifest as increased susceptibility to mucosal-surface pathogens.

Wiskott-Aldrich syndrome (**choice E**) is a complex immune deficiency with a triad of symptoms: eczema, thrombocytopenia, and immunodeficiency. It is inherited in an X-linked recessive fashion. These patients are prone to development of malignant lymphomas and have inability to respond to polysaccharide antigens.
- The correct answer is B.** Unusual frequency or severity of *Neisseria* infections should always lead to a suspicion of a terminal complement component deficiency (C5, C6, C7, or C8). *Neisseria* seem to be highly susceptible to complement-mediated lysis, so any failure of production of the membrane attack complex predisposes the patient to recurrent bacteremias with these organisms.

Common variable immunodeficiency (**choice A**) is a condition that usually appears in the late teens or early twenties. It is believed to be an autoimmune disease and is associated with the disappearance of immunoglobulin isotypes over time.

DiGeorge syndrome (**choice C**) is a condition in which there is failure of formation of the third and fourth pharyngeal pouches. Diagnosed in infancy, these individuals have facial abnormalities, failure of formation of the parathyroids, and cardiac defects, as well as an absence of T-lymphocyte development. This condition predisposes to early viral and fungal infections.

Selective IgA deficiency (**choice D**) would be expected to result in respiratory and gastrointestinal tract infections, autoimmune disease, and allergies.

Severe combined immunodeficiency (**choice E**) typically presents with early susceptibility to viral and fungal agents. It is most frequently diagnosed in infancy, after the disappearance of maternally derived IgG antibodies.

4. **The correct answer is E.** The description of painful abdominal edema and edema in the oral mucosa are typical of hereditary angioedema. This is a genetic deficiency of complement C1 inhibitor. When this important control protein is missing, there is excessive use of the classic complement pathway components, especially C4. This causes abnormal inflammation along the mucosal surfaces.

Abnormal superoxide anion production by neutrophils (**choice A**) would result in predisposition to infections with extracellular pathogens.

Abnormal T-cell function (**choice B**) would result in predisposition to infections with viral and fungal pathogens, not edema of the mucosal surfaces.

Abnormal T-cell numbers (**choice C**) would result in predisposition to infections with viral and fungal pathogens, not edema of the mucosal surfaces.

Defective neutrophil chemotaxis (**choice D**) would result in neutrophilia and failure to produce pus and abscesses in response to extracellular bacterial invasion.

5. **The correct answer is D.** The infections of this child with catalase-positive bacteria are characteristic of chronic granulomatous disease (CGD). While two thirds of CGD patients are male, one third has the autosomal recessive form of NADPH oxidase deficiency and can be female.

Adenosine deaminase deficiency (**choice A**) produces a severe combined immunodeficiency. The infections seen are likely to be the result of T-cell deficiency (viral and fungal agents). In the absence of adenosine deaminase, deoxyadenosine phosphate builds up in T cells and is toxic to them.

C1 inhibitor (**choice B**) is not an enzyme, and its absence does not predispose to infections. It is absent in the condition known as hereditary angioedema, represented by recurrent, painful bouts of mucosal edema.

Myeloperoxidase (**choice C**) deficiency is normally without clinical symptoms. This is an enzyme that is important in intracellular killing in phagocytes because it causes formation of toxic halide radicals. However, because oxygen radicals are more important in intracellular killing, no symptoms of MPO deficiency will be observed.

Superoxide dismutase (**choice E**) deficiency has not been described in leukocytes, and its absence would not be likely to predispose to infection.

6. **The correct answer is B.** The negative result on the nitroblue tetrazolium dye reduction test indicates a failure of oxygen radical generation inside phagocytic cells. It is a common diagnostic, along with the neutrophil oxidative index, to diagnose chronic granulomatous disease. This is a genetic defect in the production of a subunit of NADPH oxidase and is usually diagnosed when children develop recurrent infections with catalase-positive organisms.

Bruton X-linked agammaglobulinemia (**choice A**) is caused by a mutation in a thymidine kinase gene, which is important in B-cell maturation. The bone marrow becomes hypercellular with cells that cannot progress from the pro-B to the pre-B stage, while the peripheral blood lacks mature B lymphocytes. Because the child in this case has normal immunoglobulin levels, this diagnosis is not possible.

DiGeorge syndrome (**choice C**) is a condition in which there is failure of formation of the third and fourth pharyngeal pouches. Children with this defect are diagnosed in infancy and would have an absence of T lymphocytes and deficiencies of calcium and parathyroid hormone.

Severe combined immunodeficiency disease (**choice D**) usually manifests first as a T-lymphocyte defect. The child would be susceptible to viral and fungal pathogens and have depressed levels of immunoglobulins, decreased counts of both B and T lymphocytes, and normal nitroblue tetrazolium dye reduction.

Wiskott-Aldrich syndrome (**choice E**) is a complex immune deficiency with a triad of symptoms: eczema, thrombocytopenia, and immunodeficiency. Because the mother in this case is not aware of any bleeding dyscrasias or eczema, this would be an unlikely diagnosis.

7. **The correct answer is B.** This is a case of X-linked agammaglobulinemia, or Bruton agammaglobulinemia. During the early 1990s, the gene responsible for this condition was cloned. The normal counterpart of the mutant gene encodes a protein tyrosine kinase (Bruton tyrosine kinase, *Btk*), which is important in B-cell signaling. When it is absent or altered, B lymphocytes are unable to progress from the pro-B to the pre-B cell stage in the bone marrow. Thus, the bone marrow becomes hypercellular, while the peripheral blood is lacking mature B lymphocytes. Persons with this condition are unable to mount a normal antibody response; therefore, symptoms appear after the disappearance of maternal antibodies, and susceptibility to extracellular, encapsulated pathogens such as *Streptococcus pneumoniae* and *Haemophilus influenzae* is profound.

Adenosine deaminase deficiency (**choice A**) is an example of a severe combined immunodeficiency disease (SCID). When this enzyme is absent, toxic metabolites build up in B and T lymphocytes and cause a general failure of the immune response. It would have clinical manifestations of both B- and T-lymphocyte defects, and not exclusively B lymphocytes, as described in this case history.

A defect of the *SAP* gene (**choice C**) is believed to cause X-linked proliferative disease, in which uncontrolled T-cell proliferation follows infection with Epstein-Barr virus. *SAP* stands for SLAM-associated protein, and SLAM (signaling lymphocytic activation molecule) is a potent T-cell coactivator.

Defect of the *WAS* gene (**choice D**) causes Wiskott-Aldrich syndrome, in which a defect in CD43 (a cytoskeletal protein) causes defects in T cells and platelets. Patients with Wiskott-Aldrich syndrome display a triad of signs: thrombocytopenia, eczema, and immunodeficiency.

ICAM-1 deficiency (**choice E**) would cause defects of antigen recognition and activation of lymphocytes. ICAM-1 is an adhesion molecule in the immunoglobulin superfamily of genes and is bound by LFA-1 integrin.

8. **The correct answer is B.** This is a case of hereditary angioedema, caused by a deficiency in an important complement regulatory protein, C1-INH. When it is absent, the early components of the classical complement cascade are overused. It is normally diagnosed by the finding of depressed levels of complement component C4 in the blood.

Depressed C3 (**choice A**) would not be a correlate of C1-INH deficiency. There are separate regulatory controls on abnormal complement activation that operate at the C3 level, so this condition is rarely found.

Depressed C5 (**choice C**) would not be a correlate of C1-INH deficiency. There are separate regulatory controls on abnormal complement activation that operate at the C5 level, so this condition is rarely found.

Elevated C1 (**choice D**) would not be found in this case because the condition results in the overuse of early components of the classical complement cascade. Therefore, serum levels of C1, C4, and C2 would be decreased from normal values.

Elevated C1, C4, and C2 (**choice E**) would not be found in this case because the condition results in the overuse of early components of the classical complement cascade. Therefore, serum levels of C1, C4, and C2 would be decreased from normal values.

Acquired Immunodeficiency Syndrome

12

What the USMLE Requires You to Know

- The receptors and coreceptors of HIV
- The immunologic results of systematic TH-cell eradication
- The mechanisms through which HIV evades the immune response

Of all the newly emerging pathogens of the past century, none so efficiently and inexorably eradicates the functioning of the immune system as the **human immunodeficiency virus (HIV)**. This D-type retrovirus attaches to CD4 receptors on host cells (TH cells, macrophages, and microglia) and utilizes several chemokine receptors on these cells as **coreceptors**. Early in the infection, the virus uses the **CCR5 chemokine receptor** and is thus predominantly **macrophage-tropic**, whereas late in the infection, the virus uses the **CXCR4 chemokine receptor** and becomes **T cell-tropic**.

HIV infection ultimately results in impaired function of both adaptive and innate immune systems. The most prominent defects arise in cell-mediated immunity, but because TH cells and macrophages are infected and destroyed, all aspects of immunity are eventually affected.

In A Nutshell

- HIV infects CD4+ cells.
- CXCR4 and CCR5 chemokine receptors are coreceptors.

Table II-12-1. Mechanisms of Immune System Destruction by the Human Immunodeficiency Virus

Virus Characteristic	Immunologic Result
Multiplication in activated lymphocytes and macrophages	Reproduces virus Increases viral load
Direct cytopathic effect on lymphocytes and macrophages	Eliminates cell- and antibody-mediated immunity
<i>Nef</i> gene product downregulates class I MHC expression	Makes infected cells less susceptible to CTL killing
<i>Tat</i> gene product	Inhibits cytokine synthesis in both infected and uninfected cells
Destruction of TH cells	Eliminates immune enhancement
Immune deviation toward TH2 response	Inhibits potentially protective CMI responses and produces antibodies that can mediate ADCC, resulting in further elimination of TH cells
Antigenic drift of the gp120	Evades antibody-mediated effector mechanisms and exhausts individual's immune capacity
Heavy glycosylation of gp120	Hides potentially protective epitopes from immune recognition

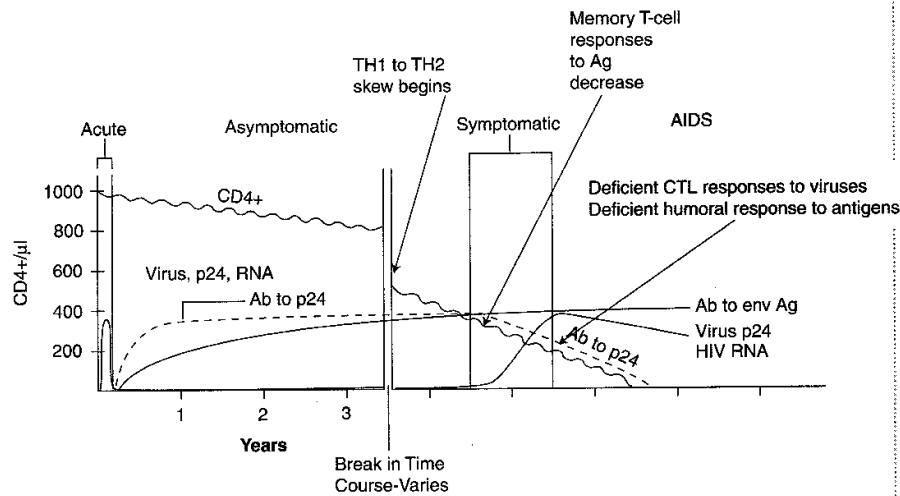


Figure II-12-1. The Course of Untreated HIV Infection and the Sequence of Immune Response Destruction

Macrophages and dendritic cells are infected with HIV, but they are resistant to its cytopathic effects. Nevertheless, they serve as reservoirs of infection, delivering virus particles into the secondary lymphoid organs and throughout the body, and their antigen-presenting function may be impaired by the infection.

HIV-infected patients exhibit both cell-mediated and humoral responses to the pathogen, but these responses are unable to eradicate the entire virus, and the infection eventually overwhelms the immune response. CD8+ cytotoxic cells are generated to kill infected CD4+ cells, and antibodies are generated to many viral antigens within 6 to 9 weeks after infection. These antibodies are used as the basis of the diagnostic tests for the infection (ELISA and Western blot, *see* Chapter 16), but there is no evidence that they play a protective role in the infection. The initial, massive CD8+ response may be responsible for the control of viremia and transition into clinical latency, but ultimately, because CD4+ cells are important in amplifying the cytotoxic response as helper numbers diminish, the amplification loop for stimulation of CD8+ cytotoxicity will be eliminated.

In A Nutshell

- Macrophages serve as reservoirs of infection.
- The immune response to the virus kills infected cells, but thereby eradicates all help.

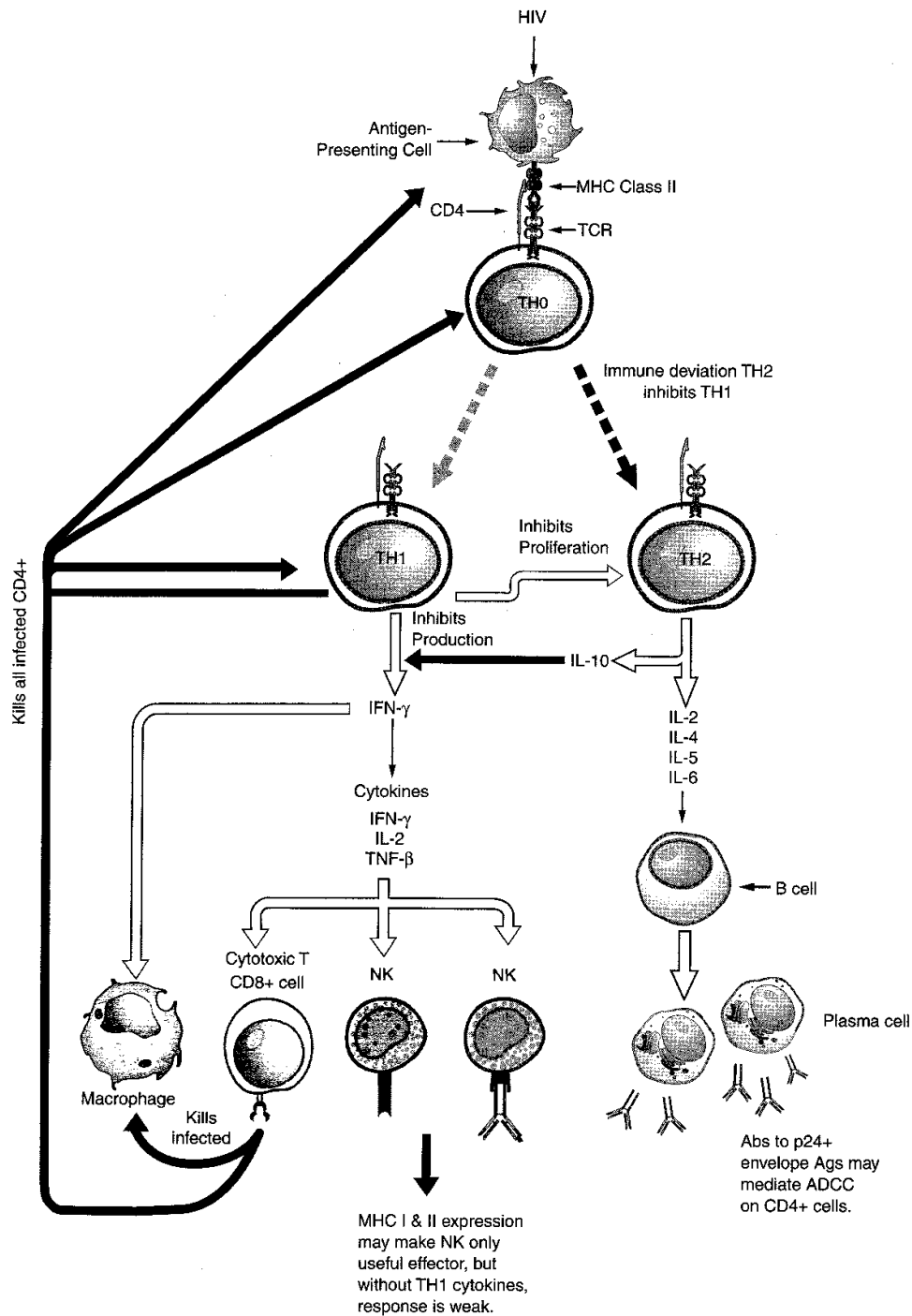


Figure II-12-2. The Global Effects of HIV on the Immune Response

Chapter Summary

- HIV infects CD4+ cells (TH cells and macrophage lineages) using CCR5 and CXCR4 chemokine coreceptors.
- HIV destroys cell-mediated and humoral immune responses by eradicating TH cells.
- HIV evades the immune response through glycosylation of surface antigens and antigenic drift.
- HIV is diagnosed using ELISA and Western blot for anti-HIV antibodies.

Review Questions

1. A 27-year-old woman who was 5 months pregnant presented for prenatal care. A routine evaluation was performed, which included testing for HIV antibody. The patient was reported to be negative for RPR but positive for HIV antibody by the EIA assay. The HIV Western blot only demonstrated the presence of an antibody to the HIV p24 antigen. How should this patient be counseled?
 - (A) That she and her baby are both infected with HIV
 - (B) That she is negative for HIV because the RPR is more specific
 - (C) That she is positive for the HIV virus
 - (D) That she should have an HIV polymerase chain reaction (PCR) test performed
 - (E) That this is a confirmed false-positive HIV result

2. A neonate born to a known HIV-positive mother is evaluated for HIV status. What would be the assay of choice?
 - (A) CD4 cell count
 - (B) EIA for the detection of HIV antibody
 - (C) Polymerase chain reaction (PCR) for the detection of HIV genome
 - (D) p120 antigen assay
 - (E) Western blot

3. The attachment of HIV to a TH cell is initiated by the binding of the HIV gp120 molecule to the CD4 receptor. Then gp120 undergoes a conformational change and binds to a second molecule. What is the second molecule on the surface of the CD4 lymphocyte that acts as a coreceptor and binds with the lymphotropic HIV?
 - (A) The chemokine receptor CCR5 expressed on dendritic cells
 - (B) The chemokine receptor CXCR4 expressed on T cells
 - (C) The chemokine receptor CXCR5 expressed on T cells
 - (D) The complement receptor CR2 (CD21) on B cells
 - (E) The cytokine receptor CCR5 expressed on T cells

4. A male exotic dancer in San Francisco has been HIV-positive for 10 years. Because of his low income, he can only irregularly afford the antiretroviral drugs prescribed to him at the AIDS clinic. When he presents to the clinic for a checkup, his CD4 cell count is found to be $390/\text{mm}^3$. Which of the following immunologic parameters is likely to be most profoundly depressed at this stage of his infection?
- (A) Cell proliferation in response to pokeweed mitogen
 - (B) Delayed-type hypersensitivity to *Candida* antigens
 - (C) IgA production
 - (D) Intracellular killing of bacteria
 - (E) Rejection of allogeneic skin grafts
5. Patients with HIV infection begin to display an immunologic deviation toward TH2 responses relatively early in the course of their disease. Which of the following cytokines would be increased in a patient with a CD4 count of $500/\text{mm}^3$?
- (A) IFN- γ
 - (B) IL-1
 - (C) IL-2
 - (D) IL-10
 - (E) TNF- β

Answers and Explanations

1. **The correct answer is D.** A patient who is HIV EIA-positive must always have the result confirmed by a confirmatory assay, e.g., HIV Western blot. The HIV Western blot is considered positive when the patient demonstrates the presence of antibody to at least two of three important HIV antigens, which are gp120, gp41, and p24. If no reaction is observed, then the patient is considered negative, but any reaction that is not consistent with a positive is reported as indeterminate. This patient is considered indeterminate. The physician can wait 6 months and retest by the EIA assay, with a reflex to the Western blot assay if the EIA result is positive. If the Western blot results are identical to the previous, then the patient is reported as negative or the patient can be tested by another confirmatory test such as the PCR assay. A negative PCR in this situation would classify this patient as negative; however, it would be wise to retest the patient in 3 to 6 months if she has risk factors. That she and her baby are both infected with HIV (**choice A**) is not correct. Approximately 30% of the babies from untreated and <8% from treated HIV-positive mothers will be infected.

That she is negative for HIV because the RPR is more specific (**choice B**) is not correct because RPR is not a test for HIV but a test for syphilis.

That she is positive for the HIV virus (**choice C**) is not correct because her Western blot was indeterminate, and a confirmatory test (e.g., Western blot or PCR) must confirm the EIA assay before the patient can be reported positive.

That this is a confirmed false-positive HIV result (**choice E**) may eventually be proven to be correct, but with the present data one does not know if the patient has been detected in the early stages of seroconversion or simply a patient who is false positive.

2. **The correct answer is C.** The neonate could not be tested for anti-HIV antibodies because the known HIV-positive mother passively transmits anti-HIV IgG antibodies to the baby through the placenta. All babies born to HIV-positive mothers will be anti-HIV antibody-positive. Because of the sensitivity of HIV antibody testing, an HIV antibody test on a specimen from a baby born to an HIV-positive mother should not be believed until 15 months of age. The test of choice is to detect the genome of the virus by PCR.

CD4 cell count (**choice A**) is not correct because this assay is used to measure the progression of the disease and the destruction of the target CD4+ cells. It is not a diagnostic test for the presence of the virus or antiviral antibodies.

EIA for the detection of HIV antibody (**choice B**) is incorrect because maternal antibodies can be detected in babies who are from known HIV-positive mothers for up to 12 months after birth. The HIV EIA assay is an antibody-screening test, whereas the HIV Western blot is a confirmatory test for the presence of anti-HIV antibody.

p120 antigen assay (**choice D**) is incorrect because there is no such test as an HIV p120 antigen assay; however, there is an HIV p24 antigen assay.

Western blot (**choice E**) is incorrect because maternal antibodies can be detected in babies who are from known HIV-positive mothers for up to 12 months after birth. The HIV EIA assay is an antibody-screening test, whereas the HIV Western blot is a confirmatory test for the presence of anti-HIV antibody.

3. **The correct answer is B.** The HIV envelope, derived from the host-cell membrane, displays viral glycoproteins gp120 and gp41. The gp120 glycoprotein has a high affinity for CD4 and is thus important in the attachment of the virus to the cell. All cells expressing CD4 are potential targets for infection with HIV. After the gp120 of the HIV binds to the CD4, it undergoes a conformational change and must then bind to a second molecule, a coreceptor, on the surface of the target cell. The variation of the gp120 molecule determines the tropism of the HIV and thus dictates which CD4+ cell can be infected.

Lymphotropic HIV virions use the CXCR4 chemokine receptor (coreceptor) expressed on T cells and require a high density of CD4 on the surface of the cell. Macrophage-tropic HIV uses the CCR5 chemokine receptor and requires only a low level of CD4 expression on the target cell.

The CCR5 chemokine receptor expressed on dendritic cells (**choice A**) is not correct. The question is addressing the infection of TH, not dendritic, cells.

The CXCR5 chemokine receptor expressed on T cells (**choice C**) is not correct. The chemokine receptor on T cells is CXCR4, not CXCR5.

The complement receptor CR2 (CD21) on B cells (**choice D**) is not correct. CR2 (CD21) is found on B cells and is not the coreceptor on TH cells for the attachment of HIV. CR2 is the receptor for EBV.

The CCR5 cytokine receptor expressed on T cells (**choice E**) is not correct. Chemokines are small cytokines of relatively low molecular weight released from a variety of cells involved in inflammatory responses; however, the receptor on the T cells is CXCR4. The CCR5 is found on macrophages and dendritic cells.

4. **The correct answer is B.** At this stage of this patient's infection, TH cell numbers are depressed to the point that TH-dependent immune responses will be depressed. Because delayed-type hypersensitivity is a purely TH1-mediated response, it will be impaired at this stage of infection.

Cell proliferation in response to pokeweed mitogen (**choice A**) is incorrect because pokeweed mitogen is a B-cell mitogen. These responses will remain within normal limits until the end of life in an HIV-positive patient.

IgA production (**choice C**) is not correct because these antibodies will not begin to fall until later in the infection. IgA is a TH2-dependent antibody, but at this stage of infection, there should still be B memory cells capable of mounting this response.

Intracellular killing of bacteria (**choice D**) is mediated by phagocytic cells. Although TH1 cytokines contribute to phagocyte activation, the intracellular killing mechanisms themselves should be unaffected in this patient.

Rejection of allogeneic skin grafts (**choice E**) is a CD8+ cell-mediated response and should not begin to be depressed until later in the course of HIV infection. Although cytotoxic T lymphocytes are stimulated by the cytokines of TH1 cells, this response should remain relatively normal at this stage of the disease.

5. **The correct answer is D.** Most patients will begin to show immune deviation of TH responses toward the TH2 arm around 600 CD4+ cells/mm³. As the TH2 response is augmented, IL-4 and IL-10 will be produced, which reciprocally further inhibit the TH1 response. Over time this leads to more and more profound skewing of the immune response toward a humoral response and more and more production of TH2 cytokines, such as IL-10 and IL-4.

Interferon- γ (**choice A**) is a product of TH1 cells and stimulates the effector cells of cell-mediated immunity and reciprocally inhibits the TH2 cell. This cytokine would be decreasing in the patient at this point.

IL-1 (**choice B**) is a product of activated macrophages. Macrophages are activated by TH1 cytokines, so interferon- γ production should begin to decrease at this time.

IL-2 (**choice C**) is a product of many TH cells. Because it can be produced in both humoral and cell-mediated arms, its levels would probably be unaffected at this point in an infection.

Tumor necrosis factor- β (**choice E**) is a product of TH1 cells and stimulates the effector cells of cell-mediated immunity and reciprocally inhibits the TH2 cell. This cytokine would be decreasing in the patient at this point.

Diseases Caused by Immune Responses: Hypersensitivity and Autoimmunity

13

What the USMLE Requires You to Know

- The immunologic mechanisms involved in the four types of hypersensitivity reactions
- The molecule, cell, or tissue targeted in the major autoimmune diseases

Hypersensitivity diseases are conditions in which tissue damage is caused by immune responses. They may result from uncontrolled or excessive responses against **foreign** antigens or from a **failure of self-tolerance**, in which case they are called **autoimmune diseases**. The two principal factors that determine the clinical and pathologic consequences of such conditions are the type of immune response elicited and the nature and location of the inciting antigen.

Hypersensitivity diseases are classified on the basis of the effector mechanism responsible for tissue injury, and four types are commonly recognized.

In A Nutshell

Hypersensitivity Diseases

- Excessive responses to foreign antigens
- Failure of self-tolerance (autoimmunity)

Table II-13-1. Classification of Immunologic Diseases

Type of Hypersensitivity	Immune Mechanisms	Mechanisms of Tissue Injury
Immediate (type I)	IgE	Mast cells and their mediators
Antibody-mediated (type II)	IgM and IgG Abs against cell or tissue Ags	Opsonization and phagocytosis of cells; complement- and Fc receptor-mediated recruitment and activation of neutrophils and macrophages; abnormalities in cellular functions (hormone/receptor signaling)
Immune complex-mediated (type III)	Immune complexes of circulating Ags and IgM or IgG Abs	Complement- and Fc-receptor-mediated recruitment and activation of leukocytes
T cell-mediated (type IV)	CD4+ T cells (delayed-type hypersensitivity)	Macrophage activation, cytokine-mediated inflammation
	CD8+ CTLs (T-cell-mediated cytotoxicity)	Direct target cell killing, cytokine-mediated inflammation

In A Nutshell

In all hypersensitivity reactions:

- First exposure sensitizes.
- Subsequent exposures damage.
- Reaction is Ag-specific

In A Nutshell

Type I Hypersensitivity

- IgE-mediated
- Protective response to helminths
- Atopic/allergic individuals develop this response to inappropriate stimuli

What the hypersensitivity reactions have in common:

- The first exposure to the antigen "sensitizes" lymphocytes.
- Subsequent exposures elicit a damaging reaction.
- The response is specific to a particular antigen or a cross-reacting substance.

Type I (Immediate) Hypersensitivities

This is the only type of hypersensitivity mediated by IgE antibodies and mast cells. It is manifested within minutes of the reexposure to an antigen. The IgE response is the normal protective response against many metazoan parasites, which are too large to be phagocytized or killed by other cytopathic mechanisms. Twenty percent of all individuals in the United States, however, display this immune response against harmless environmental antigens, such as pet dander or pollen: We call these responses **atopic** or **allergic** responses.

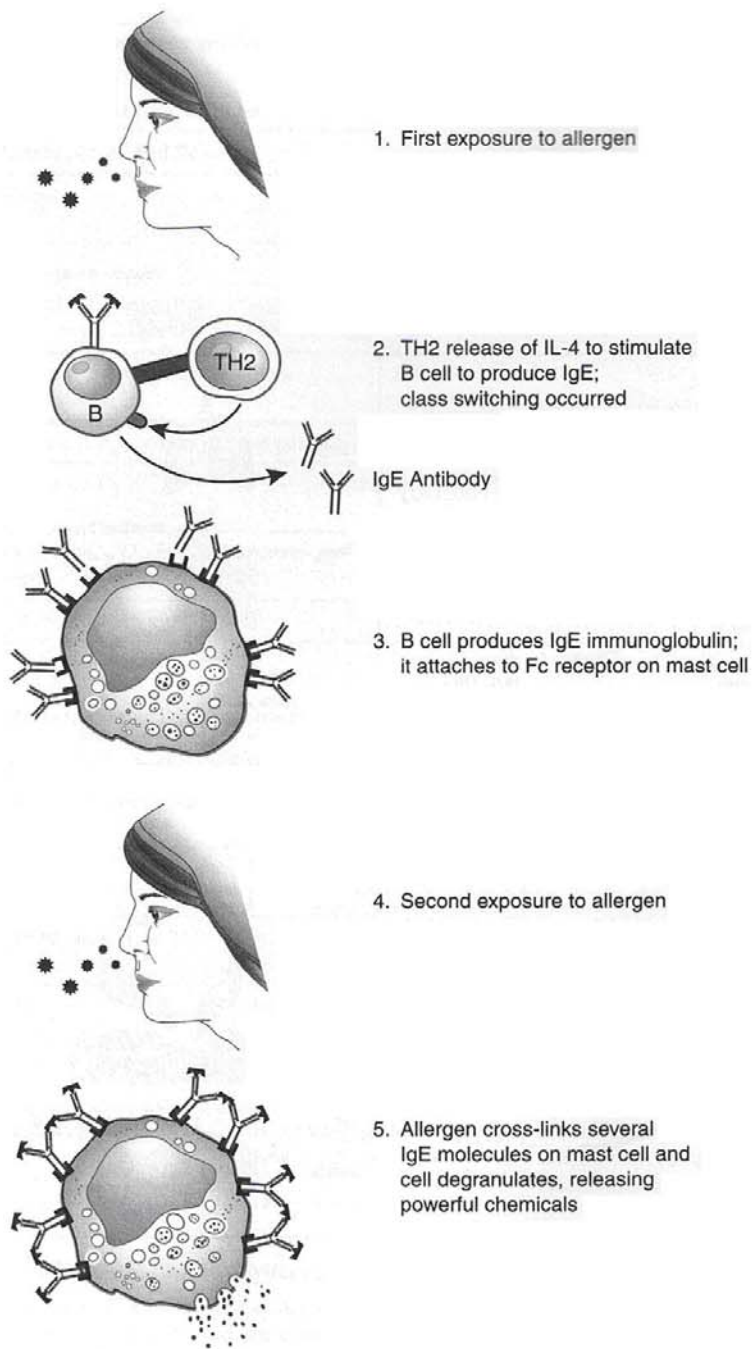


Figure II-13-1. Development of the Immediate-Hypersensitivity Reaction

In A Nutshell

Type I Effector Cells

- Mast cells
- Basophils
- Eosinophils

The effector cells of the immediate hypersensitivity reaction are mast cells, basophils, and eosinophils. The soluble substances they release into the site cause the symptoms of the reaction.

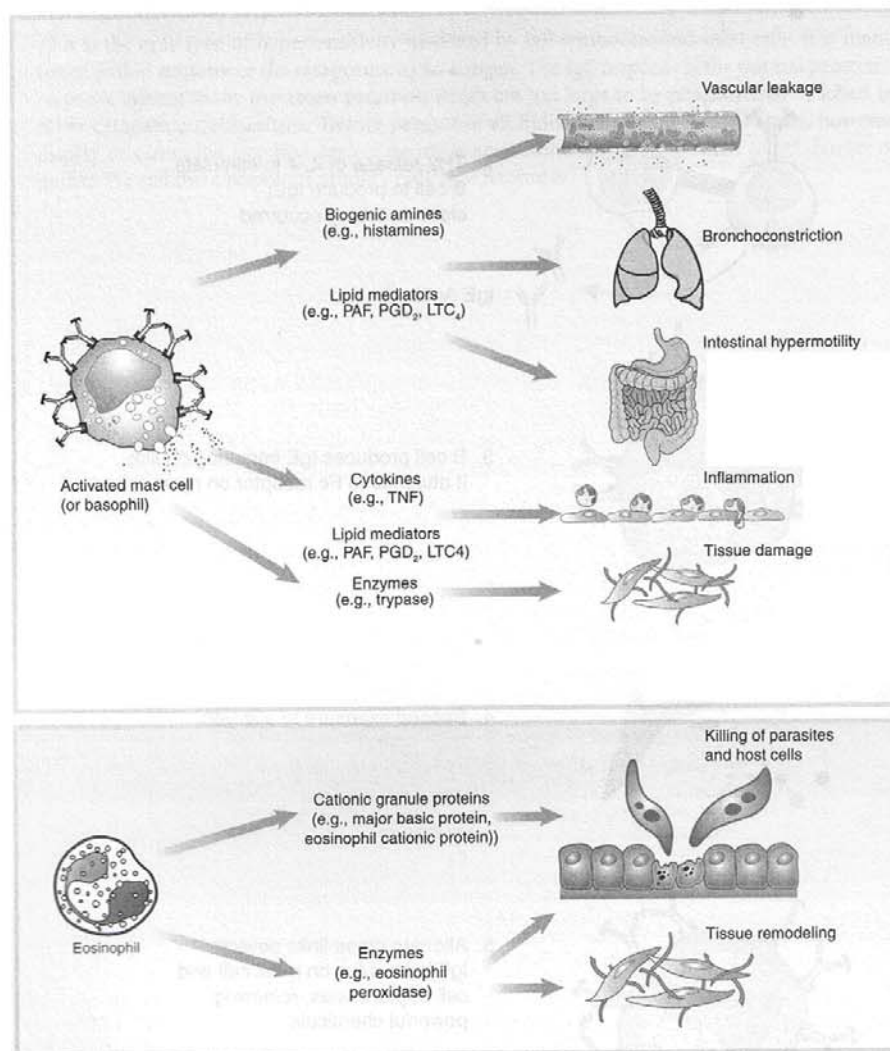


Figure II-13-2.

Two to four hours after the immediate response to release of these mediators, a **late-phase reaction** is mediated by products of the arachidonic acid cascade.

In A Nutshell

Late-phase reaction—
arachidonic acid cascade

Table II-13-2. Mast Cell Mediators

Mediators Stored and Released	Effect
Histamine	Smooth muscle contraction; increased vascular permeability
Heparin	Anticoagulant
Eosinophil chemotactic factor A (multiple chemokines)	Chemotactic
Mediators Newly Synthesized from Arachidonic Acid	Effect
Prostaglandin E ₂ (cyclooxygenase pathway)	Increased pain and vascular permeability
Prostaglandin D ₂	Increased smooth muscle contraction and vascular permeability
Leukotrienes C ₄ , D ₄ , E ₄ (lipoxygenase pathway)	Increased smooth muscle contraction and vascular permeability
Leukotriene B ₄	Chemotactic for neutrophils

Table II-13-3. Allergic Diseases Due to Specific Allergens and Their Clinical Manifestations

Allergic Disease	Allergens	Clinical Findings
Allergic rhinitis (hay fever)	Trees, grasses, dust, cats, dogs, mites	Edema, irritation, mucus in nasal mucosa
Food allergies	Milk, eggs, fish, cereals, grains	Hives and gastrointestinal problems
Wheal and flare	Insect stings, in vivo skin testing for allergies	Local skin edema, reddening, vasodilation of vessels
Asthma	Inhaled materials	Bronchial and tracheal constriction, edema, mucus production, massive inflammation
Systemic anaphylaxis	Insect stings, snake venoms, drug reactions	Bronchial and tracheal constriction, complete vasodilation and death

Type II (Antibody-Mediated) Hypersensitivities

Antibodies against cell or matrix antigens cause diseases that are **specific to the tissues** where those antigens are present and are usually not systemic. In most cases, these antibodies are **autoantibodies**, but they may be produced against a foreign antigen that is cross-reactive with self components of tissues. These antibodies can cause tissue damage by three main mechanisms:

- The antibodies may opsonize cells or activate the complement system.
- The antibodies may recruit neutrophils and macrophages that cause tissue damage.
- The antibodies may bind to normal cellular receptors and interfere with their function.

In A Nutshell

Type II Hypersensitivities

- Tissue-specific autoantibodies
- Opsonize or activate complement
- Recruit inflammatory cells
- Interfere with cellular function

Table II-13-4. Type II Hypersensitivities

Disease	Target Antigen	Mechanism of Pathogenesis	Clinical Manifestations
Autoimmune hemolytic anemia	RBC membrane proteins (Rh, I Ags)	Opsonization, phagocytosis, and complement-mediated destruction of RBCs	Hemolysis, anemia
Autoimmune thrombocytopenic purpura	Platelet membrane proteins	Ab-mediated platelet destruction through opsonization and complement activation	Bleeding
Goodpasture syndrome	Noncollagenous protein in basement membranes of kidney glomeruli and lung alveoli	Complement- and Fc-receptor-mediated inflammation	Nephritis, lung hemorrhage
Acute rheumatic fever	Streptococcal cell-wall Ag; Ab cross-reacts with myocardial Ag	Inflammation, macrophage activation	Myocarditis, arthritis
Myasthenia gravis	Acetylcholine receptor	Ab inhibits acetylcholine binding, downmodulates receptors	Muscle weakness, paralysis
Graves disease	TSH receptor	Ab-mediated stimulation of TSH receptors	Hyperthyroidism followed by hypothyroidism
Type II (non-insulin-dependent) diabetes	Insulin receptor	Ab inhibits binding of insulin	Hyperglycemia, ketoacidosis
Pernicious anemia	Intrinsic factor of gastric parietal cells	Neutralization of intrinsic factor, decreased absorption of vitamin B ₁₂	Abnormal erythropoiesis, anemia

In some types of type II hypersensitivities, complement is activated and/or ADCC is active (e.g., hemolytic disease of the newborn). In other types of type II hypersensitivities, cell function is altered in the absence of complement activation and ADCC (e.g., myasthenia gravis and Graves disease).

An important example of type II hypersensitivity is hemolytic disease of the newborn (HDNB), also known as erythroblastosis fetalis. In the fetus, this disease is due to transport of IgG specific for one of the Rhesus (Rh) protein antigens (RhD) across the placenta. About 85% of people are Rh+. If a pregnant woman is Rh- and the father is Rh+, there is a chance that the fetus will also be Rh+. This situation will pose no problem in the first pregnancy, as the mother's immune system will not usually encounter fetal blood cell antigens until placental separation at

the time of birth. At that time, however, Rh+ fetal red blood cells will enter the maternal circulation and stimulate a T-dependent immune response, eventually resulting in the generation of memory B cells capable of producing IgG antibody against RhD. In a subsequent pregnancy with another Rh+ fetus, this maternal IgG can be transported across the placenta, react with fetal Rh+ red cells, and activate complement, producing hemolytic disease. Hemolytic disease of the newborn can be prevented by treating the Rh- mother with RhoGAM™, a preparation of human anti-RhD IgD antibody, at 28 weeks of gestation and again with a human anti-RhD IgG antibody within 72 hours after birth. This antibody effectively eliminates the fetal Rh+ cells before they can generate RhD-specific memory B cells in the mother. Anti-RhD antibody should be given to any Rh- individual following any termination of pregnancy.

In A Nutshell

Hemolytic Disease of the Newborn (HDNB)

- Erythroblastosis fetalis
- Anti-RhD+ IgG
- Formed in Rh- mother carrying Rh+ child
- First pregnancy sensitizes
- Ab crosses placenta and injures subsequent fetuses
- Prevent with RhoGAM™

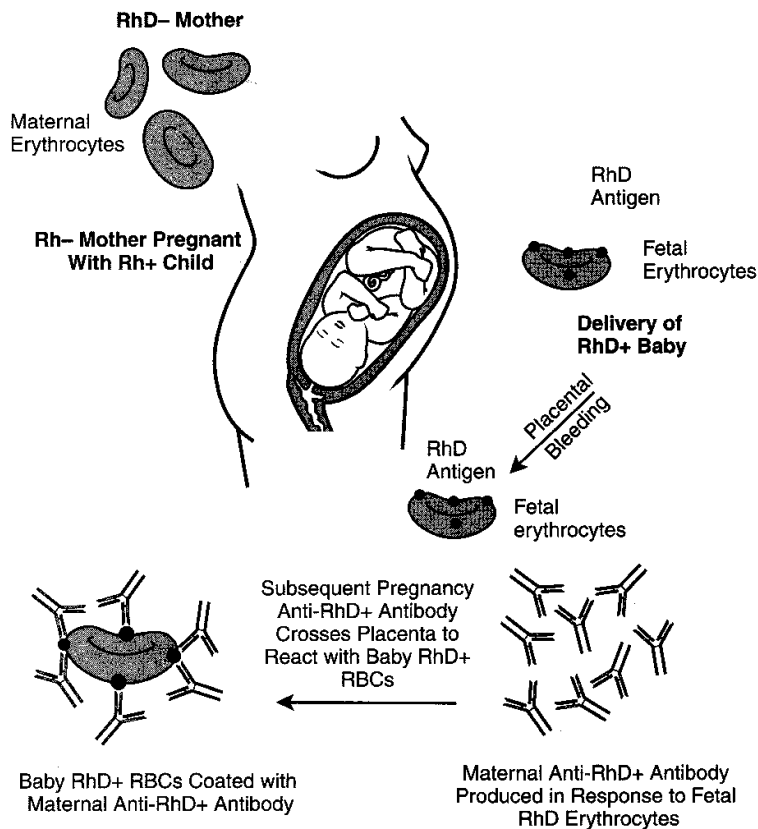


Figure II-13-3. Hemolytic Disease of the Newborn

In A Nutshell

Graves Disease

- Anti-TSH receptor antibodies
- Hyperthyroidism early
- Hypothyroidism late

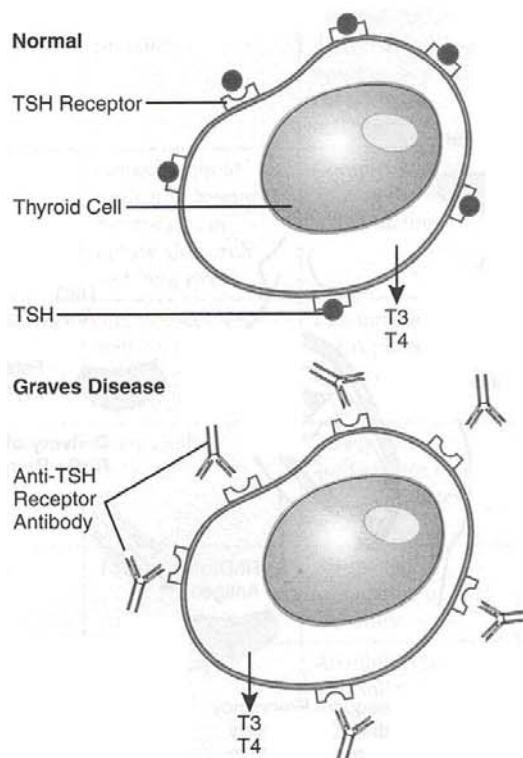


Figure II-13-4. Antibodies to the TSH Receptor in Hyperthyroidism

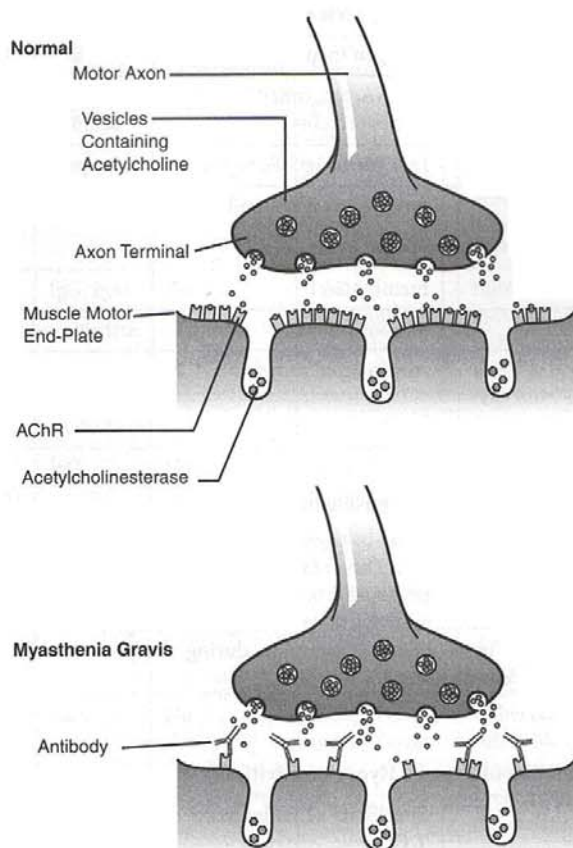


Figure II-13-5. Antibodies Against Acetylcholine Receptors in Myasthenia Gravis

Type III (Immune Complex) Hypersensitivities

The immune complexes that cause disease may either involve self or foreign antigens bound to antibodies. These immune complexes are filtered out of the circulation in the small vasculature, so their sites of ultimate damage do not reflect their sites of origin. These diseases tend to be **systemic**, with little tissue or organ specificity.

In A Nutshell

Type III Hypersensitivities

- Systemic damage
- Immune complexes activate complement
- Self or foreign Ags

Table II-13-5. Examples of Type III Hypersensitivities

Disease	Antigen Involved	Clinical Manifestations
Systemic lupus erythematosus	dsDNA, Sm, other nucleoproteins	Nephritis, arthritis, vasculitis, rash
Rheumatoid arthritis	IgM versus IgG Fc region	Joint pain, erosions
Poststreptococcal glomerulonephritis	Streptococcal cell wall Ags (may be "planted" in glomerular basement membrane)	Nephritis
Serum sickness	Various proteins	Arthritis, vasculitis, nephritis
Arthus reaction	Any injected protein	Local pain and edema

In A Nutshell

Type IV Hypersensitivities

- Delayed-type (48 to 72 hours)
- CD4+ TH cells mediate
- Activate macrophages
- Cause inflammation
- Common in chronic intracellular infections

Type IV (T-Cell-Mediated) Hypersensitivities

T lymphocytes may cause tissue injury by triggering delayed-type hypersensitivity (DTH) reactions or by directly killing target cells. These reactions are elicited by CD4+ TH1 cells and CD8+ cells, which secrete cytokines that activate macrophages (IFN- γ) and induce inflammation (TNF). These T cells may be autoreactive or specific against foreign protein antigens bound to tissues. T-cell-mediated tissue injury is common during the protective immune response against persistent intracellular microbes.

Table II-13-6. Examples of Type IV Hypersensitivities

Disease	Specificity of Pathogenic T Cells	Clinical Manifestations
Insulin-dependent diabetes mellitus (type I)*	Islet-cell antigens, insulin, glutamic acid decarboxylase, others	Polydipsia, polyuria, polyphagia, ketoacidosis
Multiple sclerosis	Myelin basic protein, proteolipid protein	Progressive demyelination, blurred vision, paralysis
Contact dermatitis	Nickel, poison ivy/oak catechols, hapten/carrier	Vesicular skin lesions, pruritus, rash
Peripheral neuritis	P2 protein of peripheral nerve myelin	Ascending paralysis
Guillain-Barré syndrome*	Peripheral nerve myelin or gangliosides	Ascending paralysis, peripheral nerve demyelination
Hashimoto's thyroiditis*	Unknown Ag in thyroid	Hypothyroidism

*Diseases classified as type IV pathologies in which autoantibodies are present and used as clinical markers

Table II-13-7. Hypersensitivity Summary

Type	Antibody	Complement	Effector Cells	Examples*
I (Immediate)	IgE	No	Basophil, mast cell	Hay fever, atopic dermatitis, insect venom sensitivity , anaphylaxis to drugs, some food allergies, allergies to animals and animal products, asthma
II (Cytotoxic)	IgG, IgM	Yes	PMN, macrophages, NK cells	Autoimmune or drug-induced hemolytic anemia, transfusion reactions, HDNB , hyperacute graft rejection, Goodpasture disease, rheumatic fever
II (Noncytotoxic)	IgG	No	None	Myasthenia gravis , Graves disease , type 2 diabetes mellitus
III (Immune complex)	IgG, IgM	Yes	PMN, macrophages	SLE , RA, polyarteritis nodosa, poststreptococcal glomerulonephritis, Arthus reaction, serum sickness
IV (Delayed, DTH)	None	No	CTL, TH1, macrophages	Tuberculin test , tuberculosis, leprosy, Hashimoto thyroiditis, poison ivy (contact dermatitis), acute graft rejection, GVHD , IDDM

*Most high-yield diseases are bolded.

Abbreviations: HDNB, hemolytic disease of the newborn; SLE, systemic lupus erythematosus; RA, rheumatoid arthritis; GVHD, graft-versus-host disease; IDDM, insulin-dependent diabetes mellitus

The Pathogenesis of Autoimmunity

The key factor in the development of autoimmunity is the recognition of self-antigens by autoreactive lymphocytes, which then become activated, proliferate, and differentiate to produce effector cells and cytokines that cause tissue injury. Autoimmunity must initially result from a failure of mechanisms of **self-tolerance**, as cells are "educated" in the bone marrow and thymus. Genetic factors contribute to this susceptibility, as well as environmental and hormonal triggers. Infections and tissue injury may alter the way that self-antigens are presented to lymphocytes and serve as an inciting factor in the development of disease. Because autoimmune reactions against one self-antigen may injure other tissues, exposing other potential self-antigens for recognition, autoimmune diseases tend to be chronic and progressive.

Among the strongest genetic associations with the development of autoimmune disease are the **class II MHC genes**.

In A Nutshell

Autoimmunity

- Failure of self-tolerance
- Genetics (class II MHC)
- Environment (infections)
- Hormones

Table II-13-8. Examples of HLA-Linked Immunologic Diseases

Disease	HLA Allele
Rheumatoid arthritis	DR4
Insulin-dependent diabetes mellitus	DR3/DR4
Multiple sclerosis	DR2
Systemic lupus erythematosus	DR2/DR3
Ankylosing spondylitis	B27
Celiac disease	DQ2 or DQ8

In A Nutshell

Infections can trigger autoimmunity through:

- Activating bystanders
- Molecular mimicry
- Inflammatory damage

In A Nutshell

Therapies for immune diseases:

- Inhibit proliferation
- Inhibit function
- Kill T cells
- Antagonize damaging products

In A Nutshell

Immunologic Therapies

- Desensitization
- Plasmapheresis
- Intravenous IgG
- Tolerance induction

Infections may trigger autoimmune responses by:

- **Bystander activation**—immune responses may recruit leukocytes and increase expression of costimulators, which activate T lymphocytes that are not specific for the infectious pathogen.
- **Molecular mimicry**—antigens of a microbe cross-react with or mimic self antigens.
- **Inflammation** and associated damage may expose self antigens that are normally concealed from immune cells.

Strategies for the Therapy of Hypersensitivity and Autoimmune Diseases

The mainstay of therapy for hypersensitivity diseases is directed at modification of T-cell function:

- Inhibit proliferation (cyclosporine)
- Inhibit function (corticosteroids)
- Kill T cells (cyclophosphamide)
- Antagonists to proinflammatory cytokines or costimulatory molecules (monoclonals or binding proteins)

These drugs are directed toward reducing tissue injury due to the effector phases of these responses. Other therapeutic regimens being used to inhibit pathologic responses include:

- Injection of small doses of allergens to reduce specific IgE production over time. The mechanism through which this procedure works is not clear. One theory is that increasing small doses of antigen leach preformed IgE out of the tissues. Another is that continuous small-dose exposures may stimulate TH cell populations that do not favor further IgE production.
- Plasmapheresis may be used to decrease circulating levels of antibody in diseases caused by antibodies or immune complexes.
- High doses of intravenous IgG may bind to Fc receptors and inhibit antibody synthesis.
- Attempts to induce tolerance by high-dose administration of antigens or altered forms of self antigens (myelin basic protein).

Chapter Summary

- There are four types of hypersensitivities: immediate, antibody-mediated (cytotoxic, blocking, enhancing), immune complex-mediated and T cell-mediated.
- Hypersensitivity reactions require initial sensitization, and subsequent exposures to the same or cross-reactive antigens cause the damage.
- Type I hypersensitivities (immediate) involve IgE antibodies and mast cells, show symptoms in minutes, and are mounted against harmless environmental antigens in atopic or allergic individuals.
- Initial tissue damage in immediate hypersensitivities is due to release of mast cell mediators, and late-phase reactions involve products of the arachidonic acid cascade.
- Examples of type I hypersensitivities include hay fever, asthma, food allergies, and systemic anaphylaxis.
- Type II (antibody-mediated) hypersensitivities are tissue specific and involve autoantibodies that opsonize or activate complement. Some noncytotoxic forms (myasthenia gravis, Graves disease, type II diabetes) cause interference with cellular function.
- Examples of cytotoxic type II hypersensitivities include autoimmune hemolytic anemia, hemolytic disease of the newborn, autoimmune thrombocytopenic purpura, Goodpasture syndrome, rheumatic fever, and pernicious anemia.
- Type III (immune complex) hypersensitivities cause systemic damage by activating complement wherever immune complexes of antibodies against self or foreign antigens are filtered from the circulation.
- Examples of type III hypersensitivities include systemic lupus erythematosus, rheumatoid arthritis, poststreptococcal glomerulonephritis, serum sickness, and the Arthus reaction.
- Type IV hypersensitivities are delayed-type (manifesting symptoms in 48 to 72 hours after reexposure); are caused by TH1 cells, CD8+ cells, and macrophages; and are common results of infection with persistent intracellular microbes.
- Examples of type IV hypersensitivities include insulin-dependent diabetes mellitus, multiple sclerosis, contact dermatitis, Guillain-Barré syndrome, and Hashimoto thyroiditis.
- Autoimmune diseases may associate with specific class II MHC haplotypes, environmental factors, or hormonal factors.
- Therapies for immune diseases involve inhibition of proliferation or function of T cells, killing of T cells, or antagonism of the damaging products of T cells.

Review Questions

1. A 43-year-old woman is seen by her physician with complaints of painful, swollen joints. On examination, her hands appear to be disfigured at the joints with apparent subcutaneous nodule formation. Her left knee is grossly enlarged, and 100 mL of fluid is withdrawn from the joint capsule. An examination of this fluid should reveal which of the following?
 - (A) Activated T lymphocytes
 - (B) Antibodies against type IV collagen
 - (C) Antibodies against double-stranded DNA
 - (D) Antibodies against microsomal antigens
 - (E) IgM antibodies reactive with the Fc region of IgG
2. A 36-year-old farmer has been exposed to poison ivy on several different occasions, and he usually gets very severe skin lesions. A pharmaceutical company is developing cytokines by recombinant DNA technology and formulating them in a fashion that they are readily absorbed through the skin. Which of the following cytokines administered topically could inhibit the severity of this reaction?
 - (A) γ -Interferon
 - (B) IL-2
 - (C) IL-3
 - (D) IL-8
 - (E) IL-10
3. In the 1960s, it was quickly ascertained that Peace Corps workers sent to schistosome-endemic areas were exposed to massive initial doses of cercariae before any protective immunity existed. In these individuals, IgG antibodies developed in response to the developing worms, and when the adults began their prodigious release of eggs into the circulation, the patients suffered acute and potentially life-threatening symptoms of fever, edema, arthralgia, and rash. Which of the following is another condition that arises by a similar immunologic mechanism?
 - (A) Arthus reaction
 - (B) Atopic allergy
 - (C) Goodpasture syndrome
 - (D) Tuberculin reaction
 - (E) Transfusion reaction

4. A young, newly married woman goes to her physician with concerns about a "hereditary problem" described to her by her mother that was associated with her own birth in 1968. Her family was poor, and her mother received no medical prenatal care before she was born "blue" and covered with "splotches" and "bruises." Although an earlier sibling had been born apparently normal, the patient required multiple transfusions before her condition stabilized, and two further pregnancies of her mother ended in stillbirths. The patient is concerned about the potential for development of similar problems in her own pregnancies. Blood tests ordered by the physician confirm his suspicions. How should he advise this patient?
- (A) Her husband should be tested for Rh incompatibility
 - (B) She is RhD- and should be treated postpartum with RhoGAM™
 - (C) She is RhD-; there is no risk to a fetus
 - (D) She is RhD+ and should be treated postpartum with RhoGAM™
 - (E) She is RhD+; there is no risk to a fetus
5. In native Egyptian populations, children are exposed to the cercariae of the fluke *Schistosoma mansoni* in early childhood when they wade in irrigation ditches throughout the Nile Delta. On first exposure, the cercariae penetrate the skin and become schistosomula, which enter the circulation and eventually mature in the mesenteric veins. On subsequent exposures, schistosomula are frequently killed within minutes by an immune response in the skin manifested by intense itching, stinging, and urticaria. What is this protective immune response a manifestation of?
- (A) Arthus reaction
 - (B) Contact dermatitis
 - (C) Passive cutaneous anaphylaxis
 - (D) Serum sickness
 - (E) Type I hypersensitivity

Answers and Explanations

1. **The correct answer is E.** This woman has the signs of rheumatoid arthritis, which is an autoimmune disease characterized by the development of IgM antibodies against one's own IgG (rheumatoid factor). These antibodies complex with their antigen (IgG) and precipitate out of the blood in the joints and vasculature, where they cause complement activation. Complement lysis is responsible for the damage to the tissues. This is an example of an immune complex-mediated (type III) hypersensitivity reaction.

Activated T lymphocytes (**choice A**) would be found in the circulation in some cases of infection, but are not a major feature of the type III hypersensitivities. In multiple sclerosis, which is a T-cell-mediated autoimmune disease, activated T lymphocytes sensitized to myelin can be found in the cerebrospinal fluid.

Antibodies against type IV collagen (**choice B**) would be found in Goodpasture disease. This is an example of a type II (cytotoxic antibody) hypersensitivity. In this disease, linear deposition of IgG and complement occurs in the alveolar and glomerular basement membranes and causes destruction of the underlying tissues.

Antibodies against double-stranded DNA (**choice C**) occur in the plasma of patients with systemic lupus erythematosus. This is a type III hypersensitivity disease, like rheumatoid arthritis, so the underlying principle of pathogenesis is similar, but patients with lupus have a characteristic facial rash, and symptoms would have their onset between 20 and 40 years of age. Although SLE can be associated with arthritis, it is not generally disfiguring as described here.

Antibodies against microsomal antigens (**choice D**) and thyroglobulin would be found in Hashimoto thyroiditis. In addition, T cells sensitized to thyroid antigens are found, so this disease is generally categorized as a type IV (delayed-type) hypersensitivity response. The age group and sex of the patient are correct for this diagnosis, but the symptoms would include hyperthyroidism, not joint inflammation.

2. **The correct answer is E.** IL-10 is produced by TH2 cells and inhibits TH1 cells. Because the response to poison ivy is a delayed-type hypersensitivity response and therefore is mediated by TH1 cells and macrophages, inhibiting their activity would minimize the severity of the reaction.

γ -Interferon (**choice A**) is a product of TH1 cells, CTLs, and NK cells. It inhibits the proliferation of TH2 cells and therefore would skew the immune response toward a more potent cell-mediated arm. This is not a cytokine that would help this patient: It would make his condition worse.

IL-2 (**choice B**) is a product of TH cells that induces the proliferation and enhances the activity of antigen-primed TH cells and CTLs. It would tend to increase the symptoms of this patient, not ameliorate them.

IL-3 (**choice C**) is a product of TH cells and NK cells. It acts on hematopoietic cells to encourage myeloid cell development. It would neither hinder nor help this man's condition.

IL-8 (**choice D**) is a product of macrophages and endothelial cells and acts on neutrophils to attract them to areas of inflammation. It would neither hinder nor help resolve the symptoms of a delayed-type hypersensitivity response.

3. **The correct answer is A.** The condition described here is an immune complex-mediated pathology. When large amounts of antigen are added into a situation where there is pre-existence of a large amount of antibody, the precipitation of those complexes in the small vasculature causes a type III hypersensitivity response. The only syndrome on this list that also has a type III etiology is the Arthus reaction.

Atopic allergy (**choice B**) is a type I hypersensitivity, mediated by IgE antibodies and mast cells. Although many parasitic diseases elicit IgE and ADCC by eosinophils, the question stem clearly stipulates the presence of IgG antibodies, so a type I hypersensitivity reaction is ruled out.

Goodpasture syndrome (**choice C**) is an example of a type II, cytotoxic antibody hypersensitivity response. In these cases, antibody binds to cells or tissues of the body and elicits complement activation in those locations. The result is a tissue- or organ-specific pathology, and not a systemic problem as described here.

The tuberculin reaction (**choice D**) is a type IV hypersensitivity response. Delayed-type hypersensitivities are mediated by TH1 cells and have no contribution whatsoever from antibodies.

The transfusion reaction (**choice E**) is an example of a type II, cytotoxic antibody hypersensitivity response. In these cases, antibody binds to cells or tissues of the body and elicits complement activation in those locations. The result is a tissue- or organ-specific pathology, and not a systemic problem as described here.

4. **The correct answer is E.** From the description in the vignette, this patient was born with hemolytic disease of the newborn, or erythroblastosis fetalis. This occurs when an Rh⁻ woman becomes pregnant with an Rh⁺ fetus. During a first pregnancy (the patient's older sibling), no problems are encountered, but the mother is sensitized against the RhD antigens. In all subsequent pregnancies, anti-Rh IgG antibodies from the mother can cross the placenta and cause hemolysis of the child's RBCs. This means that the patient is RhD⁺. It is irrelevant what the Rh type of any fetus she would carry is: She will make no immunologic response.

That her husband should be tested for Rh incompatibility (**choice A**) is not correct because the patient is Rh⁺. She will not recognize any Rh product as foreign.

That she is RhD⁻ and should be treated postpartum with RhoGAMTM (**choice B**) is not true because she is RhD⁺. Women who are RhD⁻ should be treated with RhoGAM (IgD or IgG anti-Rh antibodies) during and following the termination of any pregnancy.

That she is RhD⁻; there is no risk to a fetus (**choice C**) is not true because she is RhD⁺. Women who are RhD⁻ should be treated with RhoGAM (IgD or IgG anti-Rh antibodies) during and following the termination of any pregnancy.

That she is RhD⁺ and should be treated postpartum with RhoGAM (**choice D**) is not true because there is no need to treat RhD⁺ women in any way following termination of pregnancies.

5. **The correct answer is E.** The description in the vignette is a type I hypersensitivity reaction—the only type of hypersensitivity that can be manifested in minutes. These are IgE-mediated responses and are an important protective response against helminth parasites that migrate through the tissues.

The Arthus reaction (**choice A**) is an example of a type III (immune complex-mediated) pathology. These hypersensitivities develop when antigen is added to pre-existing antibody and immune complexes are filtered out of the circulation in the small vasculature. Complement is activated in these locations, and the underlying tissue is damaged. In this vignette, the killing is occurring in the skin and is associated within minutes with stinging, itching, and urticaria.

Contact dermatitis (**choice B**) is an example of a type IV (delayed-type) hypersensitivity response. After the sensitizing exposure, symptoms of this type of hypersensitivity will occur in 48 to 72 hours (not minutes, as described here).

Passive cutaneous anaphylaxis (**choice C**) is an example of a type I hypersensitivity reaction, but it is used to diagnose these conditions using passive transfer of serum. It is not a mechanism of protection, but a diagnostic technique.

Serum sickness (**choice D**) is an example of a type III (immune complex-mediated) pathology. These hypersensitivities develop when antigen is added to pre-existing antibody and immune complexes are filtered out of the circulation in the small vasculature. Complement is activated in these locations, and the underlying tissue is damaged. In this vignette, the killing is occurring in the skin and is associated within minutes with stinging, itching, and urticaria.

Transplantation Immunology

14

What the USMLE Requires You to Know

- The different types of tissue transplantation performed in medicine
- The mechanisms and timing of graft rejection phenomena
- The pathogenesis of graft-versus-host disease
- The techniques for tissue compatibility testing
- The therapeutic strategies to inhibit graft rejection

Transplantation is the process of taking cells, tissues, or organs (a **graft**) from one individual (the **donor**) and implanting them into another individual or another site in the same individual (the **host** or **recipient**). **Transfusion** is a special case of transplantation and the most frequently practiced today, in which circulating blood cells or plasma are infused from one individual into another. As we have seen in previous chapters, the immune system is elaborately evolved to recognize minor differences in self antigens that reflect the invasion of harmful microbes or pathologic processes, such as cancer. Unfortunately, it is this same powerful mechanism of self-protection that thwarts tissue transplantation because tissues derived from other individuals are recognized as “**altered-self**” by the educated cells of the host’s immune system.

Several different types of grafts are used in medicine:

- **Autologous** grafts (or **autografts**) are those where tissue is moved from one location to another in the same individual (skin grafting in burns or coronary artery replacement with saphenous veins).
- **Syngeneic** grafts are those transplanted between genetically identical individuals (monozygotic twins).
- **Allogeneic** grafts are those transplanted between genetically different members of the same species (kidney transplant).
- **Xenogeneic** grafts are those transplanted between members of different species (baboon heart into human child).

In A Nutshell

Transplantation

- Tissues taken from donor given to host
- Transfusion most common

In A Nutshell

Types of Grafts

- Autografts
- Syngeneic grafts
- Allogeneic grafts
- Xenogeneic grafts

Note

MHC alleles are expressed codominantly.

In A Nutshell

Graft Rejection Effectors

- CTLs
- Macrophages
- Ab-mediated

The recognition of transplanted cells as self or foreign is determined by the extremely polymorphic genes of the major histocompatibility complex, which are expressed in a **codominant** fashion. This means that each individual inherits a complete set or **haplotype** from each parent and virtually assures that two genetically unrelated individuals will have distinctive differences in the antigens expressed on their cells. The net result is that all grafts except autografts are ultimately identified as foreign invading proteins and destroyed by the process of **graft rejection**. Even syngeneic grafts between identical twins can express recognizable antigenic differences due to somatic mutations that occur during the development of the individual. For this reason, all grafts except autografts must be followed by some degree of lifelong immunosuppression of the host to attempt to avoid rejection reactions.

The time sequence of allograft rejection differs depending on the tissue involved but always displays specificity and memory. As the graft becomes vascularized, CD4+ and CD8+ cells that migrate into the graft from the host become sensitized and proliferate in response to both major and minor histocompatibility differences. In the **effector phase** of the rejection, TH cytokines play a critical role in stimulating macrophage, cytotoxic T cell, and even antibody-mediated killing. Interferons and TNF- α and - β all increase the expression of class I MHC molecules in the graft, and IFN- γ increases the expression of class II MHC as well, increasing the susceptibility of cells in the graft to MHC-restricted killing.

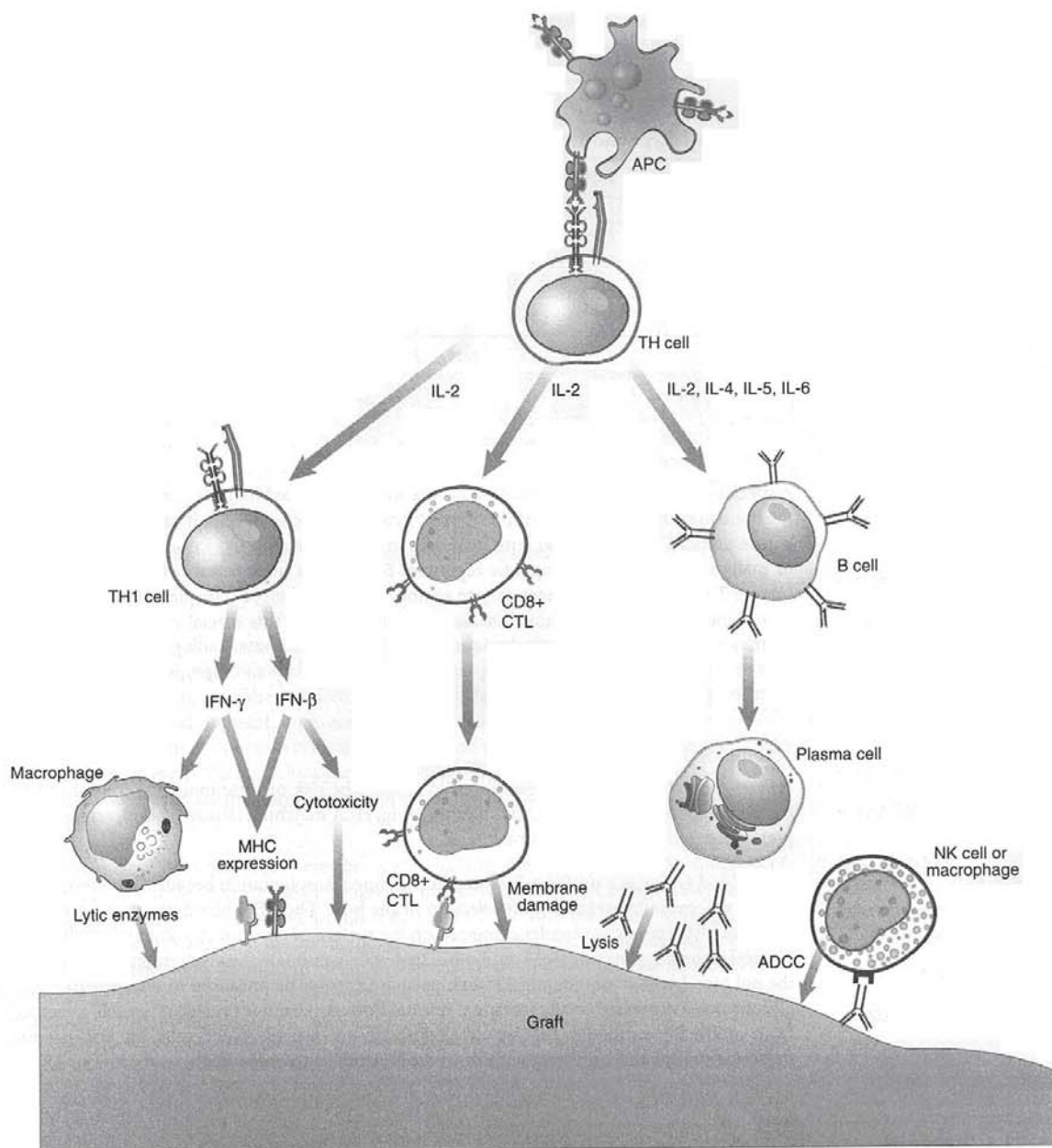


Figure II-14-1. Mechanism of Graft Rejection

In A Nutshell

Graft Rejection

- Hyperacute
- Accelerated
- Acute
- Chronic

In A Nutshell

Graft-Versus-Host Disease

- Bone marrow transplant
- Grafted T cells attack host
- Rash, jaundice, diarrhea, gastrointestinal hemorrhage

In A Nutshell

Tissue Compatibility Testing

- ABO blood typing
- Mixed lymphocyte reaction (class II)
- Microcytotoxicity test (class I)

Four different classes of allograft rejection phenomena are classified according to their time of activation and the type of effector mechanism that predominates.

Table II-14-1. Type and Tempo of Rejection Reactions

Type of Rejection	Time Taken	Cause
Hyperacute	Minutes to hours	Preformed antidonor antibodies and complement
Accelerated	Days	Reactivation of sensitized T cells
Acute	Days to weeks	Primary activation of T cells
Chronic	Months to years	Causes are unclear: antibodies, immune complexes, slow cellular reaction, recurrence of disease

A special case of tissue transplantation occurs when the grafted tissue is bone marrow. Because the bone marrow is the source of pluripotent hematopoietic stem cells, it can be used to reconstitute myeloid, erythroid, and lymphoid cells in a recipient who has lost these cells as a result of malignancy or chemotherapeutic regimens. Because the bone marrow is a source of some mature T lymphocytes, it is necessary to remove these cells before transplantation to avoid the appearance of **graft-versus-host disease** in the recipient. In this special case of rejection, any mature T cells remaining in the bone marrow inoculum can attack allogeneic MHC-bearing cells of the recipient and cause widespread epithelial cell death accompanied by rash, jaundice, diarrhea, and gastrointestinal hemorrhage.

Testing for Tissue Compatibility

Several clinical laboratory tests are used to reduce the risk of immunologic rejection during transplantation. These include ABO blood typing, HLA matching (tissue typing), screening for preformed antibodies, and crossmatching.

ABO blood typing is a uniform first step in all tissue transplantation because ABO incompatibilities will cause hyperacute graft rejection in the host. The ABO blood group antigens are a group of glycoprotein molecules expressed on the surface of erythrocytes and endothelial cells. Natural isohemagglutinins (IgM antibodies that will agglutinate the glycoprotein molecules on the red blood cells of nonidentical individuals) are apparently produced in response to similar glycoproteins expressed on the intestinal normal flora. A person is protected (by self-tolerance) from producing antibodies that would agglutinate his own red blood cells, but will produce those agglutinins that will react with the red cells from other individuals.

		Blood Types			
		A	B	O	AB
Serum from:	A				
Contains:	Anti-B Antibodies	N	A	N	A
Serum from:	B				
Contains:	Anti-A Antibodies	A	N	N	A
Serum from:	O				
Contains:	Anti-A and Anti-B Antibodies	A	A	N	A
Serum from:	AB				
Contains:	No Antibodies to A or B	N	N	N	N

A = Agglutination

N = No Agglutination

Figure II-14-2. Agglutination Test for Blood Typing

In cases where a living donor may be used for tissue transplantation (kidney and bone-marrow transplants), **tissue typing** to match the HLA antigens of recipient and donor is the next step. In general, the larger the number of matched MHC alleles between donor and recipient, the better the graft survival. In heart, lung, and liver transplantation, this step is often not considered because recipients are often in critical condition, and these organs cannot be stored indefinitely while tests are performed. Routine HLA typing focuses only on **HLA-A, HLA-B, and HLA-DR** because these are the only loci that appear to predict the likelihood of rejection of the transplant.

Two different laboratory techniques may be used to identify the histocompatibility antigens of donor and recipient. To identify class I antigens, a microcytotoxicity test using antisera against specific class I antigens is performed. In this test, lymphocytes from the donor or recipient are mixed with different antisera. If the antibodies recognize their specific epitope on the cells, they will be bound there, and addition of complement will result in cell lysis. The lysis of cells is monitored by adding a dye that will penetrate cells whose membranes have become leaky from the actions of complement.

In A Nutshell

Routine HLA Typing

- HLA-A
- HLA-B
- HLA-DR

In A Nutshell

Class I compatibility testing—microcytotoxicity test

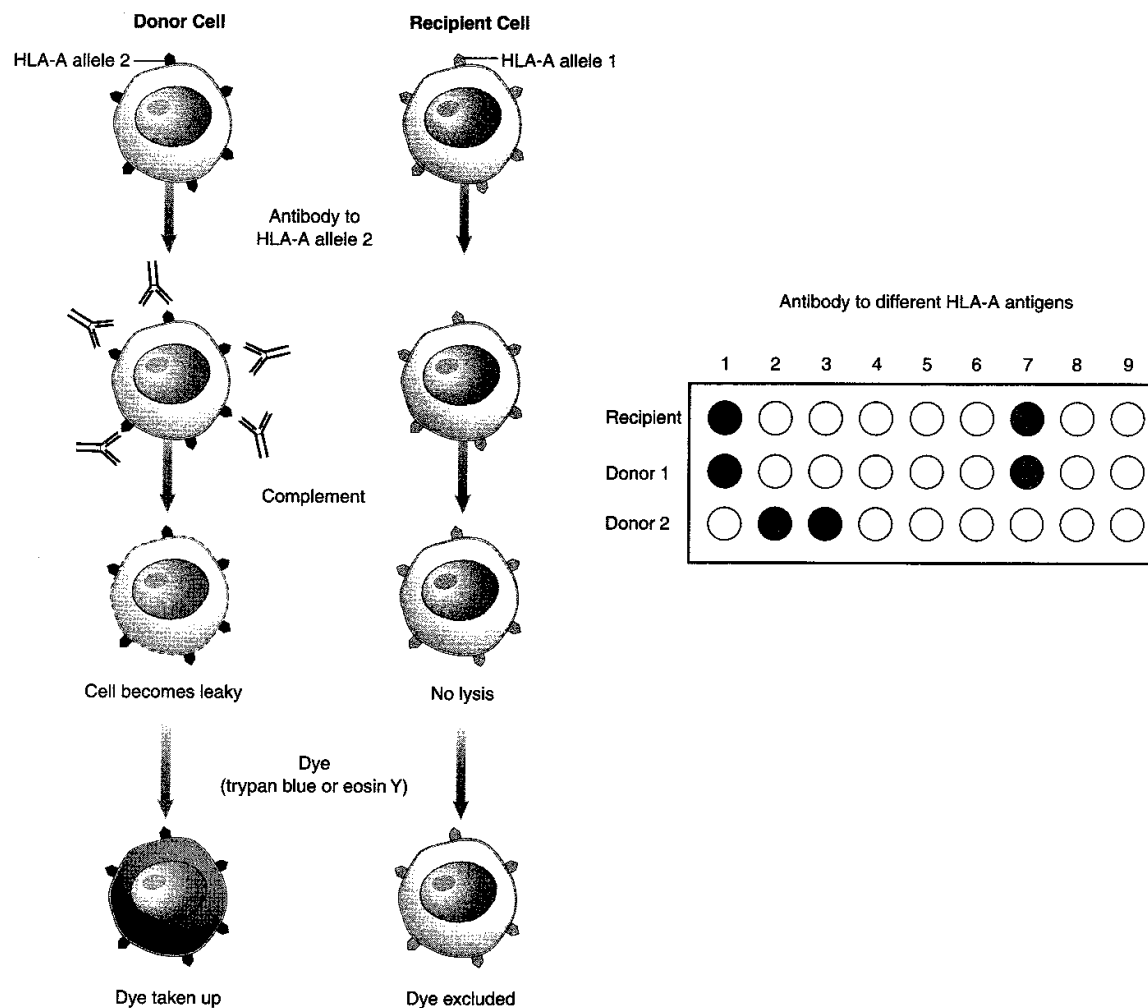


Figure II-14-3. Microcytotoxicity Assay

In A Nutshell

Class II compatibility testing—
mixed lymphocyte reaction

To identify **class II antigens**, a microcytotoxicity test (see Figure 14-3) or the **mixed lymphocyte reaction** (MLR) may be performed. In the MLR, lymphocytes from one individual being tested are irradiated so that they cannot proliferate, but will act as **stimulator cells** for the presentation of MHC antigens. The other individual's cells are added to the culture, and uptake of tritiated thymidine is used as an indicator of cell proliferation. If the MHC class II antigens are different, proliferation will occur. If they are the same, no proliferation will occur.

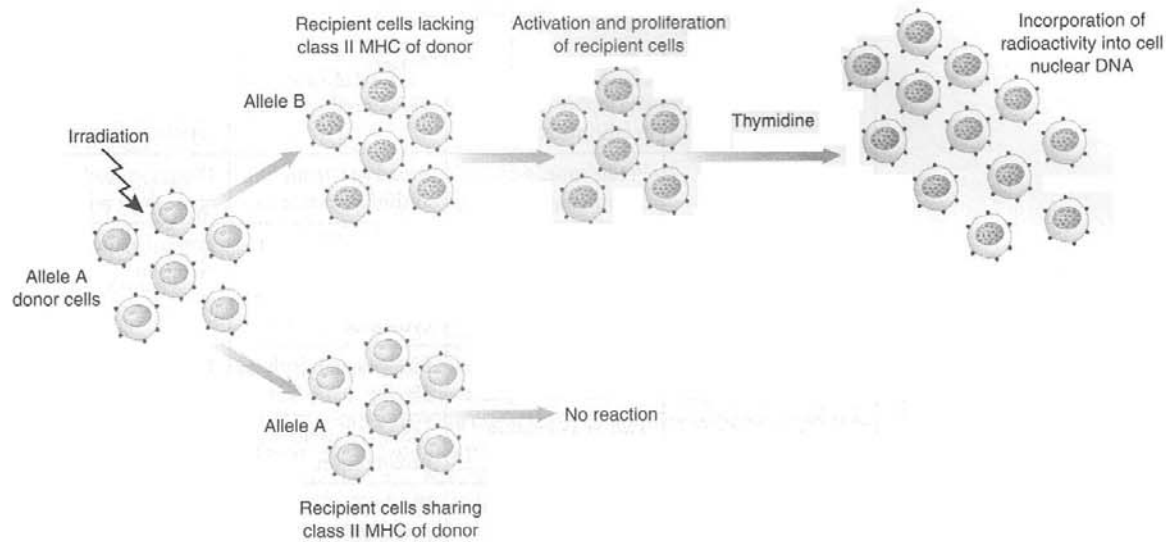


Figure II-14-4. The One-Way Mixed Lymphocyte Response

Patients awaiting organ transplants are screened for the presence of **preformed antibodies** reactive with allogeneic HLA molecules. These can arise because of previous pregnancies, transfusions, or transplantations and can mediate hyperacute graft rejection if they exist. This test is performed as a microcytotoxicity test (as above) with patient sera tested against batteries of donor cells reflective of the potential donor pool. If a potential donor is identified, then **crossmatching** is performed, testing serum from the recipient against the lymphocytes of the potential donor.

Prevention and Treatment of Allograft Rejection

Allogeneic and even syngeneic transplantation require some degree of **immunosuppression** for the transplant to survive. Most of the treatments currently in use suffer from lack of specificity: They result in generalized immunosuppression, which leaves the host susceptible to increased risk of infection.

In A Nutshell

Immunosuppression

- Necessary for all except autografts
- Inhibits T-cell mitosis
- Inhibits T-cell function
- Kills T cells
- Blocks cytokines or costimulatory molecules

Table II-14-2. Therapies to Prevent Graft Rejection

Category of Agent	Drug	Action	Use
Mitotic inhibitors	Cyclophosphamide, methotrexate	Block proliferation	Prevent acute graft rejection
Corticosteroids	Prednisone, dexamethasone	Broad spectrum antiinflammatories	During episodes of acute graft rejection
Fungal metabolites	Cyclosporin A	Block proliferation of TH cells and production of their cytokines	Prevent acute graft rejection
Experimental monoclonals	Anti-CD3	Binds to and depletes T cells	Prevent acute graft rejection
	Anti-IL2 receptor	Inhibits T-cell proliferation	
	Anti-CD40 ligand	Blocks T-cell proliferation	

Chapter Summary

- In transplantation, tissues are taken from a donor and given to a host or recipient.
- Autografts are grafts transplanted from one location to another in the same individual.
- Syngeneic grafts are grafts transplanted between monozygotic twins.
- Allogeneic grafts are grafts transplanted between nonidentical members of the same species.
- Xenogeneic grafts are grafts transplanted from one species to another.
- During graft rejection, MHC allele products are recognized as foreign by CTLs, macrophages, and antibodies, and the graft is destroyed.
- Graft rejection is hyperacute when preformed antidonor antibodies and complement destroy the graft in minutes to hours.
- Graft rejection is accelerated when sensitized T cells are reactivated to destroy the graft in days.
- Graft rejection is acute when T cells are activated for the first time and destroy the graft in days to weeks.
- Graft rejection is chronic when antibodies, immune complexes, or cytotoxic cells destroy the graft in months to years.
- Graft-versus-host disease occurs when mature T cells inside bone marrow transplants become activated against the MHC products of the graft recipient.
- Tissue compatibility testing involves ABO blood typing, the mixed lymphocyte reaction (for class II compatibility), and the microcytotoxicity test (for class I compatibility).
- Immunosuppression is required to ensure the survival of all grafts, except autografts.
- The goals of immunosuppression are to block cell proliferation (cyclophosphamide and methotrexate); to stop inflammation (prednisone, dexamethasone); or to block proliferation of TH cells and production of their cytokines (cyclosporin A).

Review Questions

1. A 42-year-old auto mechanic has been diagnosed with end-stage renal disease. His twin brother is HLA identical at all MHC loci and volunteers to donate a kidney to his brother. What type of graft transplant terminology is correct in this situation?
 - (A) Allograft
 - (B) Autograft
 - (C) Heterograft
 - (D) Syngeneic graft
 - (E) Xenograft

2. A patient with acute myelogenous leukemia (AML) undergoes irradiation and chemotherapy for his malignancy while awaiting bone marrow transplantation from a closely matched sibling. Six months after the transplant, the immune response appears to be reconstituting itself well—until 9 months postinfusion, when symptoms of generalized rash with desquamation, jaundice, and bloody diarrhea begin to appear. A second, more closely matched bone marrow donor is sought unsuccessfully, and 10 months after the transfer, the patient dies. What is the immunologic effector mechanism most closely associated with this rejection reaction?
 - (A) Activated macrophages
 - (B) Antibodies and complement
 - (C) CD8+ lymphocytes
 - (D) LAK cells
 - (E) NK cells

3. A 45-year-old welder develops a severe corneal ulcer, which requires treatment with corneal transplantation. A suitable cadaver cornea is available and is successfully engrafted. What is the appropriate postsurgical treatment for this patient?
 - (A) Corticosteroids, such as prednisone, for life
 - (B) Fungal metabolites, such as cyclosporin A, for life
 - (C) Mitotic inhibitors, such as cyclophosphamide, for life
 - (D) Monoclonal anti-IL-2 receptor for life
 - (E) No treatment required

4. A child who requires a kidney transplant has been offered a kidney by both parents and three siblings. A one-way mixed lymphocyte reaction between prospective donors and recipient is performed, and the stimulation indices are shown. The stimulation index is the ratio of proliferation (measured by [^3H]-thymidine incorporation) of the experimental group versus the negative control group. Which of the prospective donors would be the best choice?

Responder Cells	Irradiated Stimulator Cells					
	Recipient	Father	Mother	Sibling 1	Sibling 2	Sibling 3
Recipient	1.0	4.1	2.3	1.1	8.3	8.5
Father	5.3	1.0	12.3	5.6	4.9	5.9
Mother	3.2	12.6	1.0	4.5	3.9	4.8
Sibling 1	1.6	6.5	5.5	1.0	4.4	6.0
Sibling 2	7.6	5.9	4.9	4.4	1.0	7.8
Sibling 3	9.0	5.7	4.4	7.0	8.9	1.0

- (A) Father
(B) Mother
(C) Sibling 1
(D) Sibling 2
(E) Sibling 3
5. In heart-lung transplantation, where the critical illness of the transplant recipient and the inability to preserve tissues from brain-dead donors often precludes tissue typing prior to transplantation surgery, a variety of experimental immunosuppressive protocols are under study. In one such experimental protocol, patients are treated with anti-CD28 antibody Fab fragments at the time of transplantation and at monthly intervals thereafter. What would be the goal of such therapy?
- (A) To destroy T cells
(B) To induce anergy to transplanted tissues
(C) To inhibit mitosis in B cells
(D) To inhibit mitosis in T cells
(E) To stop inflammation

Answers and Explanations

1. **The correct answer is D.** A syngeneic graft is performed between **genetically identical** individuals. In human medicine, these are performed between **monozygotic twins**. In reality, even these "identical" individuals are not identical because **minor mutations can** occur during development. These sorts of grafts still require **immunosuppression for success**. They are, however, the best chance for success other than autografts.

An allograft (**choice A**) is a transplant between two members of the same species who are not genetically identical. These are the most common types of transplants used in medicine, but in this vignette, the twins are described as having identical MHC haplotypes.

An autograft (**choice B**) is a transplant from one location in the body to another. This is the only form of transplantation that will succeed without immunosuppression.

"Heterograft" (**choice C**) is not a word that is used in transplantation immunology.

A xenograft (**choice E**) is a transplant that is performed across species barriers.

2. **The correct answer is C.** Graft-versus-host disease (GVHD) is primarily a manifestation of sensitization of transplanted T cells against recipient tissues. The killing of mucosal and other epithelial cells is largely mediated through recognition of MHC class I incompatibility by transferred cytotoxic cells or their precursors. However, eventually, continuous priming by the host's own tissues will elicit immune responses at the level of all the cells of the immune system.

Activated macrophages (**choice A**) are involved in the delayed-type hypersensitivity response, but are not stimulated by MHC incompatibility, so if they become involved in pathology, it has to be secondary to TH stimulation.

Antibodies and complement (**choice B**) are not involved in GVHD. Because bone marrow is a cellular transplant, it is the cells inside it that start the problem, not accidentally transferred antibodies or complement.

LAK cells (**choice D**) are believed to be involved in the rejection of bone marrow transplants by the recipient (host-versus-graft), but not in GVHD.

NK cells (**choice E**) are believed to be involved in the rejection of bone marrow transplants by the recipient (host-versus-graft), but not in GVHD.

3. **The correct answer is E.** Because the eye is an immunoprivileged site, corneal transplantation is unique amongst allogeneic transplantation techniques practiced in human medicine in that it does not require immunosuppression. Other immunoprivileged sites in the human include the uterus, the testes, the brain, and the thymus. What these sites have in common is that they do not possess lymphatic vessels. For this reason, the alloantigens of the graft are not able to sensitize the recipient's lymphocytes, and the graft has an increased likelihood of acceptance.

Corticosteroids, such as prednisone, for life (**choice A**) are required for most types of transplantation. Corticosteroids act as broad-spectrum antiinflammatories, which are particularly important in treatment of episodes of acute graft rejection.

Fungal metabolites, such as cyclosporin A, for life (**choice B**) would be necessary for most types of transplantation. These agents act by blocking proliferation of TH cells and production of their cytokines.

Mitotic inhibitors, such as cyclophosphamide, for life (**choice C**) would be necessary for most types of transplantation. These agents act by blocking proliferation of cells.

Monoclonal anti-IL-2 receptor for life (**choice D**) is an experimental protocol that would inhibit T-cell proliferation. It would not be necessary in the case of corneal transplantation.

4. **The correct answer is C.** The lowest stimulation index (and the lowest amount of proliferation) is shown between sibling 1 and the prospective recipient, both when the donor cells are used as stimulators and as responders. This means (most importantly) that the recipient will make little response to the graft and (less importantly, except in graft-versus-host disease) that the donor will make little response against the recipient.

The father (**choice A**) is not the best choice of donors because the recipient makes four times the proliferative response to his cells as to those of sibling 1.

The mother (**choice B**) is not the best choice of donors because the recipient makes twice the proliferative response to her cells as to those of sibling 1. She would be the second-best choice, unless sibling 1 had an incompatible ABO blood group.

Sibling 2 (**choice D**) is not the best choice of donors because the recipient makes eight times the proliferative response to his cells as to those of sibling 1.

Sibling 3 (**choice E**) is not the best choice of donors because the recipient makes eight times the proliferative response to his cells as to those of sibling 1.

5. **The correct answer is B.** If a patient were treated with the Fab portions of antibodies to the CD28 molecule, this would block the binding of CD28 (on T cells) to B7 on antigen-presenting cells. Because this costimulatory signal is necessary as a second signal following TCR binding, the cell receives no second signal and becomes unresponsive (anergic) to that antigen.

To destroy T cells (**choice A**) is the goal of treatment with experimental monoclonals such as anti-CD3 antibodies. These bind to and deplete T cells, but act in a nonspecific fashion, so there is increased susceptibility to infection.

To inhibit mitosis in B cells (**choice C**) is not a goal of any of the therapies against graft rejection. T cells are at the root of all types of graft rejection, with the possible exception of hyperacute rejection based on ABO blood group incompatibilities.

To inhibit mitosis in T cells (**choice D**) is the goal of agents such as cyclophosphamide and methotrexate.

To stop inflammation (**choice E**) is the goal of corticosteroids such as prednisone and dexamethasone. These are broad-spectrum antiinflammatories used during periods of acute graft rejection.

Cancer Immunology

15

What the USMLE Requires You to Know

- The immunologic mechanisms of surveillance against tumors
- The major mechanisms of tumor immune evasion
- The rationales of major tumor immunotherapy techniques

Tumors express antigens that are recognized by the immune system, but most tumors are weakly immunogenic, and immune responses frequently fail to stop their growth. These antigens may be:

- Products of oncogenes (Ras) or tumor suppressor genes (p53)
- Mutants of cellular genes not involved in tumorigenesis
- Products of genes that are silent in most normal tissues
- Products of overexpressed genes
- Products of oncogenic viruses (E6 and E7 proteins of HPV)
- Oncofetal antigens (α -fetoprotein)
- Differentiation antigens normally present in the tissue of origin (e.g., prostate-specific antigen)

The recognition of tumor antigens by CTLs is the major effector mechanism in antitumor immunity, but other effectors such as antibody, macrophages, and NK cells may be important in some cases. Cytotoxic CD8⁺ cells perform a surveillance function by recognizing tumor peptides presented in the groove of the class I MHC molecule. If tumor-specific peptides are presented by APCs that have ingested them, then costimulatory molecules are engaged, and the response becomes more powerful. CD4⁺ TH cells probably also play a role by providing cytokines that stimulate CTL development and cytokines such as TNF and IFN- γ , which upregulate expression of MHC molecules on the surface of tumor cell targets.

Antibodies may kill tumor cells by activating complement or assisting ADCC, but this arm of the immune response is clearly secondary to the generation of tumor-specific CTLs.

In cases where tumor cells have reduced expression of class I MHC molecules, NK cells may play a major protective role. Macrophages activated by IFN- γ from TH1 cells have increased tumoricidal capacity and produce TNE, which has been shown to induce thrombosis in tumor blood vessels.

In A Nutshell

Tumors are weakly immunogenic.

In A Nutshell

Antitumor Effector Mechanisms

- CTLs
- Antibody (with complement or ADCC)
- Macrophages
- NK cells

In A Nutshell**Tumor Immune Evasion**

- Decreased MHC expression
- Antigen-loss variants
- TGF- β inhibits effector functions
- FasL induces apoptosis
- Recognition of antigen by TCR without costimulatory signal induces tolerance
- Antigens masked by mucopolysaccharides

In A Nutshell**Tumor Immunotherapy**

- Vaccination
- Increase costimulatory molecules
- Stimulate T cells
- Antitumor Abs
- Immunotoxins
- Expand cytotoxic cells in culture

Clinical Correlate

The hepatitis B vaccine was technically the first antitumor vaccine to be placed on the market.

Immune Evasion by Tumors

Several mechanisms exist by which tumors seem to evade the development or function of anti-tumor immune responses.

- Class I MHC expression may be downregulated.
- Antigen-loss variants emerge in rapidly growing tumors.
- Tumors may not stimulate CTLs if they lack costimulatory molecules (causing tolerance).
- Transforming growth factor- β secreted by some tumors inhibits proliferation and effector functions in lymphocytes and macrophages.
- Fas ligand expressed on some tumors may induce apoptotic death of lymphocytes.
- The antigens of tumors may be masked by sialic acid-containing mucopolysaccharides.

Immunotherapy for Tumors

Immunotherapy for tumors is designed to augment development of an active immune response against the tumor or passively deliver tumor-specific immune effectors to patients:

- Vaccination with tumor cells or antigens
- Administration of tumor cells modified to express high levels of costimulators
- Administration of cytokines that stimulate T-cell proliferation and differentiation
- Administration of antitumor antibodies
- Administration of antibodies conjugated to toxic drugs (immunotoxins)
- Administration of tumor-reactive T cells and NK cells expanded in culture by growth factors (LAK cells)

Clinical Correlate**The Therapeutic Use of Interferons**

Since the first description of interferons almost 50 years ago, a multitude of dramatic immunomodulatory roles have been discovered for this group of proteins. As a group, interferons induce increases in the expression of class I and II MHC molecules and augment NK cell activity. They increase the efficiency of presentation of antigens to both cytotoxic and helper cell populations. Cloning of the genes that encode α , β , and γ interferons has made it possible to produce amounts of these products, which make their use clinically practical.

Interferon has well-known antiviral activity and has been used in the treatment of hepatitis B and C infections. Within cancer therapy, IFN- α has shown promise in treatment of hairy B-cell leukemia, chronic myelogenous leukemia, and Kaposi sarcoma.

Interferon- β was the first drug shown to have a positive effect on young adults with multiple sclerosis. Patients treated with IFN- β enjoy longer periods of remission and reduced severity of relapses.

Interferon- γ is being used in the treatment of chronic granulomatous disease (CGD). This molecule is a potent inducer of macrophage activation and a promoter of inflammatory responses. Its application appears to significantly reverse the CGD patient's inability to generate toxic oxygen metabolites inside phagocytic cells.

The side effects of IFN therapy are fortunately mild and can be managed with acetaminophen. They include headache, fever, chills, and fatigue and diminish with continued treatment.

Chapter Summary

- Most tumors express antigens that are weakly immunogenic.
- CTLs are the major antitumor effector cell.
- Antibodies can act with complement or assist in ADCC to kill tumor cells.
- Macrophages activated by TH1-cell-produced IFN- γ and NK cells can kill tumor cells in some cases.
- TNF and IFN- γ from TH1 cells increase expression of MHC class I on tumor cells, making them more susceptible to CTL killing.
- Tumors evade the immune response by downregulating MHC expression, lacking costimulatory molecules (and thereby inducing tolerance), inhibiting immune cell proliferation, inducing apoptosis of immune cells, and masking antigens in mucopolysaccharides.

Review Questions

1. A patient with advanced metastatic melanoma decides to join an experimental treatment protocol in the hope that it will cause regression of his tumor masses. Malignant cells are aspirated from several of his lesions and transfected in vitro with the gene encoding GM-CSF production. The transfected tumor cells are then reinfused into the patient. Mobilization of which of the following cells from the bone marrow would be likely to result from this treatment?
 - (A) Antigen-presenting cells
 - (B) B lymphocytes
 - (C) NK cells
 - (D) Plasma cells
 - (E) T lymphocytes

2. A patient with a B-cell lymphoma is referred to an oncology clinic for the analysis of his condition. The malignant cells are found to be producing IgM monomers. Which of the following therapeutic regimens is most likely to destroy the malignant cells and no others?
 - (A) Anti-CD3 antibodies plus complement
 - (B) Anti-CD19 antibodies plus complement
 - (C) Anti-CD20 antibodies plus complement
 - (D) Anti-idiotypic antibodies plus complement
 - (E) Anti- μ chain antibodies plus complement

3. An experimental protocol for the protection of high-risk populations against human papilloma virus and cervical intraepithelial neoplasia involves the production of a recombinant DNA vaccine. The genes of the virus that encode the E6 and E7 proteins are inserted into a viral vector and inoculated intramuscularly into the patient. Which of the following would be the goal of such treatment?
 - (A) To inactivate viral oncogenes
 - (B) To increase MHC class I expression on tumor cells
 - (C) To produce anti-idiotypic antibodies
 - (D) To protect natural cellular antioncogenes
 - (E) To stimulate CD8⁺ T lymphocytes

4. A woman with advanced metastatic breast cancer undergoes a radical mastectomy, followed by irradiation and chemotherapy. After a 2-year remission, a metastatic focus appears, and she enrolls in an experimental treatment protocol. In it, a sample of her aspirated bone marrow is treated with GM-CSF, TNF- α , and IL-2, then pulsed with membrane fragments of her tumor cells and reinfused. Which of the following cell subpopulations is the most directly targeted by this treatment?
- (A) B lymphocytes
 - (B) Cytotoxic T lymphocytes
 - (C) NK cells
 - (D) TH1 cells
 - (E) TH2 cells
5. A 6-year-old child from Zimbabwe is admitted to a U.S. oncology center for treatment of an advanced case of Burkitt lymphoma. Analysis of the malignant cells reveals that they are lacking MHC class I antigens on their surface. Which of the following cytokines produced by recombinant DNA technology could be injected into his solid tumor to increase this tumor cell's susceptibility to CD8+-mediated killing?
- (A) IFN- γ
 - (B) IL-1
 - (C) IL-2
 - (D) IL-10
 - (E) TNF- α

Answers and Explanations

1. **The correct answer is A.** Tumor cells transfected with the gene encoding GM-CSF would produce granulocyte/monocyte colony-stimulating factor. This is a cytokine that acts on the bone marrow to cause production and mobilization of antigen-presenting cells. The goal of such therapy would be to induce the production of antigen-presenting cells, which might increase the presentation of tumor-cell antigens to cells important in cell-mediated cytotoxicity.

B lymphocytes (**choice B**) would not be mobilized by such a treatment. The cytokine that favors development of lymphoid precursors in the bone marrow is IL-7.

NK cells (**choice C**) would not be mobilized by such a treatment. Although NK cells are granular, they are derived from lymphoid, not granulocyte/monocyte, precursors. The cytokine that favors development of lymphoid precursors in the bone marrow is IL-7.

Plasma cells (**choice D**) are produced in the secondary lymphoid organs and submucosa. IL-7, which stimulates lymphoid precursors in the bone marrow, would have an indirect effect on plasma cell production, but they are not mobilized from the bone marrow.

T lymphocytes (**choice E**) would not be mobilized by such a treatment. The cytokine that favors development of lymphoid precursors in the bone marrow is IL-7.

2. **The correct answer is D.** Because malignant cells are clonal in origin, all the cells in this patient's lymphoma should be producing IgM monomers of a single idiotype. Treatment with anti-idiotype antibodies plus complement, therefore, would specifically kill only malignant cells, and leave all other B lymphocytes unharmed.

Anti-CD3 antibodies plus complement (**choice A**) would kill all T lymphocytes in the body. This lymphoma is clearly of B-cell origin because it is bearing IgM monomers.

Anti-CD19 antibodies plus complement (**choice B**) would kill all B lymphocytes in the body. It would not specifically target malignant cells.

Anti-CD20 antibodies plus complement (**choice C**) would kill all B lymphocytes in the body. It would not specifically target malignant cells.

Anti- μ chain antibodies plus complement (**choice E**) would kill all mature and naive B cells and immature B cells that had completed VDJ rearrangement of their heavy chain genes. It would not be specific for malignant cells.

3. **The correct answer is D.** In human papilloma virus infections, production of viral proteins E6 and E7 inactivates two cellular tumor suppressor genes, p53 and p110. The vaccine in this case is designed to create an immune response against these viral proteins, and in so doing, protect the function of two normal cellular antioncogenes.

To inactivate viral oncogenes (**choice A**) is not correct because proteins E6 and E7 inactivate cellular antioncogenes.

To increase MHC class I expression on tumor cells (**choice B**) is not correct because proteins E6 and E7 do not affect MHC class I expression. IFN treatment has been shown to increase expression of MHC class I products on cells.

To produce anti-idiotype antibodies (**choice C**) is not the goal of this vaccine because anti-idiotype antibodies can only be produced by repeated inoculations of monoclonal antibodies. This vaccine is inoculating E6 and E7 viral proteins, not antibodies.

To stimulate CD8⁺ T lymphocytes (**choice E**) would be an indirect benefit of this vaccine because it should generate cells capable of responding to the viral E6 and E7 proteins. Because the E6 and E7 proteins combine with and inactivate cellular anti-oncogenes, however, CTLs alone would not be sufficient to protect against malignant transformation.

4. **The correct answer is D.** The goal of this therapy is to provide an increased population of activated antigen-presenting cells primed with tumor cell antigens so that these can be presented to the TH cells involved in stimulation of cell-mediated immunity. The TH1 cell is the first cell listed which would be activated by such a treatment.

B lymphocytes (**choice A**) would not be stimulated by such treatment. B lymphocytes bind to and are activated by unprocessed (not cell-bound) antigens.

Cytotoxic T lymphocytes (**choice B**) would be indirectly stimulated by this treatment. Cytokines secreted by the activated TH1 cells would have the effect of increasing the number and cytotoxic activity of killer cells.

NK cells (**choice C**) would be indirectly stimulated by this treatment. Cytokines secreted by the activated TH1 cells would have the effect of increasing the number and cytotoxic activity of killer cells.

TH2 cells (**choice E**) would be stimulated by this treatment, but this is not the major goal of such therapy. TH2 cells stimulate humoral immunity, which is not the most important protective mechanism against tumor cells.

5. **The correct answer is A.** IFNs of all types increase cellular expression of MHC class I and II products. Because CD8+ cells recognize their targets by MHC class I-dependent mechanisms, increases in the amount of these antigens on tumor-cell targets would increase susceptibility to cytotoxic killing.

IL-1 (**choice B**) does not increase MHC class I molecule expression. The endogenous pyrogen is responsible for alteration of the hypothalamic temperature set point during acute inflammatory events.

IL-2 (**choice C**) does not increase MHC class I molecule expression. It is produced by TH cells and causes proliferation of many classes of lymphocytes.

IL-10 (**choice D**) does not increase MHC class I molecule expression. It is a product of TH2 cells and inhibits TH1 cells; thus, it inhibits the cell-mediated arm of the immune response.

TNF- α (**choice E**) does not increase MHC class I molecule expression. It may act directly on tumor cells to cause their necrosis and decrease angiogenesis. It is a product of TH1 cells that stimulates the effector cells of cell-mediated immunity.

Laboratory Techniques in Immunology

16

What the USMLE Requires You to Know

- The procedures for quantification of antigen–antibody complexes
- The general procedures and applications of agglutination, fluorescent antibody, radioimmunoassay, ELISA, Western blot, and flow cytometry
- How to interpret data from these tests to diagnose immunologic or microbial diseases

Many diagnoses in infectious disease and pathology would not be possible without laboratory procedures that identify antibodies or antigens in the patient. This chapter will illustrate the most commonly used techniques.

Antigen–Antibody Interactions

Interaction of antigen and antibody occurs *in vivo*, and in clinical settings it provides the basis for all immunologically based tests. The formation of immune complexes produces a visible reaction that is the basis of precipitation and agglutination tests. Agglutination and precipitation are maximized when multiple antibody molecules share the binding of multiple antigenic determinants, a condition known as **equivalence**. *In vivo*, the precipitation of such complexes from the blood is critical to the trapping of invaders and the initiation of the immune response in the secondary lymphoid organs, as well as the initiation of the pathologic phase of many immune complex–mediated diseases. *In vitro*, the kinetics of such reactions can be observed by titration of antigen against its specific antibody.

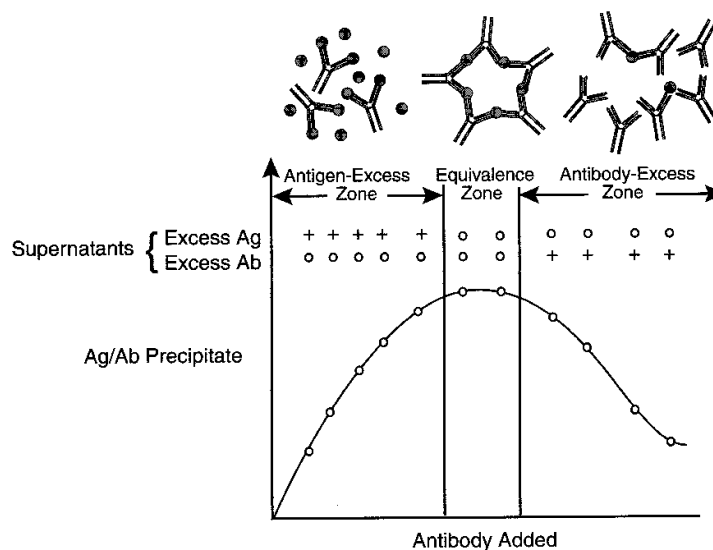


Figure II-16-1. Precipitation of Ag-Ab Complexes During Titration of Ag with Ab

In A Nutshell

- Ag excess—early in infection
- Equivalence—all available Ag complexed with Ab = window period
- Ab excess—late in infection

Figure II-16-1 demonstrates the normal progression of the antibody response during many infectious diseases. At the start of the infection, the patient is in a state of **antigen excess** because the pathogen is proliferating in the host and the development of specific antibodies has not yet begun. As the patient begins to make an adequate antibody response, he enters the **equivalence zone**, when all available antigen is complexed with antibody, and neither free antigen nor free antibody can be detected in the serum. Finally, as the infection is resolved, the patient enters the **antibody excess** zone, when more antibody is being produced than is necessary to precipitate all available antigen. The clinical demonstration of this phenomenon is most elegantly seen in our use of the serologic diagnosis of hepatitis B infection. Early in the course of this infection, HBsAg and HBeAg are easily detectable in the blood. The patient is in the antigen excess zone. As the patient enters the **window period** (the equivalence zone), the antigens disappear from the circulation because they are being removed by antibody precipitation. The respective antibody titers (HBsAb and HBeAb) rise in the serum as the patient enters the antibody excess zone and resolves the infection. Although the “window period” in the hepatitis B infection is used exclusively to connote the absence of HBsAg and HBsAb from the serum (because it is the only antigen-antibody response that has a clinical significance in the prognosis of disease), an equivalence zone, or “window period,” is a universal stage in the development of any antibody-antigen interaction.

Agglutination

Agglutination tests are widespread in clinical medicine and are simply a variation on precipitation reactions in which the antigen is a particle and not a soluble material. The two particles most commonly used in medicine for this purpose are RBCs and latex beads, and both will settle out of suspension in the form of a carpet of antibody-bound particles in the presence of appropriate antibodies. Latex bead agglutination tests are available for the diagnosis of cerebrospinal infections such as *Haemophilus*, pneumococcus, meningococcus, and *Cryptococcus neoformans*. In each of these cases, antibodies against these organisms are conjugated to latex beads, and the presence of pathogen antigens in the CSF is detected by the subsequent agglutination of those

beads. RBC agglutination reactions are important in defining ABO blood groups (see Chapter 14), diagnosing Epstein-Barr virus infection (the monospot test), and the Coombs test for Rh incompatibility.

Two variations of the Coombs test exist. The direct Coombs is designed to identify maternal anti-Rh antibodies already bound to infant RBCs or antibodies bound to RBCs in patients with autoimmune hemolytic anemia. The indirect Coombs test is used to identify Rh-negative mothers who are producing anti-Rh antibodies of the IgG isotype, which may be transferred across the placenta to harm Rh-positive fetuses.

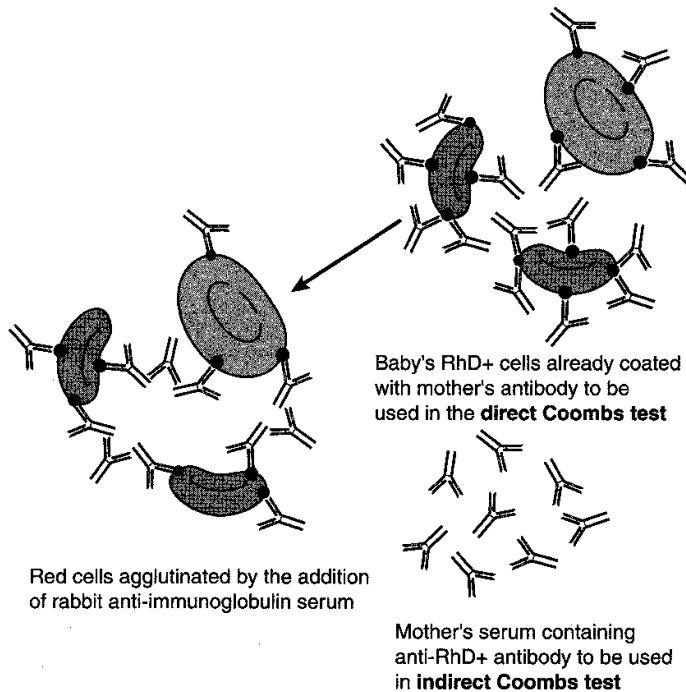


Figure II-16-2. Coombs Test

Fluorescent Antibody Tests

The direct fluorescent antibody test (DFA) is used to detect and localize antigen in the patient. The tissue sample to be tested is treated with antibodies against that particular antigen that have been labeled with a fluorescent dye. If the antigen is present in the tissues, the fluorescent-labeled antibodies will bind, and their binding can be detected with a fluorescence microscope. Variations of this test are used to diagnose respiratory syncytial virus, herpes simplex-1 and -2, and *Pneumocystis* infections.

In A Nutshell

Direct Coombs—detect antibodies bound to RBC

Indirect Coombs—detect production of anti-RBC antibodies

In A Nutshell

DFA identifies Ag in tissues.

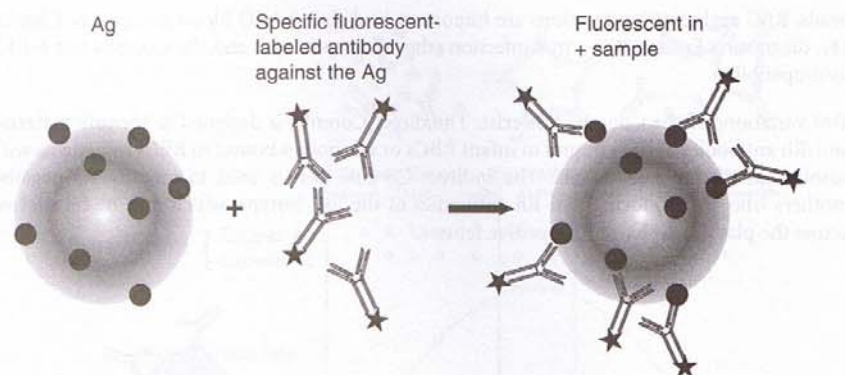


Figure II-16-3. DFA Test

In A Nutshell

IFA detects Abs in the patient.

The **indirect fluorescent antibody test (IFA)** is used to detect pathogen-specific antibodies in the patient. In this case, a laboratory-generated sample of infected tissue is mixed with serum from the patient. A fluorescent dye-labeled anti-immunoglobulin (raised in animals) is then added. If binding of antibodies from the patient to the tissue sample occurs, then the fluorescent antibodies can be bound, and fluorescence can be detected in the tissue by microscopy. This technique is used to detect antinuclear antibodies, anti-ds DNA antibodies, antithyroid antibodies, antiglomerular basement-membrane antibodies, and anti-Epstein-Barr virus viral-capsid antigen antibodies.

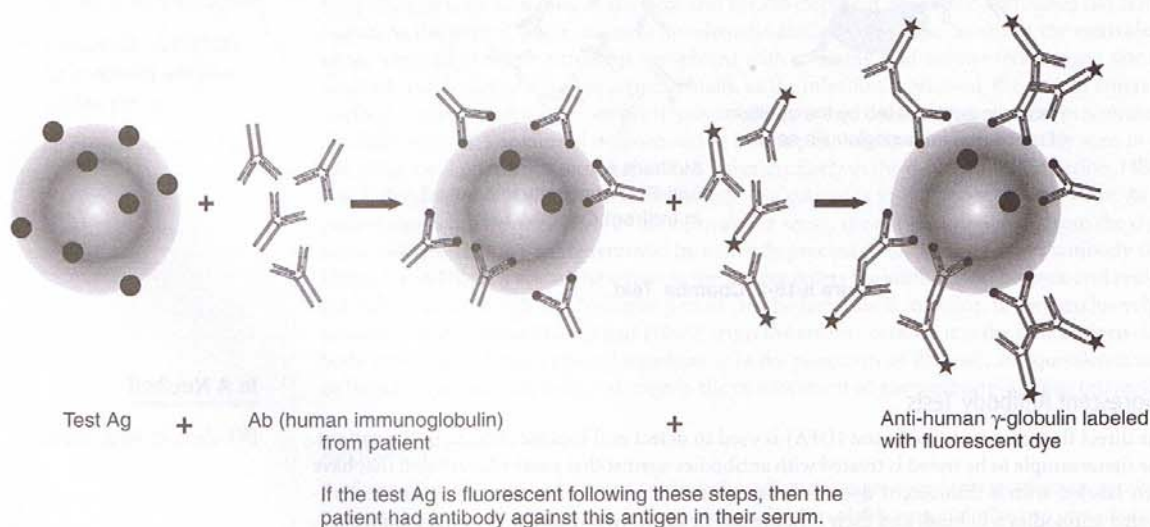


Figure II-16-4. Indirect FA Test

Radioimmunoassay (RIA) and Enzyme-Linked Immunoabsorbent Assay (EIA or ELISA)

RIA and ELISA are extremely sensitive tests (as little as 10^{-9} g of material can be detected) that are common in medical laboratories. They can be used to detect the presence of hormones, drugs, antibiotics, serum proteins, infectious disease antigens, and tumor markers. Both tests are conducted similarly, but the RIA uses the detection of a radiolabeled product and the ELISA detects the presence of enzyme-mediated color changes in a chromogenic substrate.

In the screening test for HIV infection, the ELISA is used, with the p24 capsid antigen from the virus coated onto microtiter plates. The serum from the patient is then added, followed by addition of an enzyme-labeled antihuman immunoglobulin. Finally, the enzyme substrate is added, and the production of a color change in the well can be observed if all reagents bind one another in sequence.

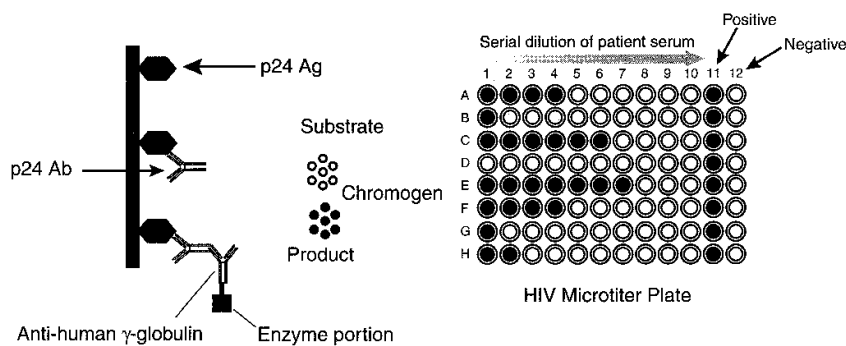


Figure II-16-5. ELISA Test

Western Blot or Immunoblot

The Western blot is primarily used in medicine for confirmation of HIV infection in a patient who has seroconverted and is thus positive by the ELISA test. Because the ELISA suffers from the detection of some false-positives, this follow-up step is essential for the diagnosis of HIV infection. In this test, the antigens of the virus are separated in an electric field and blotted onto nitrocellulose paper. The serum of the patient is then added and allowed to bind to any antigens that it recognizes. Next, antihuman immunoglobulin antibodies conjugated to either enzyme or radioactive labels are added, to bind to the previously bound patient's antibodies (if any). The result of color change in the case of an enzyme system or radioactivity in the case of radiolabeling can then be visualized. An HIV Western blot is judged to be "reactive" if the patient has reacted to at least two of three gene products (p24, gp41, gp120).

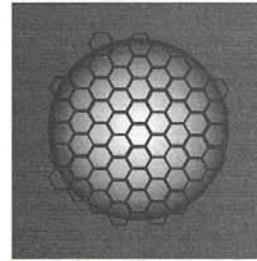
In A Nutshell

ELISA and RIA

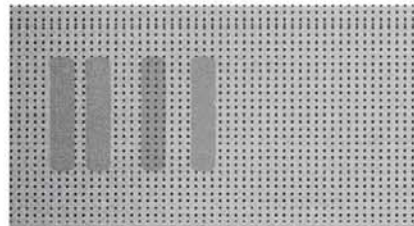
- Extremely sensitive
- ELISA is screening test for HIV

In A Nutshell

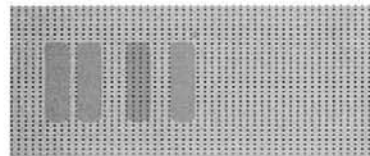
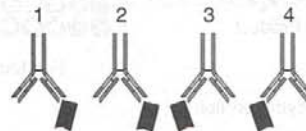
Western blot is a confirmatory test for HIV.



Viral organism separated into constituent proteins



Proteins separated in SDS-PAGE (sodium dodecyl sulfate-polyacrylamide gel electrophoresis)



Separated proteins are transferred to nitrocellulose sheet and reacted with patient's serum. Next, antihuman γ -globulin labeled with an enzyme is reacted for color development and identification. Four different antibodies were identified from this patient.

Figure II-16-6. Western Blot Test

In A Nutshell

Flow cytometry analyzes cell populations in a complex mixture.

This procedure is used to rapidly analyze cell types in a complex mixture and sort them into different populations based on their binding to specific fluorescent dyes. By using antibodies against specific cell-surface markers conjugated to different fluorescent dyes, it is possible to analyze the relative numbers of cells present in a specific tissue location. The apparatus then generates a computerized histogram, reflecting the intensity of the fluorescence and the numbers of cells that have borne that marker. The histogram is produced as a scatter diagram, with cells expressing increasing fluorescence with one dye plotted along the x -axis, and increasing fluorescence with the other dye along the y -axis. Cells with high fluorescence from both dyes will therefore be found in the top right quadrant.

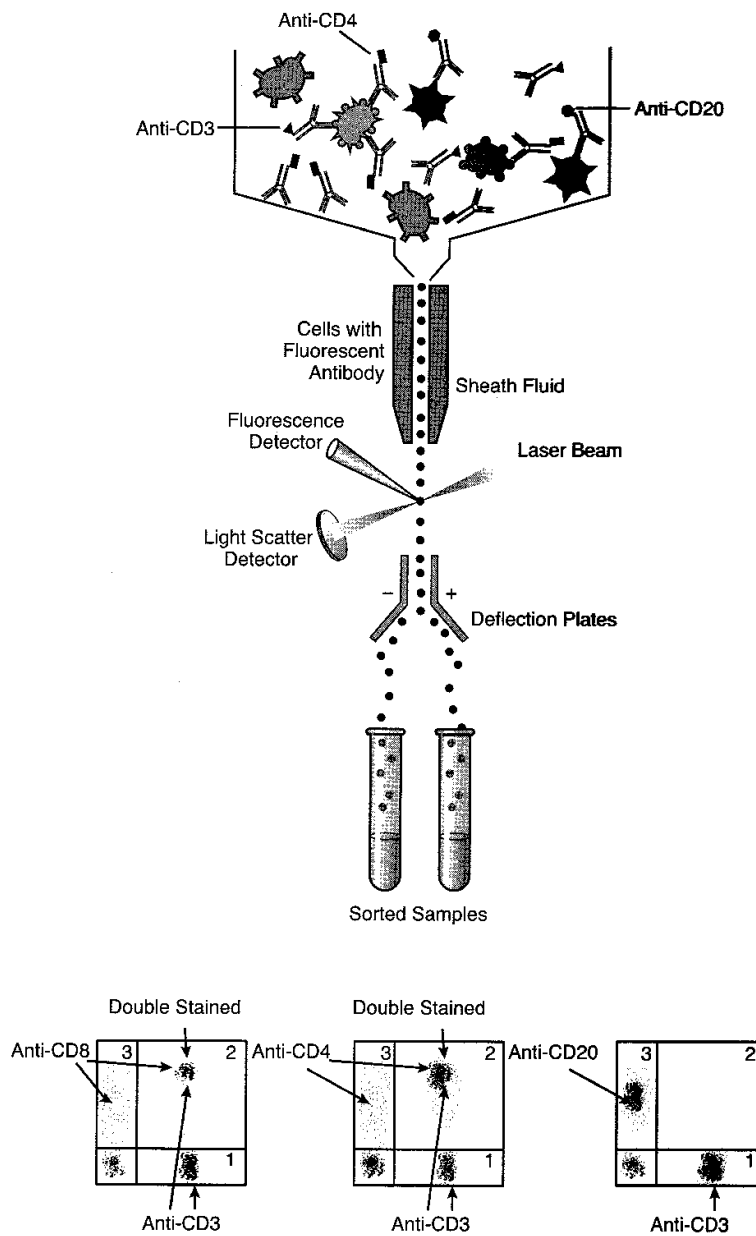


Figure II-16-7. Flow Cytometric Analysis

Chapter Summary

- Antigen-antibody interactions can be visualized in vitro and serve as the basis of many medical diagnostic tests.
- Early in infection when antigen is in excess, only the pathogen's antigens can be detected in patient serum. As antibodies begin to be produced, complexes are formed that precipitate out of the circulation, and the patient enters the equivalence zone. Rising titers of antibody are measured as the patient progresses into the antibody excess zone.
- Agglutination tests are used to measure antibodies that can cause clumping of particles (RBCs and latex beads).
- The direct Coombs test is an agglutination test that detects infants at risk for developing erythroblastosis fetalis. The indirect Coombs test is used to diagnose the presence of antibody in mothers who are at risk of causing this condition in their children.
- The direct fluorescent antibody test is used to detect and localize antigen in patient tissues. The indirect fluorescent antibody test is used to detect antibody production in a patient.
- The radioimmunoassay and enzyme-linked immunoabsorbent assay are extremely sensitive tests that can be modified to detect antigens or antibodies. The ELISA is used as a screening test for HIV infection.
- The Western blot is the confirmatory test for HIV infection.
- Flow cytometry is used to analyze and separate cell types out of complex mixtures.

Review Questions

1. A 16-year-old runaway heroin user visits a family planning/STD clinic irregularly to receive birth control pills. In April 2004, the standard HIV screen performed by this clinic reports back that her test was positive. What does the primary test for HIV infection use?
 - (A) Electrophoresis of HIV antigens in polyacrylamide gel
 - (B) HIV antigen covalently coupled to RBC, patient serum, and anti-immunoglobulin
 - (C) HIV antigen covalently coupled to RBC, patient serum, and complement
 - (D) HIV antigen, patient serum, anti-immunoglobulin serum, and enzyme-substrate ligand
 - (E) HIV antigen, patient serum, anti-immunoglobulin serum, and radioactive ligand
2. A direct Coombs test was performed on a baby in its seventh month of gestation. The mother has had trouble with two earlier pregnancies, and she has never received RhoGAM™. The physician is concerned about the possibility of erythroblastosis fetalis. What ingredients would be necessary to perform this procedure?
 - (A) Mother's serum plus RhoGAM plus Coombs reagent
 - (B) Mother's serum plus Rh-RBCs plus Coombs reagent
 - (C) RhoGAM plus Rh+ RBCs from the baby
 - (D) Rh+ RBCs from the baby plus Coombs reagent
 - (E) Rh+ RBCs plus mother's serum plus Coombs reagent
3. A patient with Chediak-Higashi syndrome is analyzed for ability to mobilize NK cells into the peripheral blood. His peripheral blood leukocytes are treated with fluorescent-labeled antibodies to CD3, CD56, and CD20 before they are passed through a fluorescence-activated cell sorter. The computer-generated results of this process are shown. In which quadrant of which panel would the natural killer cells be found?

Panel 1

Panel 2

- (A) Panel 1, quadrant A
- (B) Panel 1, quadrant B
- (C) Panel 1, quadrant C
- (D) Panel 1, quadrant D
- (E) Panel 2, quadrant A
- (F) Panel 2, quadrant B
- (G) Panel 2, quadrant C
- (H) Panel 2, quadrant D

4. In both ABO blood typing and the Coombs test for detection of hemolytic disease of the newborn, agglutination of coated erythrocytes is a positive test result. Why is addition of Coombs reagent not a necessary step in ABO blood typing?
- (A) All antibodies made in response to blood glycoproteins are IgG
 - (B) Complement-mediated lysis is not important in ABO incompatibilities
 - (C) Coombs serum identifies only anti-Rh antibodies
 - (D) IgM pentamers are large enough to agglutinate erythrocytes directly
 - (E) The high titer of natural isohemagglutinins makes Coombs reagent unnecessary
5. A young woman is in the care of an infertility specialist for evaluation of her inability to conceive since her marriage 5 years ago. As a first step in her examination, cervical scrapings are tested for the possibility of undiagnosed infection with *Chlamydia trachomatis*, which could cause fallopian tube scarring. Which of the following diagnostic tests could be used to identify chlamydial antigens in this specimen?
- (A) Direct fluorescent antibody test
 - (B) Enzyme-linked immunosorbent assay (ELISA)
 - (C) Indirect fluorescent antibody test
 - (D) Radioimmunoassay
 - (E) Western blot

Answers and Explanations

1. **The correct answer is D.** The standard screening test for HIV infection is the enzyme-linked immunosorbent assay, or ELISA. In this test, the virus p24 antigen is coated onto microtiter plates. Serum from the test subjects is added, followed by antihuman-immunoglobulin, which is labeled with an enzyme. When the substrate for the enzyme is added, if the antibodies listed have bound in sequence, there will be a color change in that microtiter well.

Electrophoresis of HIV antigens in polyacrylamide gel (**choice A**) describes the Western blot, which is used as a confirmatory test of HIV infection.

HIV antigen covalently coupled to RBC, patient serum, and anti-immunoglobulin (**choice B**) describes an erythrocyte agglutination test. There is no such test in use for diagnosis of HIV. The indirect Coombs test, which is used to detect Rh- mothers who have become sensitized to the Rh antigens of their fetuses, operates on this principle, however.

HIV antigen covalently coupled to RBC, patient serum, and complement (**choice C**) describes either a complement-fixation or complement-mediated hemolysis assay. There is no such test in use for the diagnosis of HIV.

HIV antigen, patient serum, anti-immunoglobulin serum, and radioactive ligand (**choice E**) describes a radioimmunoassay. This is not used in the standard screening for HIV.

2. **The correct answer is D.** If the child is developing hemolytic disease of the newborn, then his erythrocytes will already be coated with maternal anti-Rh antibodies. Adding Coombs serum (antihuman gammaglobulin) to the baby's RBCs then will cause agglutination. This is the direct Coombs test.

Mother's serum plus RhoGAM plus Coombs reagent (**choice A**) is not a set of reagents that will accomplish any diagnosis. RhoGAM is anti-RhD immunoglobulin, which is given to Rh- mothers at the termination of any Rh+ pregnancy. If the mother is sensitized, she is making IgG antibodies of the same specificity. Adding these three reagents together would tell you nothing of the baby's condition.

Mother's serum plus Rh-RBCs plus Coombs reagent (**choice B**) is not a set of reagents that will accomplish any diagnosis. If the mother is Rh-, she will not make a response to Rh- RBCs, and addition of Coombs reagent will accomplish nothing.

RhoGAM plus Rh+ RBCs from the baby (**choice C**) is not a set of reagents that will accomplish any diagnosis. RhoGAM will bind to Rh+ RBCs from the baby by definition, but adding these reagents together would tell you nothing of the baby's condition.

Rh+ RBCs plus mother's serum plus Coombs (**choice E**) is the set of reagents necessary for the performance of the indirect Coombs test. This is a test used to determine if the mother is making IgG anti-Rh antibodies, which could cross the placenta and harm a fetus. The question asks about the direct Coombs test, not the indirect.

3. **The correct answer is D.** The cell surface marker that would be useful to identify NK cells is CD56. The cells that have the highest fluorescence with antibodies to CD56 are found in quadrant D of panel 1.

Panel 1, quadrant A (**choice A**) contains the cells with maximum fluorescence with antibodies to CD3. These would be T lymphocytes.

Panel 1, quadrant B (**choice B**) contains the cells double-labeled with CD3 and CD56. Because CD3 is the pan-T-cell marker and CD56 is an NK-cell marker, there are no double-labeled cells in this case.

Panel 1, quadrant C (**choice C**) contains the cells that have background fluorescence with both CD3 and CD56. These are non-T, non-NK cells, so they could be B lymphocytes or any other leukocyte.

Panel 2, quadrant A (**choice E**) contains the cells with maximum fluorescence with antibodies to CD3. These are T lymphocytes.

Panel 2, quadrant B (**choice F**) contains double-labeled cells, which fluoresce with both antibody to CD3 and antibody to CD20. Because CD3 is a T-cell marker and CD20 is a B-cell marker, there are no cells in this quadrant.

Panel 2, quadrant C (**choice G**) contains the cells with background fluorescence with both antibodies to CD3 and CD20. These would be non-B, non-T cells and would contain some NK cells, but other leukocytes would be included here, so this is not the best choice.

Panel 2, quadrant D (**choice H**) contains the cells with maximum fluorescence with antibody to CD20, which is a B-cell marker.

4. **The correct answer is D.** Coombs reagent is antihuman IgG. It is necessary in the direct and indirect Coombs tests because in those cases, one is looking for IgG antibodies that could be transported across the placenta to harm an unborn child. IgG is a much smaller molecule than IgM, and is not capable of agglutinating erythrocytes without the addition of a “developing” antibody. In the ABO blood typing test, the important isohemagglutinins are of the IgM variety, capable of agglutinating erythrocytes by themselves because of their sheer size.

The statement that all antibodies made in response to blood glycoproteins are IgG (**choice A**) is not true because isohemagglutinins against ABO blood group antigens are IgM.

The statement that complement-mediated lysis is not important in ABO incompatibilities (**choice B**) is not true because isohemagglutinins of the IgM variety are extremely powerful activators of complement-mediated lysis. The agglutination tests here, however, do not use complement-mediated lysis as the indicator system.

That Coombs serum identifies only anti-Rh antibodies (**choice C**) is not true. Coombs serum is antihuman IgG. It will bind to the Fc portion of any human IgG molecule, regardless of its antigenic specificity.

The statement that the high titer of natural isohemagglutinins makes Coombs reagent unnecessary (**choice E**) is not true. It is the isotype of these antibodies (IgM) and the size of that molecule that allows agglutination to proceed without a developing antibody.

5. **The correct answer is A.** The direct fluorescent antibody test is used to detect antigens in the tissues of a patient.

The enzyme-linked immunosorbent assay (**choice B**) is a test usually used to detect antibody production. It can be modified to detect antigen, but not from a tissue specimen such as this one.

The indirect fluorescent antibody test (**choice C**) is used to detect antibodies being produced in a patient. It is not used for detection of microbial antigens.

Radioimmunoassay (**choice D**) is used to detect antibodies in a patient, not antigen.

Western blot (**choice E**) is used to detect antibodies in a patient, not antigen.

CD Markers

CD Designation	Molecular Family	Cellular Expression	Known Functions
CD2 (LFA-2)	Ig superfamily	T cells, thymocytes, NK cells	Adhesion molecule
CD3	Ig superfamily	T cells, thymocytes	Signal transduction by the TCR
CD4	Ig superfamily	TH cells, thymocytes, monocytes, and macrophages	Signaling and adhesion coreceptor in Ag-induced TH-cell activation, receptor for HIV
CD8	Ig superfamily	CTLs, some thymocytes	Coreceptor for MHC class I-restricted T cells
CD14 (LPS receptor)		Monocytes, macrophages, granulocytes	Binds LPS
CD16 (Fc receptor)	Ig superfamily	NK cells, macrophages, mast cells	Immune complex-induced cellular activation, ADCC
CD18	β chain of LFA-1 family, β_2 integrin subunit	Leukocytes	Complement receptor and cell adhesion molecule
CD19	Ig superfamily	B cells	Coreceptor with CD21 for B-cell activation
CD20	Tetraspan family	Most or all B cells	Unknown role in B-cell activation
CD21 (CR2, C3d receptor)		Mature B cells, follicular dendritic cells	Receptor for complement fragment C3d, forms coreceptor complex with CD19, Epstein-Barr virus receptor

(Continued)

CD Designation	Molecular Family	Cellular Expression	Known Functions
CD28	Ig superfamily	T cells	T-cell receptor for costimulatory molecule B7
CD34	Sialomucin	Precursors of hematopoietic cells, endothelial cells in HEV	Cell-cell adhesion, binds L-selectin
CD40	TNF-R family	B cells, macrophages, dendritic cells, endothelial cells	Binds CD40L, role in T-cell-dependent B cell, macrophage, dendritic cell and endothelial cell activation
CD45 (Leukocyte common antigen)	Tyrosine phosphatase	Hematopoietic cells	Plays critical role in T- and B-cell antigen receptor-mediated signaling
CD56		NK cells	Not known

Cytokines

Cytokine	Secreted by	Target Cell/Tissue	Activity
Interleukin (IL)-1	Monocytes, macrophages, B cells, dendritic cells, endothelial cells, others	TH cells	Costimulates activation
		B cells	Promotes maturation and clonal expansion
		NK cells	Enhances activity
		Endothelial cells	Increases expression of ICAMs
		Macrophages and neutrophils	Chemotactically attracts
		Hepatocytes	Induces synthesis of acute-phase proteins
		Hypothalamus	Induces fever
IL-2	TH cells	Antigen-primed TH and CTLs	Induces proliferation, enhances activity
IL-3	TH cells, NK cells	Hematopoietic cells	Supports growth and differentiation
IL-4	TH2 cells	Antigen-primed B cells	Costimulates activation
		Activated B cells	Stimulates proliferation and differentiation, induces class switch to IgG1 and IgE
		Resting B cells	Upregulates class II MHC expression
IL-5	TH2 cells	Activated B cells	Stimulates proliferation and differentiation, induces class switch to IgA

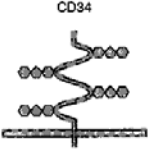
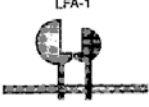
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Cytokine	Secreted by	Target Cell/Tissue	Activity
IL-6	Monocytes, macrophages, TH2 cells, bone marrow stromal cells	Proliferating B cells	Promotes terminal differentiation into plasma cells
		Plasma cells	Stimulates Ab secretion
		Myeloid stem cells	Helps promote differentiation
		Hepatocytes	Induces synthesis of acute-phase proteins
IL-7	Bone marrow, thymic stromal cells	Lymphoid stem cells	Induces differentiation into progenitor B and T cells
IL-8	Macrophages, endothelial cells	Neutrophils	Chemokine, induces adherence to endothelium and extravasation into tissues
IL-10	TH2 cells	Macrophages	Suppresses cytokine production by TH1 cells
IL-12	Macrophages, B cells	Activated CD8+ cells	Acts synergistically with IL-2 to induce differentiation into CTLs
		NK and LAK cells and activated TH1 cells	Stimulates proliferation
Interferon- α	Leukocytes	Uninfected cells	Inhibits viral replication
Interferon- β	Fibroblasts	Uninfected cells	Inhibits viral replication
Interferon- γ	TH1, CTLs, NK cells	Macrophages	Enhances activity
		Many cell types	Increases expression of classes I and II MHC
		Proliferating B cells	Induces class switch to IgG2a, blocks IL-4-induced class switch to IgE and IgG1
		TH2 cells	Inhibits proliferation
		Inflammatory cells	Mediates effects important in DTH

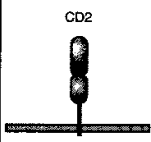
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Cytokine	Secreted by	Target Cell/Tissue	Activity
Transforming growth factor- β	Platelets, macrophages, lymphocytes, mast cells	Proliferating B cells	Induces class switch to IgA
Tumor necrosis factor- α	Macrophages, NK cells	Tumor cells	Has cytotoxic effect
		Inflammatory cells	Induces cytokine secretion, causes cachexia of chronic inflammation
Tumor necrosis factor- β	TH1 and CTL	Tumor cells	Has cytotoxic and other effects, like TNF- α
		Macrophages and neutrophils	Enhances phagocytic activity
Granulocyte colony-stimulating factor (G-CSF)	Macrophages and TH cells	Bone marrow granulocyte precursors	Induce proliferation, used clinically to counteract neutropenia following ablative chemotherapy
Granulocyte-macrophage colony-stimulating factor (GM-CSF)	Macrophages and TH cells	Bone marrow granulocyte and macrophage precursors	Induces proliferation; used clinically to counteract neutropenia following ablative chemotherapy

Adhesion Molecules

Family	Figures	Name	Tissue Distribution	Ligand
Selectins (bind carbohydrates, initiate leukocyte-endothelial interaction)		L-selectin	Leukocytes	Addressins
		P-selectin	Endothelium and platelets	Addressins
		E-selectin	Activated endothelium	Addressins
Mucin-like vascular addressins (bind to L-selectin, initiate leukocyte endothelial interaction)		CD34	Endothelial venules	L-selectin
		GlyCAM-1	High endothelial venules	L-selectin
		MAdCAM-1	Mucosal lymphoid tissue venules	L-selectin
Integrins (bind to cell-adhesion molecules and extracellular matrix; strong adhesion)		LFA-1	Monocytes, T cells, macrophages, neutrophils, dendritic cells	ICAMs
		CR3	Neutrophils, monocytes, macrophages	ICAM-1, iC3b, fibrinogen
		CR4	Dendritic cells, macrophages, neutrophils	iC3b

(Continued)

Family	Figures	Name	Tissue Distribution	Ligand
Immunoglobulin superfamily (various roles in cell adhesion, ligand for integrins)		CD2	T cells	LFA-3
		ICAM-1	Activated vessels, lymphocytes, dendritic cells	LFA-1, CR3
		ICAM-2	Resting vessels, dendritic cells	LFA-1
		ICAM-3	Lymphocytes	LFA-1
		LFA-3	Lymphocytes, antigen-presenting cells	CD2
		VCAM-1	Activated endothelium	VLA-4

Mechanisms of Resistance to Microbial Infections

Category	Pathogen	Protective Mechanism	Evasive Mechanism	Immune Pathology
Prion	CJD, kuru, etc.	No immune response	None	None
Viruses	Naked capsid (all)	Antibody blocks receptor binding		
	Rhinovirus	See above	Antigenic drift	
	Adenovirus	See above	Decrease MHC I expression	
	Enveloped (all)	Antibody blocks receptor binding, C lyses envelope, Ab and C enhances phagocytosis, TH1 cells stimulate CTLs and NK		
	Hepatitis C		Blocks IFN- α and - β	
	Hepatitis B			Immune complexes cause vasculitis
	All Herpesviridae		Nuclear membrane envelope is nonimmunogenic	
	Herpes simplex 1 and 2		Block TAP function (inhibits MHC I expression), viral glycoprotein $\downarrow$ C activation	

(Continued)

Category	Pathogen	Protective Mechanism	Evasive Mechanism	Immune Pathology
	Cytomegalovirus		Generalized immuno-suppression, ↓ MHC I and II expression, produces chemokine receptor and MHC I homologues	
	Epstein-Barr virus		Generalized immuno-suppression, produces molecule homologous to IL-10 (shuts down TH1)	
	Paramyxoviruses		Generalized immuno-suppression	
	Human Immuno-deficiency virus		Infects and kills immuno-competent cells, antigenic drift	
	Rubeola		↓ MHC II expression	
	Influenza		Antigenic shift and drift	
Extracellular bacteria	All	Ab or C enhance phagocytosis or block toxin binding		
Gram-negative extracellular bacteria		C opsonizes		LPS activates macrophages → IL-1 and TNF-α
	<i>Neisseria gonorrhoeae</i>		Antigenic variation of pili	
	<i>Pseudomonas</i>		Inactivates C3a, C5a	

(Continued)

Appendix IV: Mechanisms of Resistance to Microbial Infections

Category	Pathogen	Protective Mechanism	Evasive Mechanism	Immune Pathology
	<i>Borrelia burgdorferi</i>			Immune complexes cause rash, arthritis, neurologic symptoms
Encapsulated bacteria	All	Opsonization with Ab and C	Resist phagocytosis	
IgA protease producers	<i>Streptococcus pneumoniae</i> , <i>Haemophilus influenzae</i> , <i>Neisseria meningitidis</i> and <i>Neisseria gonorrhoeae</i>		Destroy IgA	
Coagulase-positive bacteria	<i>Staphylococcus aureus</i> <i>Yersinia pestis</i>		Inhibit phagocytosis	
Gram-positive, extracellular bacteria	<i>Streptococcus pyogenes</i>			Exotoxins act as superantigens, nonsuppurative sequelae: types II and III hypersensitivities
Catalase-positive bacteria	All		Catalase destroys reactive oxygen intermediates	
	<i>Staphylococcus aureus</i>			Exotoxins act as superantigens
Intracellular bacteria		CD4+ cells activate NK, macrophages, and CTLs		
	<i>Mycobacterium tuberculosis</i>	See above	Sulfatides inhibit phagolysosome formation	Granuloma formation is CMI-mediated
	<i>Mycobacterium leprae</i>	TH1 stimulation is protective (tuberculoid form)	TH2 stimulation results in lepromatous form	Nerve damage in tuberculoid form is DTH-mediated

(Continued)

Category	Pathogen	Protective Mechanism	Evasive Mechanism	Immune Pathology
	<i>Listeria monocytogenes</i>	DTH and CTL	Hemolysin disrupts phagosome membrane, allows escape into cytoplasm	
	<i>Chlamydia trachomatis</i>	DTH and CTLs		Scarring of fallopian tubes is DTH-mediated
Extracellular protozoa	All	Abs to surface molecules plus C cause lysis or opsonize		
	<i>Trypanosoma brucei</i> <i>rhodesiense</i> and <i>gambiense</i>	See above	Antigenic variation of variable surface glycoprotein	
	<i>Entamoeba histolytica</i>	See above	Antigen shedding	
Intracellular protozoa		TH1 cells stimulate phagocytosis, CTLs		
	<i>Plasmodium</i> spp.	Antibodies plus above	Maturation stages change antigens, antigenic variation	
	<i>Trypanosoma cruzi</i>	Amastigotes killed by CMI, trypomastigotes (extracellular) killed by Ab and C		
	<i>Leishmania</i> spp.	TH1 cells stimulate phagocytosis, CTLs		

(Continued)

Appendix IV: Mechanisms of Resistance to Microbial Infections

Category	Pathogen	Protective Mechanism	Evasive Mechanism	Immune Pathology
Helminth parasites	All	ADCC-mediated by eosinophils, macrophages		
	Filarial nematodes		Immuno-suppression secondary to lymphatic obstruction	
	<i>Schistosoma</i> spp.		Adult envelopes itself in host self glycoproteins (ABO and MHC antigens)	Granulomas in liver due to DTH response to egg antigens
Fungi	Most	Phagocytosis by PMNs and macrophages, Abs not protective		
	<i>Cryptococcus neoformans</i>	As above	Inhibits TH1, stimulates TH2, capsule protects against phagocytosis	
	<i>Histoplasma capsulatum</i>	TH1 cell stimulates CTLs and NK		

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